EC325 Workbook

EC325: Econometrics | Colby College

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Preface

Welcome to EC 325: Econometrics at Colby College! This booklet contains the lecture notes, interactive exercises, and R tutorials that will guide you through your first course in econometrics.

What is Econometrics?

Econometrics is the application of statistical methods to economic data with a particular emphasis on *causal inference*—figuring out whether one thing actually *causes* another, rather than merely being correlated with it. This distinction matters enormously for policy: if we want to know whether raising the minimum wage reduces employment, whether health insurance improves health, or whether education increases earnings, we need tools that go beyond simple correlations.

This course will teach you those tools. By the end of the semester, you'll be able to read and critically evaluate empirical research in economics, conduct your own quantitative analyses, and—most importantly—think carefully about when statistical evidence can and cannot support causal claims.

About These Notes

These notes are designed to be a *companion* to the course, not a replacement for the textbook or class participation. They aim to:

- Present core concepts in accessible, narrative form
- Provide extensive R code examples you can run yourself
- Include interactive exercises to test your understanding
- Emphasize intuition through simulations before introducing formulas

The content draws heavily from the following texts, which are required or recommended for this course:

- Wooldridge (2019) — The primary textbook, providing rigorous coverage of econometric methods

- Angrist and Pischke (2014) — An accessible introduction to causal inference and the "credibility revolution" in economics
- Bailey (2020) — A practical guide to doing econometrics in R
- Wickham, Çetinkaya-Rundel, and Grolemund (2023) — The definitive guide to data science in R with the tidyverse

Structure of the Course

The course is organized into three modules:

Module 1: Correlation, Causation, and Regression We begin by establishing why correlation doesn't imply causation and what additional assumptions we need to make causal claims. We then develop the core tool of econometrics—Ordinary Least Squares (OLS) regression—first in its simple bivariate form, then with multiple control variables.

Module 2: Statistical Inference and Real-World Regression With the mechanics of OLS in hand, we turn to quantifying uncertainty. How confident should we be in our estimates? We develop the tools of hypothesis testing, confidence intervals, and statistical significance. We then explore practical issues like functional form, dummy variables, and interactions.

Module 3: Causal Inference Methods The final module covers the major identification strategies economists use to make causal claims from observational data: instrumental variables, difference-in-differences, and regression discontinuity designs.

Roadmap Through the Midterm

The first half of the course (through the midterm) covers the foundations:

- **Chapter 1**: We begin with the central question—does correlation imply causation?—using the example of health insurance and health outcomes. You'll see why simple comparisons can be misleading and why we need econometric tools.

- **Chapter 2**: Before we move onto econometric theory, we cover best practices for data best practices and reproducible research in R

- **Chapter 3**: We introduce the "gold standard" of causal inference—Randomized Control Trials (RCTs)—and the Potential Outcomes Framework, which provides the mathematical foundation for thinking about causality.

- **Chapter 4**: We develop bivariate OLS regression, deriving the estimator, understanding when it gives us causal estimates, and diagnosing when it fails due to omitted variable bias.

- **Chapter 5**: We extend OLS to multiple independent variables, learning how to "control for" confounders and interpret coefficients in the multivariate setting.
- **Chapter 6**: We tackle statistical inference—hypothesis testing, t-statistics, p-values, and confidence intervals—learning to distinguish real effects from statistical noise.

Statistical Software

Modern econometrics is done with statistical software that you interact with through code. In this course, we use **R**, a free and open-source programming language widely used in academia, government, and industry.

Why R? Three reasons:

1. **It's free.** Unlike Stata or SAS, you can install R on any computer and use it forever at no cost.
2. **It's powerful.** R can handle everything from simple regressions to cutting-edge machine learning, beautiful data visualizations, and reproducible research documents (these notes were entirely written in R!).
3. **It's in demand.** R skills are valued by employers in consulting, tech, finance, public policy, and academic research. It regularly ranks among the most widely used programming languages.

Don't worry if you've never programmed before. We'll start from the basics. See Appendix A for instructions on getting set up.

A Note on Learning Econometrics

Econometrics is challenging. It combines economic intuition, statistical theory, and programming skills, often all at once. If you find yourself struggling (and you will), that's normal. Here's my advice:

- **Run the code yourself.** Don't just read the examples—type them out, run them, break them, fix them. You learn programming by doing.
- **Focus on intuition first.** Before memorizing formulas, make sure you understand *why* a method works.
- **Ask questions.** Come to office hours, post on the discussion board, work with classmates.
- **Be patient with yourself.** The skills you're learning will serve you well beyond this course, whether you pursue graduate study, policy work, or a career in the private sector.

About the Instructor

I am a recent PhD from UMass Amherst. My research focuses on the relationship between public policy and maternal, infant, and child health, with particular attention to racial and ethnic health disparities. In my work, I use many of the methods you'll learn in this course.

You can find out more about me and my research at www.raymondcaraher.com.

These notes are a work in progress. If you spot errors or have suggestions for improvement, please let me know!

1 Introduction

Key Questions

- What is the difference between correlation and causation?
- What does an econometrics research project entail?
- What are the main types of data formats we will work with?
- What is statistical software and why is it useful?

Suggested Readings

- Wooldridge (2019), Ch. 1
- Wickham, Çetinkaya-Rundel, and Grolemund (2023), Ch. 1

1.1 What is this Class About?

This class is obviously dedicated to the study of econometrics (endearingly shortened 'metrics). But what exactly does econometrics entail? What differentiates it from broader terms such as "statistical analysis" or "data analysis"?

From Wooldridge (2019), we get a classic definition of econometrics:

> "...statistical methods for estimating economic relationships, testing economic theories, and evaluating and implementing government and business policy."

However, my view of econometrics is at the same time, broader and more narrow:

Definition of Econometrics

The practice of using statistical, quantitative methods to study social phenomena and provide insights into public policy. More specifically, modern econometrics has focused on identifying the **causal effects** of social phenomena/policy as opposed to **correlations**.

In other words, I view the practice of econometrics as the use of quantitative methods to understand the social world around us, with a special (though not exclusive) lens focusing on separating correlation from causation.

1.2 Does Having Health Insurance *Cause* You to be Healthier?

Let's think about a critical question in public policy: *does having health insurance make someone healthier*? This is especially an important question in the context of the United States, which does not have a public health insurance system. Instead, it is a predominantly private system entangled with a complex net of social safety nets meant to provide a patchwork of coverage to the most marginalized.

However, as of 2023, 9.5% of U.S. adults do not have any health insurance coverage.[1] Would the US as a whole be better off guarantee health insurance coverage for everyone? If health insurance makes people healthier, then there may be a strong case to do so. If not, the benefits of such a system would become, at the very least, more opaque.

Let's take a look at life expectancy in the US (a course but important measure of population health), relative to similar high-income, developed countries. Figure 1.1 shows the average life expectancy from 1990 to 2021 for the US, UK, Canada, Australia, and a number of other high-income countries. As can be seen, the US performs considerably worse in terms of life expectancy, and this divergence between peer countries has gotten *larger* since the 1990s, and especially during the COVID-19 pandemic.

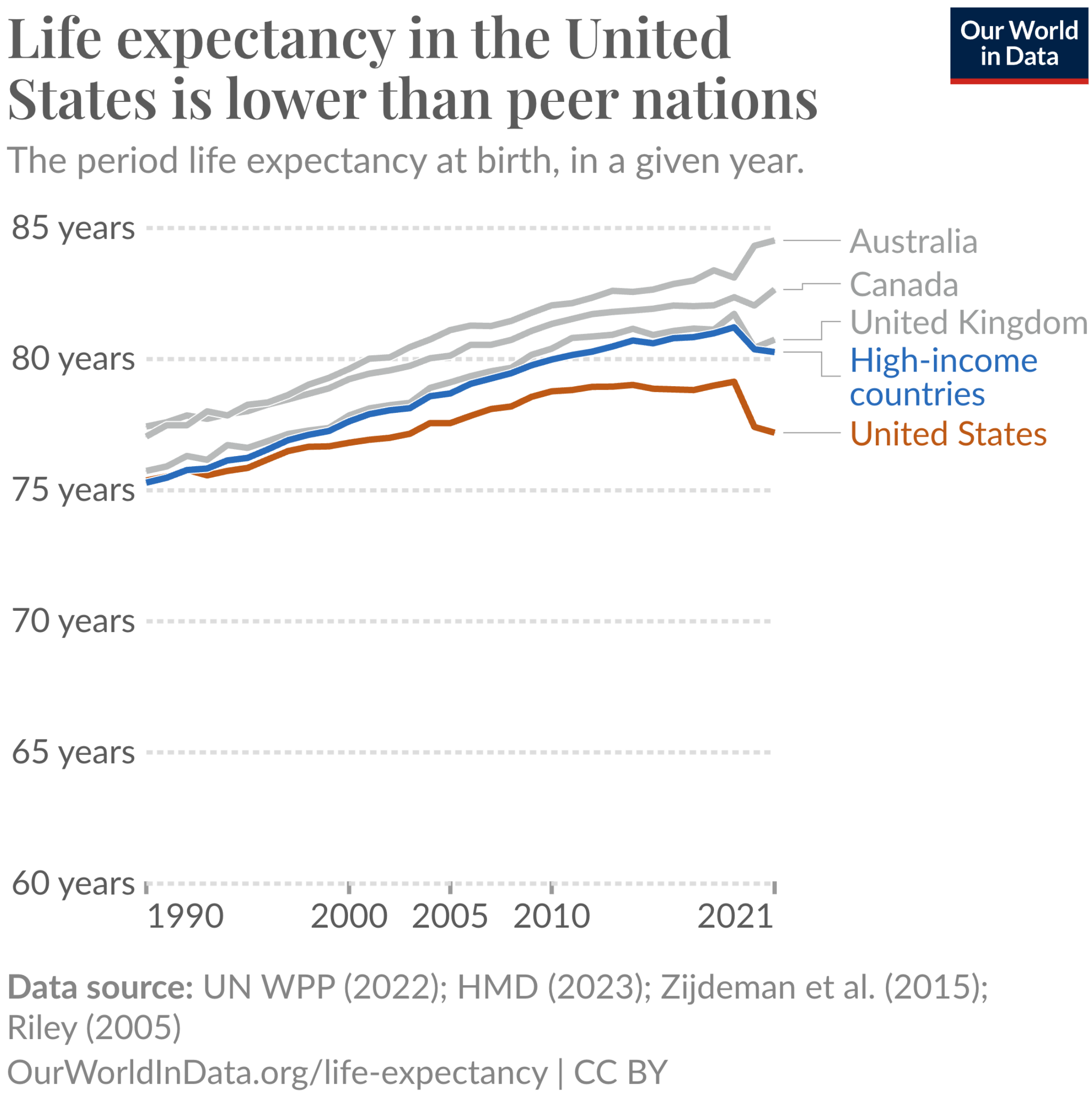
Critically—with the exception of the US—all of the countries in the peer group have some form of universal health care. In other words, all developed countries which are healthier than the US have policies which guarantee the uninsured rate is zero.

This then leads to a critical question: did universal health insurance *cause* these countries to have better health outcomes? Or is this a correlation between policy and outcomes? Certainly, one could make an argument that having health insurance does cause better health outcomes. On the other hand, there are lots of differences between the US and other high-income countries which may also explain the pattern seen in Figure 1.1. For example, maybe these countries have different regulations regarding the food quality, which may explain part of the difference in life expectancy. Or maybe because relatively more people commute to work via bicycle or walking, which may have health benefits compared to driving to work.

These other differences between groups which may *confound* our comparison between countries with universal health insurance and those without also extends to individuals. In the US at least, those with health insurance may be more likely to have a high-paying job which includes insurance as a benefit. Therefore, a comparison between those who have insurance and those who do not will be plagued with the fact that those who have insurance *also are more likely*

[1] Source: KFF

Figure 1.1: Life expectancy in the US vs. peer countries.



Source: Our World in Data

to have a high income job, who may have had better health outcomes regardless of if they had insurance or not. In other words, we may be confusing *correlation* for **causation*.

> **ℹ Correlation vs. Causation**
>
> The **correlation** between two variables describes how they move together. It means two things are *associated*. **Causation** describes a direct relationship where a change in one variable causes a change in another. It means one thing *makes the other happen.*

A key part of the content in this course is to think about when our comparisons are describing a **correlative** relationship or a **causal** relationship. We will study techniques which—if one is convinced that the requisite assumptions hold—will actually be able to recover the *causal effect* of social phenomena, and bring important insights into essential public policy questions like: what is the effect of health insurance on health outcomes? Does going to college cause you to have higher wages? Do immigrants reduce native wages? and does policing cause decreases in crime?

1.3 Broad Steps of Econometric Analysis

Let's say we want to know what the effect of on-the-job training has on your wage. What steps would we need to take to answer this question?

1.3.1 Step 1: Write the Model

Our first step would be to write an *econometric model,* with the goal of using data and econometric tools to *estimate the parameters of the model.* This model can sometimes be derived from a formal model in economics (i.e., the profit maximization of firms), but it is at least based on some intuitive logic which mathematically relates some variable (the **dependent variable**) to some **independent variable**(s).

An example of the type of econometric model you may want to run to study the effect of on-the-job training is:

$$wage = \beta_0 + \beta_1(training) + \mu, \tag{1.1}$$

where *wage* is the *outcome, left-hand side (LHS), predicted, or dependent variable* (these are synonyms for the same thing), and *training* is the *explanatory, right-hand side (RHS), predictor, or independent variable* (again, synonyms for the same thing). The β_0 and β_1 variables are the parameters that we want to *estimate the value of* using econometric methods, and they tell us *how* our *independent* variable is related mathematically to our dependent variable.

1.3.2 Step 2: Find and Explore the Data

The next thing you have to do—after we figure out the model we want to estimate, such as in Equation 1.1—is to get the data. There is a *huge* range of datasets out there, and we are especially lucky that the data-gathering infrastructure in the United States is incredibly robust. As we go through the semester, we will work with as many different datasets that we can, which cover a range of topics: from sports to health, crime, and economic development. While different datasets usually can their own quirks, broadly speaking there are three different types of data we will work with in this class:

1. **Cross-sectional data:** This is data which contains information on units (e.g., individuals, countries, firms, households, etc.) at a *single point in time.* This data is the most common, and also is generally the simplest to work with, so we will start out by working mostly with cross-sectional data.

2. **Time-series data:** This data consists of observations on a single unit (e.g., a country's Gross Domestic Product (GDP), a specific company's stock price, or the national unemployment rate) across multiple time periods (e.g., years, quarters, months, or even minutes). In time series data, a crucial characteristic is the temporal ordering of the observations. This structure is often used for forecasting and modeling how a variable's past values influence its present and future values.

3. **Panel (or Longitudinal) data**: Panel data combines both cross-sectional and time-series dimensions. It consists of observations on multiple units (e.g., individuals, firms, or countries) over multiple time periods. It essentially tracks the same set of cross-sectional units over time. This gives it more information, more variability, and allows researchers to use a rich set of methods to really hone in on the causal effects of whatever social phenomena being studied.

In all of these cases, econometric data is organized in what is called a **matrix** or a **dataframe**. It should look roughly similar to an Excel Spreadsheet, where each column represents a different variable (which is usually labelled, sometimes not), and each row represents the values that a specific unit has across these variables.

For example, the data we use to estimate the model in Equation 1.1 may look like Table 1.1,

where the **person_name** column is (obviously) each person's name, **person_id** column corresponds the unique numerical identifier associated with each person (this will be more common to see in datasets since names are usually anonymous), the **wage** column shows each person's hourly wage in dollars, and the **yrs_training** column shows the number of years each person received on-the-job training. This row-column structure of organizing data is so fundamental that the word *column* is often interchanged with the word *variable*.

When you first look at some data, it is really important to figure out what each row represents. In the simple example shown in Table 1.1, this is straightforward since each row represents a different person. However, in more complex datasets, it may not be immediately obvious what

Table 1.1: Relationship between training and wages for 10 workers

person_name	person_id	wage	yrs_training
Paul	1	18.50	1.00
Ringo	2	22.00	3.00
George	3	17.65	2.50
John	4	27.95	8.20
Linda	5	19.20	0.50
Yoko	6	24.80	6.80
Pattie	7	20.50	5.50
Maureen	8	23.75	4.20
Cynthia	9	16.40	0.80
Barbara	10	28.50	9.10

each row represents. For example, in panel data, each row may represent a specific person at a specific point in time. So row 1 may be Paul in year 2020, row 2 may be Paul in year 2021, row 3 may be Ringo in year 2020, and so on. Figuring out what each row represents is *critical* to understanding the data, as well as how to appropriately analyze it.

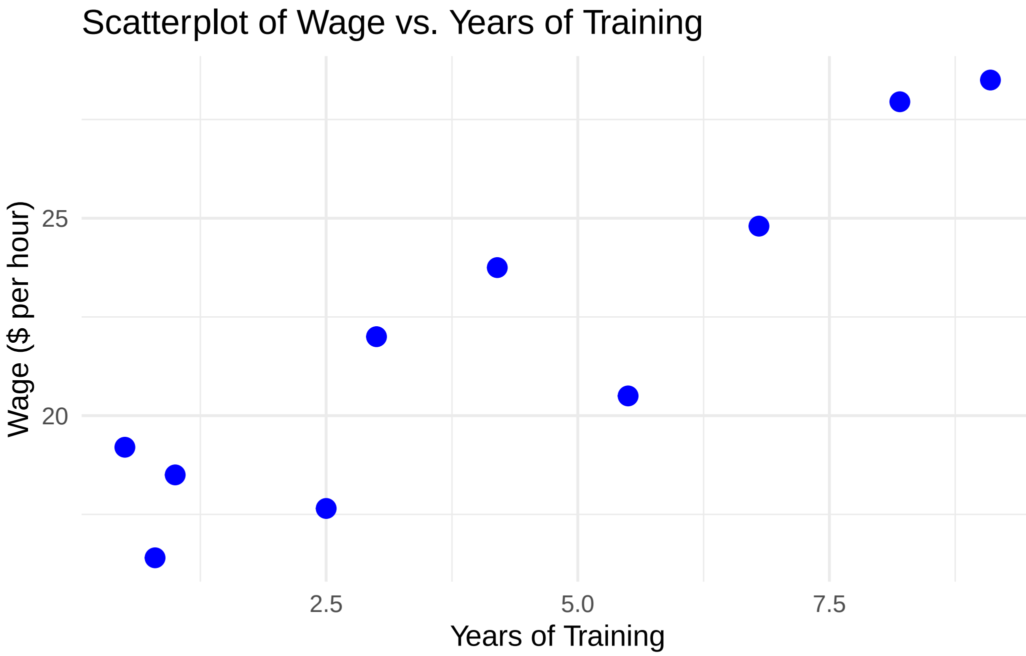
Once we have the data in hand, then we may need to do some data cleaning and organization. This may involve: - Removing missing values - Creating new variables - Changing the format of variables (e.g., from string to numeric) - Merging datasets together - Filtering the data to only include certain observations - And many other tasks.

These are all tasks that in many cases take up the bulk of the time in an econometric research project, and while not the most glamorous part of econometrics, it is absolutely essential to get right.

Before moving on to estimation, it is also important to do some **exploratory data analysis**, (which should be considered step 2.5 in the econometric research process). This involves looking at summary statistics of the data (means, medians, standard deviations, etc.), as well as visualizations (histograms, scatter plots, etc.). Exploratory data analysis is important for understanding the data, as well as for identifying potential issues with the data (e.g., outliers, missing values, etc.).

One of the most important parts of exploratory data analysis is to look at the relationship between the dependent variable and independent variable(s) in your model. One way this can be done is using a **scatterplot**, which plots each observation in the dataset using the independent and dependent variables as coordinates, with the independent variable on the x-axis and the dependent variable on the y-axis.

Figure 1.2: Relationship between training and wages for 10 workers



In Figure 1.2, we see a scatterplot of the relationship between years of training and wages for the 10 workers shown in Table 1.1. From this plot, we can see that there is a positive relationship between years of training and wages: as years of training increases, wages also tend to increase. While not good enough to draw any strong conclusions, one can see how this type of exploratory data analysis is useful for understanding the data and the relationship between variables.

1.3.3 Step 3: Estimate the Model

Once we have a model in mind as well as the appropriate data, we can then use our econometric methods to estimate the relationship. The bulk of this class is dedicated to this step in the econometric research process, but the other stages are no less important! The most common way that econometric models are estimated is with **Ordinary Least Squares** (OLS), which you may remember from your pre-req. In addition to estimating the actually value of the parameters in the model, we also want some idea of how *uncertain* we are about the estimated parameter value. This is where notions of statistical significance and confidence intervals come in.

1.3.4 Step 4: Interpret the Estimate

Once we have an estimated number for the parameter in our model, we then have to interpret the number. Here, the phrase "interpret the estimate" means two parallel things. First, there is the literal, mathematical interpretation of the model. This will change depending on the functional form we use to estimate the model, and we will spend lots of time learning about this. The second part of "interpreting" the model is about what the results mean in the context of your broader research question. Is your estimate big or small? Is it economically important? Do you actually think it reflects a causal effect? These are all types of interpretation which are a critical part of econometrics research, but for which there is not usually a formula we can use. A large part of the "art" working on an econometrics project is exactly this type of interpretation, and how you form it into a compelling narrative. While there are some tools we will cover which will help in this regard, in my opinion, nothing will help you more with this than just reading lots of good, well-written econometrics papers.

1.4 Statistical Software

Much of the econometric process is done using specialized statistical software. We will use this software to: "clean" our data and organize in the appropriate row-column format to do our econometric estimations; explore our data using descriptive statistics (means, medians, etc.) and visualizations (an increasingly important part of a compelling econometric project); estimate our model and conduct analyses of statistical uncertainty; and output our results to easy-to-interpret tables and figures.

There are a few commonly used econometrics software:

- **Stata:** Proprietary software purpose-built for econometrics
- **R:** Open-source programming language with a focus on statistical analysis
- **Python:** Open-source programming language with a focus on data science more broadly
- **EViews:** Proprietary software with a focus on forecasting and other time series applications

Each one of these software of their own pros and cons, and it is worth knowing several quite well. For the purposes of this class, we will learn the **R** programming language to do econometrics. If you haven't take the time to review Appendix A to learn more about R, how to install it, and to run your first chunk of code!

1.5 Summary and Conclusion

This chapter introduced the fundamental framework that will guide our study of econometrics throughout this course. We defined econometrics as the use of quantitative methods to understand social phenomena and inform public policy, with a particular focus on distinguishing between correlation and causation. Through the example of health insurance and health outcomes, we saw how patterns in data can be misleading. While the United States lags behind peer countries in life expectancy, and those peer countries all have universal health insurance, we cannot immediately conclude that universal coverage causes better health outcomes. The tools you will learn in this course are designed to help you identify causal relationships rather than mere associations.

1.5.1 Key Takeaways:

- **Econometrics vs. correlation:** Modern econometrics focuses on identifying causal effects, not just correlations between variables
- **Four steps of econometric research:**
  1. Write an econometric model
  2. Find and explore the appropriate data
  3. Estimate the model parameters
  4. Interpret the results
- **Three main data types:** Cross-sectional (single point in time), time-series (one unit over time), and panel data (multiple units over time)
- **Statistical software:** We will use **R** to clean data, estimate models, and present results

As we move forward, remember that econometrics is as much an art as it is a science. The mathematical tools we will learn are powerful, but they are only as good as the questions we ask and the care with which we interpret our results.

1.6 Check Your Understanding

Note: This section contains interactive content only available in the HTML version.

2 Good Data Practices

Key Questions

- Why do data errors matter for econometric analysis?
- What are descriptive statistics and why should we examine them?
- How can data visualization help us detect problems?
- What is replication and why is it fundamental to credible research?

Suggested Readings

- Bailey (2020), Ch. 2

Before we dive into sophisticated econometric methods, we need to establish good habits for working with data. Even the most advanced statistical techniques cannot rescue an analysis built on flawed or misunderstood data. This chapter focuses on the essential first steps: understanding our data through descriptive statistics and visualization, and documenting our work so that others (including our future selves) can verify and build upon it.

2.1 When Good Economists Use Bad Data

In 2010, economists Carmen Reinhart and Kenneth Rogoff published influential research on government debt and economic growth. Their analysis covered more than 3,700 annual observations from countries around the world, and their key finding was striking: when a country's government debt exceeded 90% of GDP, economic growth dropped dramatically.

The policy implications seemed clear: governments should be cautious about using deficit spending to fight unemployment. This finding influenced policy debates across the world during a period when many countries were considering how aggressively to respond to the Great Recession.

There was just one problem. The data didn't quite say what Reinhart and Rogoff thought it did.

In 2014, a graduate student named Thomas Herndon was trying to replicate their results for a homework assignment. He couldn't get the numbers to match. After obtaining the original

Excel spreadsheet from Reinhart and Rogoff, Herndon and his advisors discovered three issues: some observations had been mistakenly excluded from key calculations, others contained typos, and—most embarrassingly—some cells in the Excel spreadsheet had simply not been included in the formula calculating the average.

When the data were corrected, the dramatic "cliff" in economic growth disappeared. While growth did decline somewhat as debt increased, the relationship was much more gradual than originally reported. The corrected analysis told a very different policy story.

> **!** The Lesson
>
> Even accomplished researchers can make data mistakes. The best defense is establishing habits that minimize errors and make them easier to detect when they occur.

Reinhart and Rogoff's experience teaches us that data management is not just tedious housekeeping—it's fundamental to credible research. No amount of sophisticated econometric technique can compensate for errors in the underlying data. This chapter introduces practices that help us avoid such problems.

2.2 Know Your Data

The first rule of data analysis is simple: **know your data**. This means examining it carefully before conducting any formal analysis. We want to understand what we're working with, detect potential errors, and develop intuition about the relationships we might find.

2.2.1 Descriptive Statistics

For every variable in our dataset, we should know several key statistics: the number of observations, the mean, the standard deviation, and the minimum and maximum values. These statistics give us a feel for the data and help us spot problems.

Let's look at a simple example. Suppose we're studying the relationship between education and wages. Here's simulated data for 100 individuals:

```
# Set seed for reproducibility
set.seed(42)

# Generate simulated wage and education data
n <- 100
education <- round(rnorm(n, mean = 14, sd = 2.5))
education <- pmax(8, pmin(education, 20))  # Constrain between 8 and 20 years
```

	Unique	Missing Pct.	Mean	SD	Min	Median	Max
education	12	0	14.1	2.6	8.0	14.0	20.0
wage	100	0	44.9	7.9	23.2	45.6	60.8
age	33	0	35.3	9.4	22.0	35.0	60.0

	Unique	Missing Pct.	Mean	SD	Min	Median	Max
education	13	0	13.9	3.2	-5.0	14.0	20.0
wage	100	0	52.8	80.9	23.2	45.6	850.0
age	34	0	37.0	19.0	22.0	35.0	200.0

```r
# Generate wages with some relationship to education
wages <- 10 + 2.5 * education + rnorm(n, mean = 0, sd = 5)
wages <- pmax(wages, 8)  # Minimum wage floor

# Create age variable
age <- round(rnorm(n, mean = 35, sd = 10))
age <- pmax(age, 22)  # Minimum age

# Create a tibble
wage_data <- tibble(
  id = 1:n,
  education = education,
  wage = wages,
  age = age
)

# Display descriptive statistics
datasummary_skim(wage_data |> select(education, wage, age),
                 title = "Descriptive Statistics: Education and Wages",
                 histogram = FALSE)
```

These statistics immediately tell us several things. We have 100 observations with no missing data (N = 100 for all variables). Education ranges from 8 to 20 years, with a mean around 14 years—this seems reasonable for a sample of working adults. Wages range from about \$8 to \$63 per hour, with substantial variation (standard deviation of about \$10).

But what if our summary statistics looked like this instead?

Now we have a problem! The maximum wage of \$850 per hour seems suspicious—is this real or a data entry error? The minimum education value of -5 years is impossible. And an age of 200 is clearly wrong. These obvious errors suggest we need to investigate our data more carefully.

> **ℹ The Standard Deviation**
>
> The **standard deviation** measures how dispersed the values of a variable are around the mean. A larger standard deviation indicates more spread-out values. For a variable X with mean $\bar{X}$, the standard deviation is:
>
> $$SD(X) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(X_i - \bar{X})^2}$$

2.2.2 Examining Distributions

When a variable takes on a limited number of values, it's helpful to examine the frequency distribution. Suppose we have a variable indicating whether someone has a college degree (1 = yes, 0 = no):

```r
# Add college degree variable
wage_data <- wage_data |>
  mutate(college = ifelse(education >= 16, 1, 0))

# Create frequency table
wage_data |>
  count(college) |>
  mutate(percent = n / sum(n) * 100) |>
  knitr::kable(
    col.names = c("College Degree", "Count", "Percent"),
    digits = 1,
    caption = "Distribution of College Degree"
  )
```

Table 2.1: Distribution of College Degree

College Degree	Count	Percent
0	67	67
1	33	33

This tells us that 29% of our sample has a college degree. But imagine our frequency table looked like this:

Table 2.2: A Problematic Distribution

College Degree	Count	Percent
0	30	58.8
1	20	39.2
100	1	2.0

We have a value of 100 for the college degree variable. Either we have someone with 100 college degrees (unlikely), or—much more probably—we have a coding error. This is the kind of problem that descriptive statistics help us catch.

2.3 Visualizing Data

Numbers are useful, but graphs often reveal patterns and problems that summary statistics miss. Effective data visualization is both a diagnostic tool and a way to understand relationships in our data. In econometrics, we rely heavily on visualizations to explore data, diagnose problems, and communicate results.

! Why Visualization Matters

A good data visualization translates complex data into a clear visual format, revealing patterns, trends, and relationships that would be difficult to see in tables of numbers. The goal is to tell an accurate and compelling story that allows your audience to quickly grasp key insights.

2.3.1 The Grammar of Graphics: ggplot2

We'll use `ggplot2` (part of the tidyverse) for creating visualizations. The "gg" stands for "grammar of graphics"—a systematic way of thinking about how to build plots.

Every ggplot has two main components:

1. **The scaffolding**: titles, axis labels, legends, sources
2. **The content**: the actual data encoded as visual elements (points, lines, bars, etc.)

The key insight of ggplot2 is that we build plots **layer by layer**, adding elements one at a time until we have a complete visualization.

2.3.2 Getting Started: The Palmer Penguins

Let's use real data to learn ggplot2. The Palmer Penguins dataset contains measurements of 344 penguins from three species collected near Palmer Station, Antarctica.

```r
# Load the penguins data
library(palmerpenguins)

# Take a look at the data
glimpse(penguins)
```

```
Rows: 344
Columns: 8
$ species           <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
$ island            <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgerse~
$ bill_length_mm    <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, ~
$ bill_depth_mm     <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, ~
$ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186~
$ body_mass_g       <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, ~
$ sex               <fct> male, female, female, NA, female, male, female, male~
$ year              <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

The data includes species, island location, bill measurements, flipper length, body mass, sex, and year of observation.

2.3.3 Building a Plot: Step by Step

Let's answer a research question: **Do penguins with longer flippers weigh more?**

We start by creating an empty plot object:

```r
ggplot(data = penguins)
```

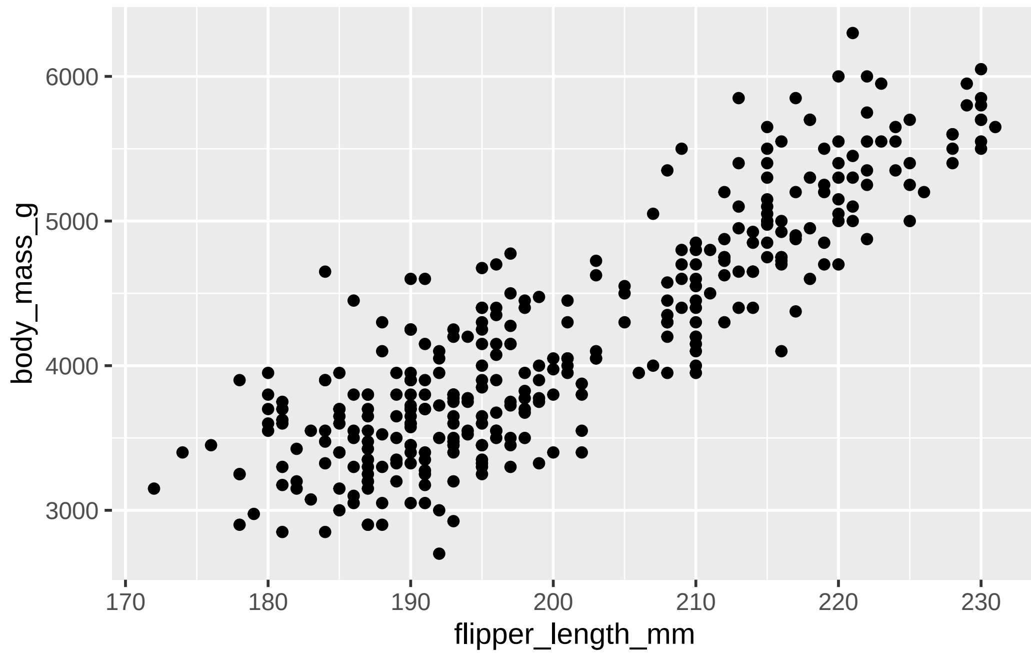
Figure 2.1: An empty ggplot object with no layers.

This creates the plotting area but doesn't show any data yet.

Now let's add a layer with points. We need to specify: - The **geometry** (what shape): `geom_point()` for a scatterplot - The **aesthetic mapping** (which variables go where): `aes(x = ..., y = ...)`

```
ggplot(data = penguins) +
  geom_point(aes(x = flipper_length_mm, y = body_mass_g))
```

Figure 2.2: A basic scatterplot showing flipper length versus body mass.



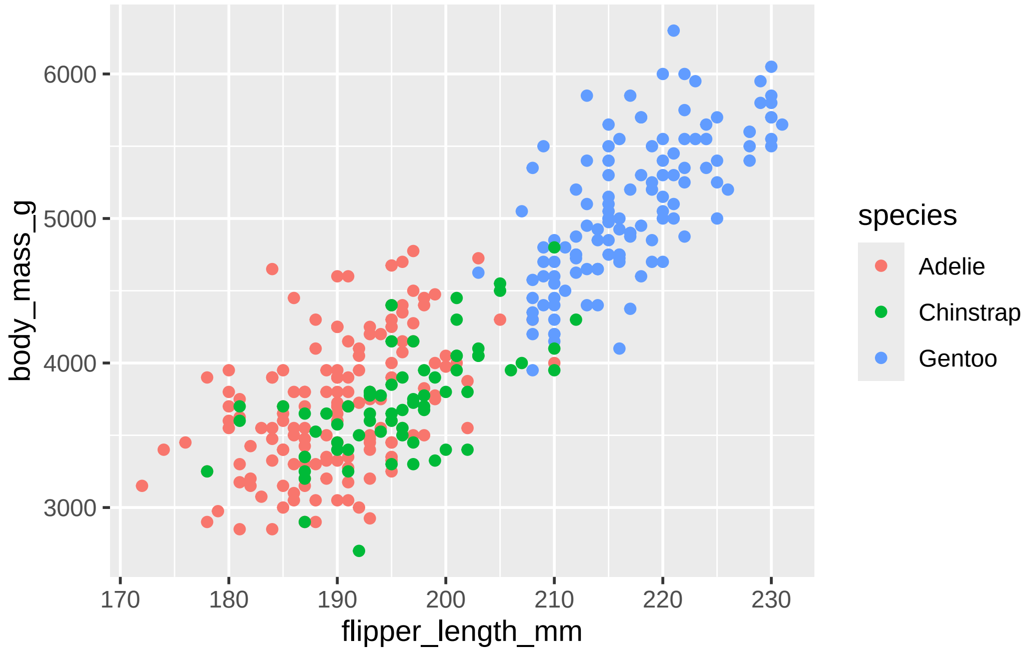
The plot clearly shows a positive relationship: penguins with longer flippers tend to be heavier.

2.3.4 Adding More Information: Color and Shape

But wait—could this relationship be driven by different penguin species? Maybe larger species have both longer flippers and greater body mass. We can encode species information using color:

```
ggplot(data = penguins) +
  geom_point(aes(x = flipper_length_mm, y = body_mass_g, color = species))
```

Figure 2.3: Scatterplot with species indicated by color. The positive relationship holds within each species.



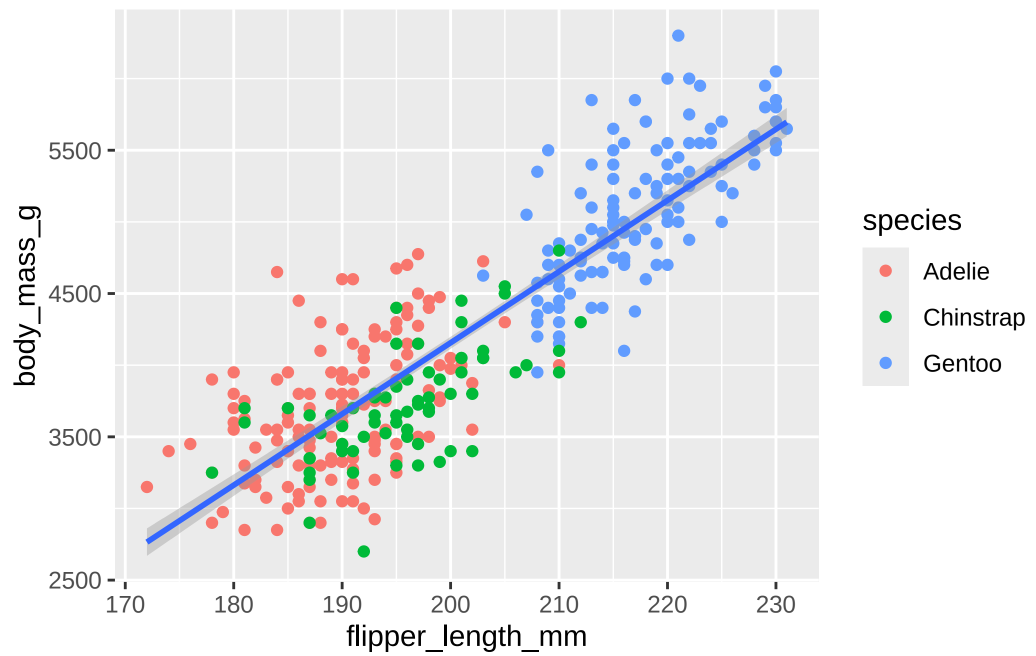
Now we can see that: - Gentoo penguins (blue) are generally larger - The positive relationship between flipper length and body mass exists **within each species** - This suggests the relationship isn't just due to species differences

2.3.5 Adding a Regression Line

Let's add the regression line to visualize the overall trend. This requires a new **layer** (not a new aesthetic):

```
ggplot(data = penguins) +
  geom_point(aes(x = flipper_length_mm, y = body_mass_g, color = species)) +
  geom_smooth(aes(x = flipper_length_mm, y = body_mass_g), method = "lm")
```

Figure 2.4: Scatterplot with regression line. The blue line shows the estimated linear relationship.



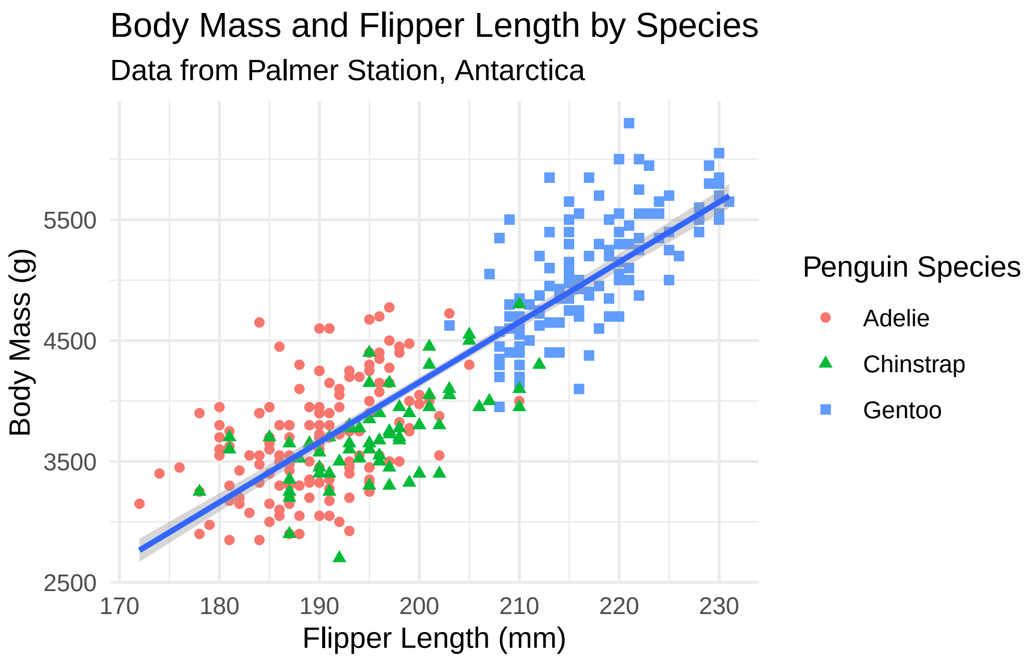
The `geom_smooth()` layer adds a line of best fit using linear regression (`method = "lm"`).

2.3.6 Proper Labeling

Our plot conveys information well, but the axis labels and title need improvement. Good labels are essential for clear communication:

```
ggplot(data = penguins) +
  geom_point(aes(x = flipper_length_mm, y = body_mass_g, color = species, shape = species)) +
  geom_smooth(aes(x = flipper_length_mm, y = body_mass_g), method = "lm") +
  labs(
    title = "Body Mass and Flipper Length by Species",
    subtitle = "Data from Palmer Station, Antarctica",
    x = "Flipper Length (mm)",
    y = "Body Mass (g)",
    color = "Penguin Species",
    shape = "Penguin Species"
  ) +
  theme_minimal()
```

Figure 2.5: A properly labeled scatterplot with clear titles and axis labels.



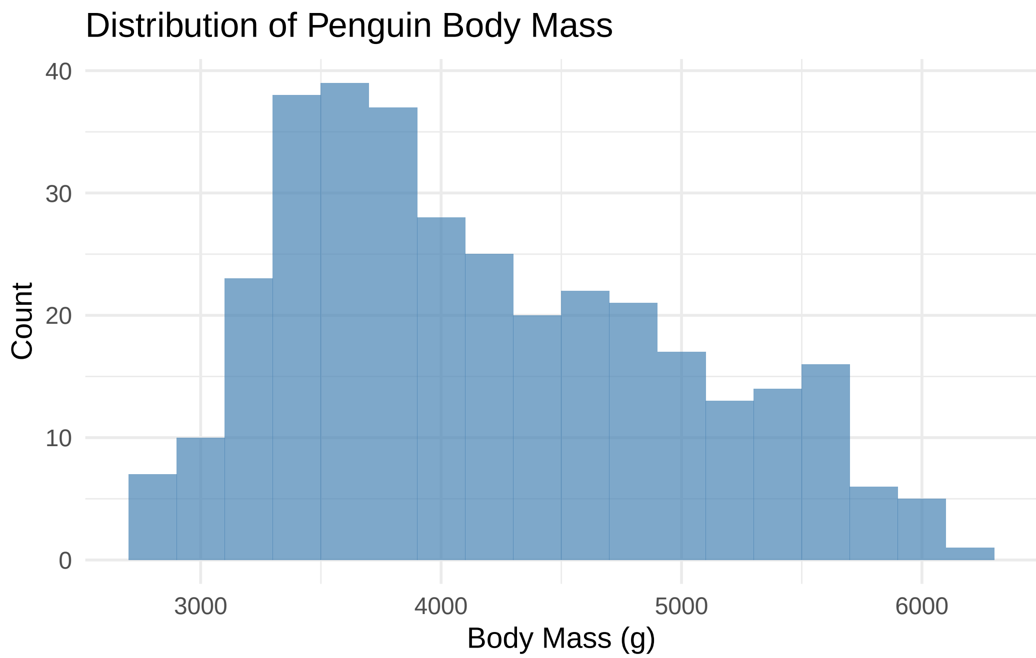
Notice how much clearer this is! The plot now includes: - A descriptive title - Proper axis labels with units - A data source in the subtitle - Readable legend labels

2.3.7 Histograms for Distributions

Histograms show the distribution of a continuous variable by dividing it into bins:

```
ggplot(penguins) +
  geom_histogram(aes(x = body_mass_g), binwidth = 200, fill = "steelblue", alpha = 0.7) +
  labs(
    title = "Distribution of Penguin Body Mass",
    x = "Body Mass (g)",
    y = "Count"
  ) +
  theme_minimal()
```

Figure 2.6: Histogram showing the distribution of penguin body mass.



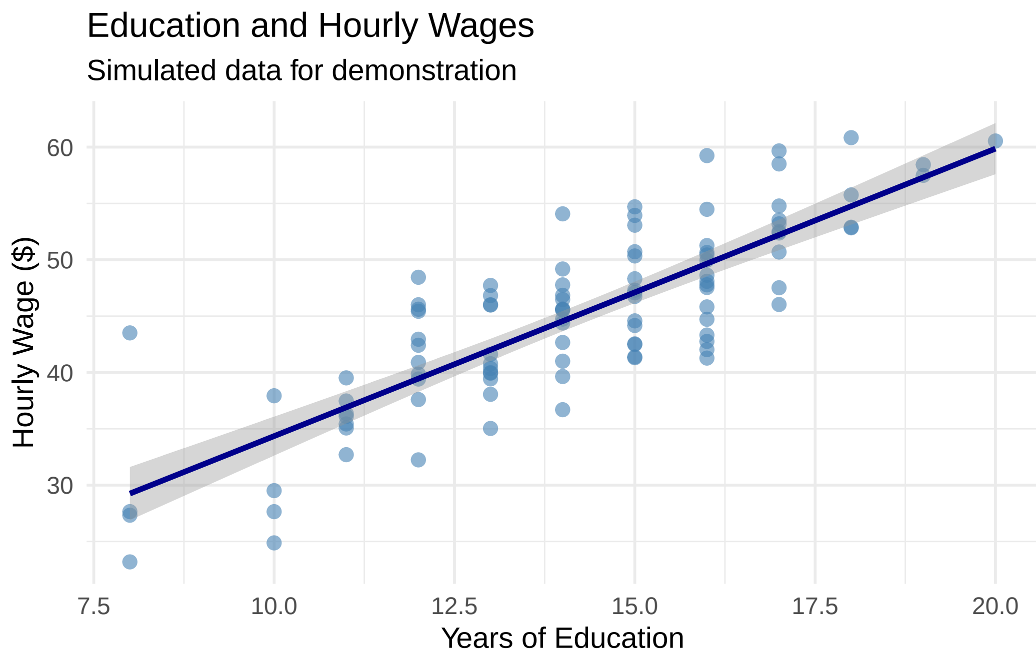
The `binwidth` parameter controls how wide each bar is. Smaller binwidths create more bars and show more detail.

2.3.8 A Simple Example with Our Wage Data

Let's apply these principles to our wage and education data:

```
ggplot(wage_data, aes(x = education, y = wage)) +
  geom_point(alpha = 0.6, size = 2, color = "steelblue") +
  geom_smooth(method = "lm", se = TRUE, color = "darkblue") +
  labs(
    title = "Education and Hourly Wages",
    subtitle = "Simulated data for demonstration",
    x = "Years of Education",
    y = "Hourly Wage ($)"
  ) +
  theme_minimal()
```

Figure 2.7: The relationship between education and wages with proper labeling and styling.



> **Building Intuition**
>
> The layered approach of ggplot2 mirrors how we think about data analysis: - Start with the data - Add visual representations - Refine with labels and formatting Experiment with different geometries, aesthetics, and layers to find the visualization that best tells your data's story.

2.4 Replication: The Foundation of Credible Research

Science depends on replication. If other researchers cannot reproduce our results using the information we provide, then our findings cannot be verified or built upon. This is not just an abstract principle—it's a practical necessity.

> **The Replication Standard**
>
> Research meets the **replication standard** when an independent researcher can exactly duplicate the results using only the information provided.

2.4.1 Why Replication Matters

Replication serves several crucial purposes. First, it allows others to check our work. The Reinhart-Rogoff example shows why this matters: only because they shared their data could Thomas Herndon discover the errors. Without replication materials, the mistakes might never have been found.

Second, replication enables others to probe our analysis. Statistical results often hinge on seemingly small decisions—which observations to include, how to handle missing data, which variables to control for. Skeptical readers (which should be everyone!) want to see whether reasonable alternative approaches produce similar conclusions. If a certain choice substantially changes our results, we need to pay attention.

Third, committing to a replication standard keeps our work honest. If we know others will check our work, we're less likely to make choices based on getting the "right" answer rather than statistical merit.

Finally, replication helps our future selves. Even a moderately complex project can involve hundreds of decisions over weeks or months. What seemed obvious at the time can become completely opaque later. Good documentation means we can return to a project after a break and immediately understand what we did.

2.4.2 What Makes Good Replication Materials?

Effective replication files have two components: a **codebook** that documents the data, and **analysis code** that shows exactly how results were produced.

The codebook describes where each variable comes from and any transformations applied. For example:

- **Variable name**: `wage`
- **Description**: Hourly wage in dollars, calculated as annual salary divided by hours worked in past year
- **Source**: National Longitudinal Survey of Youth, 1996 wave
- **Notes**: Wages above $200/hour were excluded as likely data errors

The analysis code shows the exact steps used to produce each result. This means providing the actual R code, not just descriptions of what we did. And crucially, the code should include comments explaining *why* we made certain choices:

```
# Estimate relationship between education and wages
# We control for age because it affects both education and wages
# (older workers have had more time to complete education and gain experience)
```

```r
model <- lm(wage ~ education + age, data = wage_data)
summary(model)
```

2.4.3 Good Coding Practices

As we write code for our analysis, several practices make our work easier to understand and verify:

Use descriptive variable names. Compare `x1` versus `years_education`—the second is much clearer.

Comment extensively. Explain what each section of code does and why you made certain choices. Your future self will thank you.

Organize your code logically. Start with data loading and cleaning, then move to analysis, then to creating tables and figures.

Make your code reproducible. Use `set.seed()` before any random operations so results don't change each time you run the code.

Here's an example of well-documented code:

```r
# =============================================================================
# Analysis of Education and Wages
# Author: Your Name
# Date: December 2024
# =============================================================================

# Load required packages
library(tidyverse)

# Set seed for reproducibility
set.seed(42)

# -----------------------------------------------------------------------------
# 1. LOAD AND CLEAN DATA
# -----------------------------------------------------------------------------

# Load data from National Longitudinal Survey
# Downloaded from: https://www.nlsinfo.org/content/cohorts/nlsy79
# Download date: December 1, 2024
wage_data <- read_csv("nlsy_wages.csv")

# Remove observations with missing wage data
```

```r
# This affects 23 observations (2.3% of sample)
wage_data <- wage_data |>
  filter(!is.na(wage))

# Create college degree indicator
# Defined as 16+ years of education (bachelor's degree)
wage_data <- wage_data |>
  mutate(college = ifelse(education >= 16, 1, 0))

# ------------------------------------------------------------------------------
# 2. DESCRIPTIVE STATISTICS
# ------------------------------------------------------------------------------

# Summary statistics for main variables
summary(wage_data |> select(wage, education, age))

# ------------------------------------------------------------------------------
# 3. MAIN ANALYSIS
# ------------------------------------------------------------------------------

# Estimate basic relationship between education and wages
# Simple regression without controls
model1 <- lm(wage ~ education, data = wage_data)

# Add age as control variable
# Age affects both education completion and wage levels
model2 <- lm(wage ~ education + age, data = wage_data)
```

2.5 A Quick Example: State Crime Data

Let's practice these principles with a small example. We'll look at violent crime rates across U.S. states and potential correlates.

```r
# Create simulated state crime data for demonstration
set.seed(123)
n_states <- 50

state_data <- tibble(
  state = state.abb,
  violent_crime = rnorm(n_states, mean = 400, sd = 150),
```

	Unique	Missing Pct.	Mean	SD	Min	Median	Max
violent_crime	50	0	405.2	138.9	105.0	389.1	725.3
percent_urban	49	0	72.1	13.4	35.4	72.3	100.0
percent_poverty	50	0	13.2	3.0	8.0	13.1	20.3

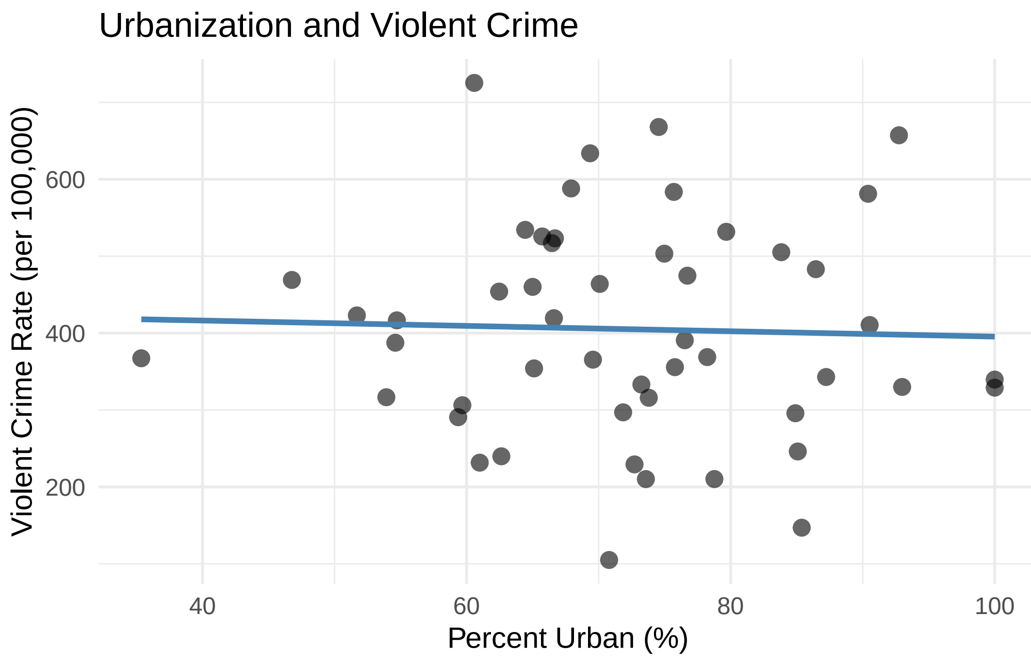
```r
  percent_urban = rnorm(n_states, mean = 70, sd = 15),
  percent_poverty = rnorm(n_states, mean = 14, sd = 3)
) |>
  mutate(
    violent_crime = pmax(violent_crime, 100),  # Keep positive
    percent_urban = pmax(20, pmin(percent_urban, 100)),  # Constrain to 0-100
    percent_poverty = pmax(8, pmin(percent_poverty, 22))  # Reasonable range
  )

# Display descriptive statistics
datasummary_skim(state_data |> select(-state),
                 title = "State Crime Data: Descriptive Statistics",
                 histogram = FALSE)
```

The descriptive statistics look reasonable. Crime rates range from about 150 to 750 per 100,000 people. The urbanization variable is measured on a 0-100 scale, while poverty is measured as a percentage (8% to 22%). Understanding these scales is important for interpreting results.

Now let's visualize the relationship between urbanization and crime:

```r
ggplot(state_data, aes(x = percent_urban, y = violent_crime)) +
  geom_point(alpha = 0.6, size = 2.5) +
  geom_smooth(method = "lm", se = FALSE, color = "steelblue") +
  labs(
    x = "Percent Urban (%)",
    y = "Violent Crime Rate (per 100,000)",
    title = "Urbanization and Violent Crime"
  ) +
  theme_minimal()
```

Figure 2.8: Relationship between urbanization and violent crime rates across U.S. states.



The scatterplot suggests a positive relationship: more urbanized states tend to have higher violent crime rates. But this is just correlation, not causation. Many factors differ between urban and rural states beyond just population density, and any of these could be driving the relationship we observe. This is precisely the kind of endogeneity problem that motivates the econometric methods we'll study in later chapters.

2.6 Summary

Good data practices are the foundation of credible econometric analysis. Before conducting any formal statistical tests, we must understand our data through descriptive statistics and visualization. These diagnostic tools help us catch errors, understand variable scales and distributions, and develop intuition about relationships in the data.

Equally important is documenting our work so that others can replicate our results. A replication file should include a codebook describing data sources and transformations, plus the complete analysis code with clear comments explaining our choices. This commitment to transparency keeps our work honest, enables others to verify and build on our findings, and helps our future selves understand what we did months or years later.

The Reinhart-Rogoff case reminds us that even accomplished researchers make data mistakes. The best defense is establishing good habits: examine your data carefully, document everything, and write code that your future self (and others) can understand.

2.7 Check Your Understanding

Note: This section contains interactive content only available in the HTML version.

3 Causality and Randomized Control Trials

Key Questions

- Why can't we interpret correlations as causal effects?
- What is selection bias and how does it arise?
- How do randomized control trials eliminate selection bias?
- What is the potential outcomes framework and how does it formalize causal inference?

Suggested Readings

- Angrist and Pischke (2014), Ch. 1

In Chapter 1, we discussed how econometrics focuses on identifying **causal effects** rather than mere correlations. We used the example of health insurance: while people with health insurance tend to be healthier than people without insurance, we cannot immediately conclude that health insurance *causes* better health. This chapter develops the conceptual and mathematical tools we need to think rigorously about causality, and introduces the randomized control trial (RCT) as the gold standard for causal inference.

3.1 The Problem with Simple Comparisons

Let's return to our health insurance example. People with health insurance are, on average, healthier than people without it. This correlation is undeniable. But can we interpret this as evidence that insurance *causes* better health?

The problem is that people who have health insurance differ from those who don't in many ways beyond just their insurance status. Those with insurance tend to be wealthier, more educated, and have more stable employment. All of these factors independently affect health outcomes. When we compare the health of insured and uninsured individuals, we're not just comparing the effect of insurance—we're comparing people who differ along many dimensions.

> **Selection Bias**
>
> **Selection bias** occurs when we compare groups that differ systematically in ways beyond just the treatment we're studying. In other words, we are comparing **apples to oranges**.

The comparison we *want* to make is between two otherwise identical people who differ only in whether they have health insurance. In other words, we want an **apples to apples** comparison. This is the essence of the *ceteris paribus* condition: **all else being equal**, the difference between treated and untreated units reveals the causal effect of treatment.

3.2 The Multiverse of Possibilities

To build intuition for causal inference, consider the television show *Rick and Morty*. In this show, the character Rick possesses a "portal gun" that allows him to travel across infinite parallel universes. In each universe, there exists a version of his grandson Morty who is identical in every way except for one characteristic—Cowboy Morty, Evil Morty, Hammer Morty, and so on. Each Morty has the same home, same parents, same sister, same *everything else*.

This multiverse provides the perfect setting for causal inference! Comparisons between different Mortys are truly *ceteris paribus* comparisons because everything except the one characteristic of interest is held constant.

3.2.1 The Causal Effect of College

Suppose we want to know the causal effect of attending college on wages. We can't simply compare the wages of college graduates to non-graduates because these groups likely differ in many unobserved ways (motivation, family background, innate ability, etc.).

But imagine we could observe two versions of Morty from parallel universes: Morty C-137 (who didn't attend college) and College Morty (who did). These two Mortys are identical in every way—same intelligence, same family, same opportunities—except that one went to college and the other didn't.

The difference in wages between College Morty and Morty C-137 would be a genuine *ceteris paribus* difference—the true causal effect of college on wages.

Figure 3.1: Rick and Morty: A show about infinite parallel universes.

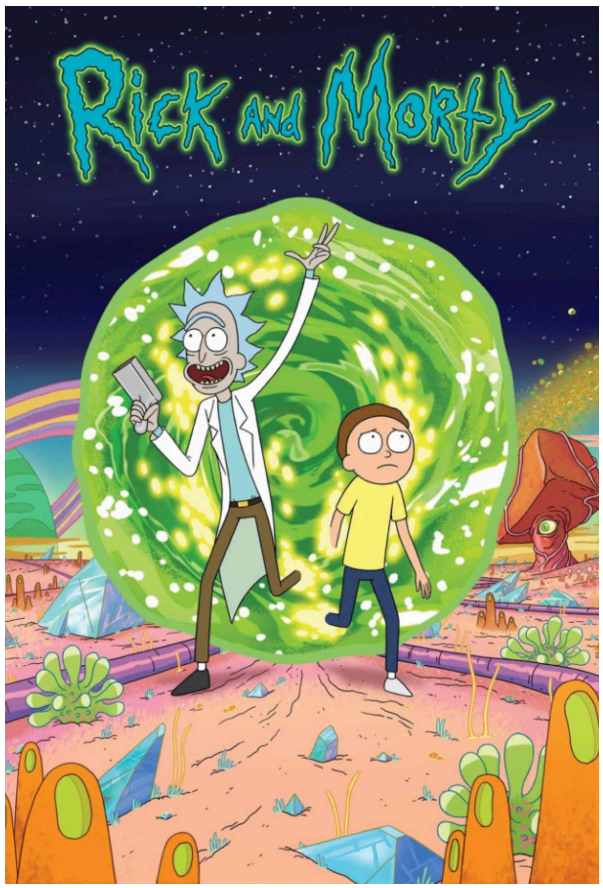
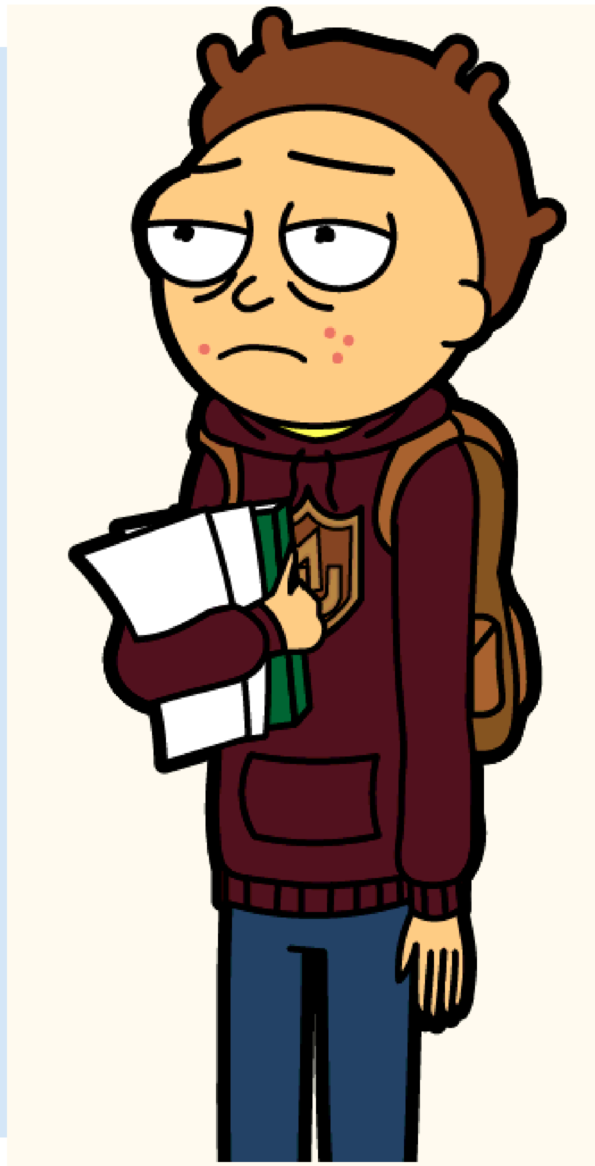


Figure 3.2: Morty C-137 (left) and College Morty (right). The wage difference between them represents the true causal effect of college.

(a) Morty C-137

(b) College Morty



3.2.2 The Fundamental Problem

Of course, the fundamental problem of causal inference is that **we don't have a portal gun**. We only observe one version of reality, not parallel universes. We can never observe what would have happened to a college graduate had they *not* gone to college, or what would have happened to a non-graduate had they attended. This unobserved "what if" scenario is called the **counterfactual**.

3.3 Randomized Control Trials: The Next Best Thing

Since we can't travel between parallel universes, we need another way to make valid causal comparisons. The solution that comes closest to achieving *ceteris paribus* conditions in the real world is the **randomized control trial** (RCT).

Consider how a pharmaceutical company tests whether a new drug works. They don't just give the drug to sick people and see if they get better—that would be subject to all sorts of selection bias. Instead, they conduct an RCT:

1. **Sample** a group of volunteers from the population of interest
2. **Randomly assign** some volunteers to receive the drug (the "treatment" group) and others to receive a placebo (the "control" group)
3. **Compare** the average outcomes between the treatment and control groups

The key insight is that **random assignment eliminates selection bias**. When treatment is assigned randomly, there is no systematic relationship between who receives treatment and their underlying characteristics. On average, the treatment group and control group will be identical in terms of age, health status, income, motivation, and every other characteristic—both observed and unobserved.

> **!** Why Randomization Works
>
> Random assignment ensures that, on average, the treatment and control groups have identical characteristics. Any difference in outcomes between the groups can therefore be attributed to the treatment itself, not to pre-existing differences between the groups.

3.4 The Law of Large Numbers

Why exactly does randomization eliminate selection bias? The answer lies in a fundamental result from probability theory: the **Law of Large Numbers** (LLN).

The LLN states that as the sample size increases, the sample average converges to the population average. Formally, if $Y_1, Y_2, ..., Y_n$ are independent, identically distributed random variables with mean μ, then:

$$\text{plim}(\bar{Y}_n) = \mu$$

where $\bar{Y}_n$ is the sample average and plim denotes the probability limit (what the value converges to as $n \to \infty$).

3.4.1 Seeing the LLN in Action: Rolling Dice

To see the Law of Large Numbers at work, consider a simple example: rolling a fair six-sided die. The expected value (average outcome) of a single roll is:

$$\mu = \frac{1+2+3+4+5+6}{6} = 3.5$$

If we roll a die just once or twice, we might get values far from 3.5 (like a 1 or a 6). But what happens if we roll it many, many times? Let's see what happens!

First, we will "pretend" rolling a dice by using the `sample()` function. This function will randomly select `n` numbers from a vector, here the numbers 1-6.

Let's first roll the dice once. We will set the `size` argument in the `sample()` function to `size = 1` to generate a single random number from our vector.

```
# Set seed for reproducibility
set.seed(1248)

# Create a vector which represents the possible outcomes of a dice roll
dice <- c(1, 2, 3, 4, 5, 6)

# Now "roll" the dice using the sample() function

roll_1 <- sample(dice, size = 1, replace = TRUE)

# Print the result of the dice roll
roll_1
```

```
[1] 2
```

With 1 dice roll, we get a value of 2. Obviously pretty far from the expected value $\mu = 3.5$.

Now, let's roll it 5 times and take the average of our 5 different rolls using the `mean()` function:

```
roll_5 <- sample(dice, size = 5, replace = TRUE)

# Print the 5 different dice rolls
roll_5
```

```
[1] 1 5 2 5 1
```

```
# Find the mean of the 5 different dice rolls
mean(roll_5)
```

```
[1] 2.8
```

With just 5 rolls, our average is a little closer to the expected value. Now, let's try to roll it 25 times, 50 times, and 100 times and find the average values of our dice roll!

```
# Continue rolling to get 10 total rolls
rolls_25 <- sample(dice, size = 10, replace = TRUE)
mean(rolls_25)
```

```
[1] 2.3
```

```
# Keep going to 50 rolls
rolls_50 <- sample(dice, size = 50, replace = TRUE)
mean(rolls_50)
```

```
[1] 3.68
```

```
# And finally 100 rolls
rolls_100 <- sample(1:6, size = 100, replace = TRUE)
mean(rolls_100)
```

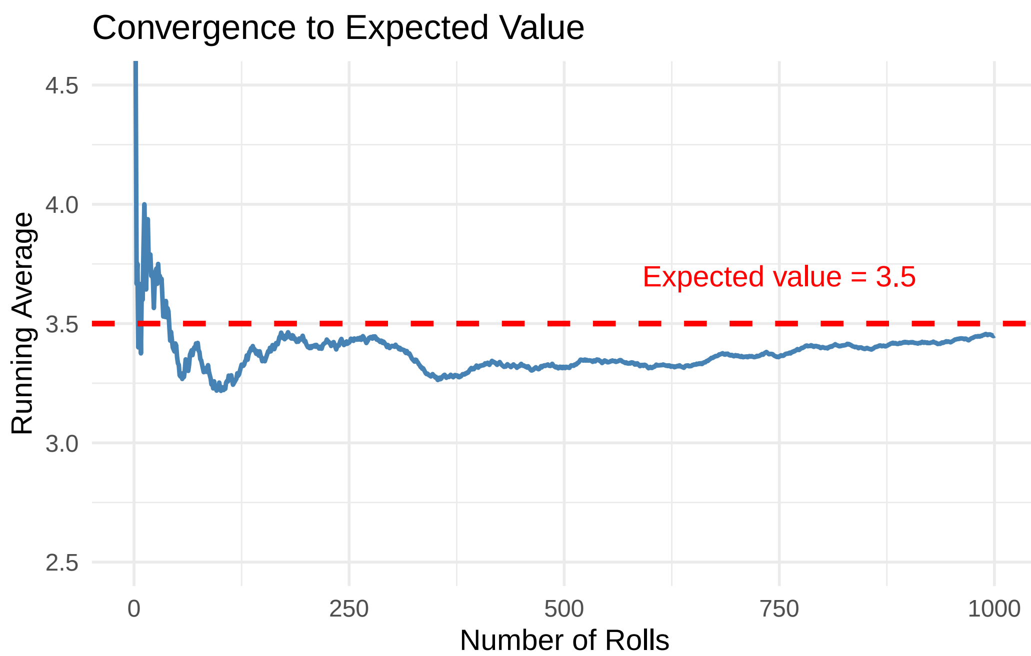
```
[1] 3.35
```

Notice how the average tends to get closer to 3.5 as we increase the number of rolls. This isn't guaranteed on any single attempt (randomness still plays a role), but the general tendency is clear.

Now let's see what happens when we roll the die many times and track the running average at each step. In Figure 3.3, I plot the *sample average* of our dice rolls on the y-axis, and the number of times I roll the dice on the x-axis. The population mean, 3.5, is plotted with a dashed red line.

```r
# Set seed for reproducibility
set.seed(12345)

# Number of dice rolls to simulate
n_rolls <- 1000

# Simulate rolling a six-sided die n_rolls times
dice_rolls <- sample(1:6, size = n_rolls, replace = TRUE)

# Calculate the running average after each roll
running_avg <- cumsum(dice_rolls) / 1:n_rolls

# Create a data frame for plotting
dice_data <- tibble(
  roll_number = 1:n_rolls,
  running_average = running_avg
)

# Plot the running average
ggplot(dice_data, aes(x = roll_number, y = running_average)) +
  geom_line(color = "steelblue", linewidth = 0.8) +
  geom_hline(yintercept = 3.5, linetype = "dashed", color = "red", linewidth = 1) +
  annotate("text", x = 750, y = 3.7, label = "Expected value = 3.5",
           color = "red", size = 4) +
  labs(
    x = "Number of Rolls",
    y = "Running Average",
    title = "Convergence to Expected Value"
  ) +
  theme_minimal() +
  coord_cartesian(ylim = c(2.5, 4.5))
```

Figure 3.3: The Law of Large Numbers in action. As we roll a die more times, the average of all rolls converges to the expected value of 3.5.



Notice several key features about Figure 3.3:

- **Early volatility**: In the first few rolls, the running average bounces around wildly. After just 10 rolls, we might be at 4.2 or 2.9.
- **Gradual stabilization**: As we accumulate more rolls (say, 100 or 200), the average starts to settle down closer to 3.5.
- **Long-run convergence**: By the time we've rolled 1,000 times, the average is very close to the theoretical expected value of 3.5.

This is the Law of Large Numbers in action: **with enough observations, sample averages converge to population averages**. The key word is "enough"—we need a sufficiently large sample for the convergence to occur.

3.4.2 How the LLN Enables Causal Inference

When we randomly assign individuals to treatment and control groups, we are essentially drawing random samples from the same underlying population. The LLN guarantees that, with a large enough sample, the average characteristics of each group will converge to the population average—and therefore, to each other.

This is why randomization eliminates selection bias: the treatment and control groups become statistically identical in all characteristics—both observed and unobserved—as the sample size grows.

3.5 How Randomization Eliminates Selection Bias: A Simulation

> **What is a Simulation?**
>
> A **simulation** is a computer-based experiment where we generate artificial data according to rules we specify. Unlike real-world data, we know exactly how the simulated data was created—including the true causal effects, which variables influence which outcomes, and how treatment was assigned.
> Simulations are powerful learning tools because they let us see what *should* happen under ideal conditions. If we set the true causal effect to zero and then compare treated and untreated groups, any difference we observe must be due to something other than a real treatment effect (like selection bias or random chance). By manipulating features of the simulation—such as sample size or how treatment is assigned—we can build intuition for how these factors affect our ability to recover causal effects. Think of simulations as a laboratory where we can run controlled experiments on statistical methods themselves.

Let's look at a more complex version of how the LLN works in the context of a randomized control trial. Consider a simulation where we have a population with three observable characteristics: light-colored eyes, dark hair, and being tall. Suppose these characteristics are correlated with both treatment status and outcomes in the observational data (i.e., without random assignment). Specifically, having light-colored eyes will be *strongly positively correlated* with the treatment and the outcome; having dark hair will be *moderately positively correlated* with the treatment and the outcome, and being tall will be *slightly negatively correlated* with the treatment and the outcome. Critically, we set the true causal effect of treatment on the outcome to **zero**.

> **A Note on the Code Below**
>
> The R code in the following sections is more advanced than what we've covered so far in the course. Don't worry if you don't understand every line! The code is provided for those who are curious and want to see how simulations work "under the hood." Even if the syntax is unfamiliar, reading through the comments can help you understand the logic of what we're doing. As you progress through the course, more of this code will start to make sense.

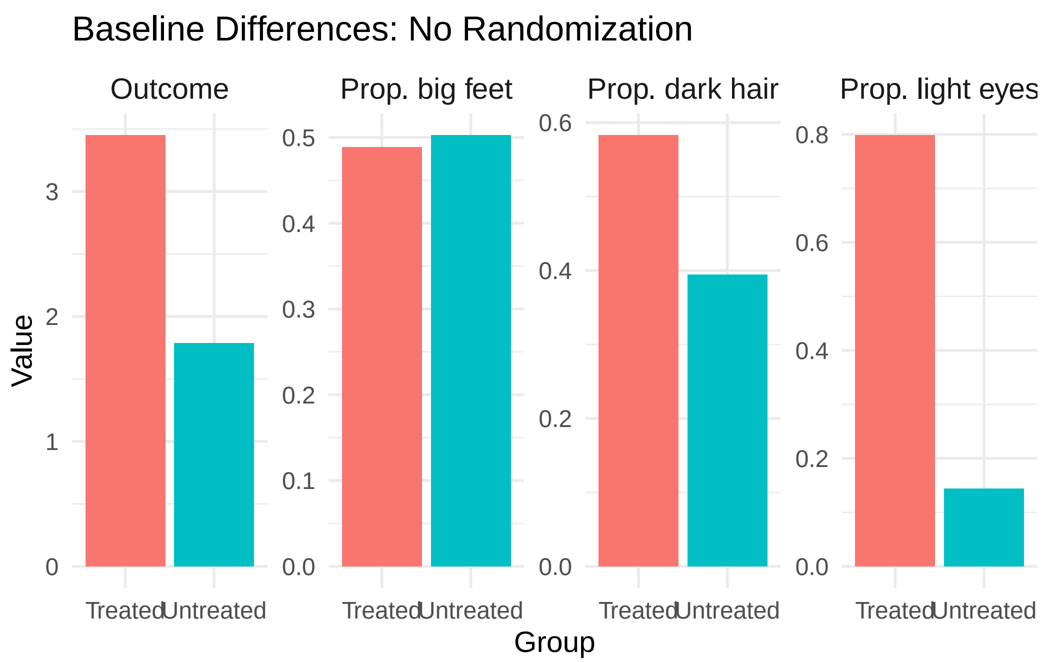
```
# Load the tidyverse package for data manipulation and plotting
library(tidyverse)

# Set a "seed" for reproducibility
# This ensures we get the same random numbers each time we run the code
set.seed(181247)

# Define the TRUE causal effect of treatment
# We're setting this to ZERO so we know any observed difference is bias!
causal_effect <- 0

# Create a population of 10,000 individuals
n_population <- 10000

# Generate three random characteristics for each person
# Each characteristic is randomly 0 or 1 (like a coin flip)
eye_light <- sample(x = c(1, 0), size = n_population, replace = TRUE)
hair_dark <- sample(x = c(1, 0), size = n_population, replace = TRUE)
big_feet <- sample(x = c(1, 0), size = n_population, replace = TRUE)

# Generate random "noise" - unexplained variation in outcomes
err <- rnorm(n_population)

# Create an underlying "propensity" for treatment
# Notice: people with light eyes are MUCH more likely to be treated (coefficient = 2.1)
# People with dark hair are moderately more likely (coefficient = 0.83)
# Big feet has a small negative effect (coefficient = -0.11)
treatment_propensity <- 1 + 2.1 * eye_light + 0.83 * hair_dark - 0.11 * big_feet + err

# Assign treatment based on this propensity (treated if propensity > 2.25)
# This creates SELECTION BIAS: treatment is correlated with characteristics
treated <- ifelse(treatment_propensity > 2.25, 1, 0)

# Create the outcome variable
# The outcome depends on the same characteristics that predict treatment!
# But notice: we add "treated * causal_effect" which equals zero
# So treatment has NO real effect on the outcome
outcome <- 1 + 2.2 * eye_light + 1.2 * hair_dark - 0.01 * big_feet + treated * causal_effect

# Combine everything into a data frame
dat <- tibble(
  outcome = outcome,
```

```r
  treated = treated,
  eye_light = eye_light,
  hair_dark = hair_dark,
  big_feet = big_feet
)
```

```r
# Calculate the mean of each variable for treated vs untreated groups
baseline_diffs <- dat |>
  # Group the data by treatment status
  group_by(treated) |>
  # Calculate summary statistics for each group
  summarise(
    Outcome = mean(outcome),
    `Prop. light eyes` = mean(eye_light),
    `Prop. dark hair` = mean(hair_dark),
    `Prop. big feet` = mean(big_feet)
  ) |>
  # Reshape data from "wide" to "long" format for plotting
  pivot_longer(
    cols = Outcome:`Prop. big feet`,
    names_to = "variable",
    values_to = "value"
  ) |>
  # Create nice labels for the treatment groups

  mutate(group = ifelse(treated == 1, "Treated", "Untreated"))

# Create the bar plot
ggplot(baseline_diffs) +
  geom_bar(
    aes(x = group, y = value, fill = group),
    stat = "identity"
  ) +
  # Create separate panels for each variable
  facet_wrap(~variable, nrow = 1, scales = "free_y") +
  labs(
    x = "Group",
    y = "Value",
    title = "Baseline Differences: No Randomization"
  ) +
  theme_minimal() +
  theme(
```

```
    legend.position = "none",
    strip.text = element_text(size = 11)
  )
```

Figure 3.4: Baseline differences between treated and untreated groups without randomization. Despite the true causal effect being zero, we observe large differences in both characteristics and outcomes.



Notice in Figure 3.4 how the treated and untreated groups look very different! The treated group has a higher proportion of people with light eyes and dark hair. Most importantly, the treated group has a higher mean outcome—even though **we know the true causal effect is zero**. This is selection bias in action: people who "select into" treatment have characteristics that also lead to better outcomes.

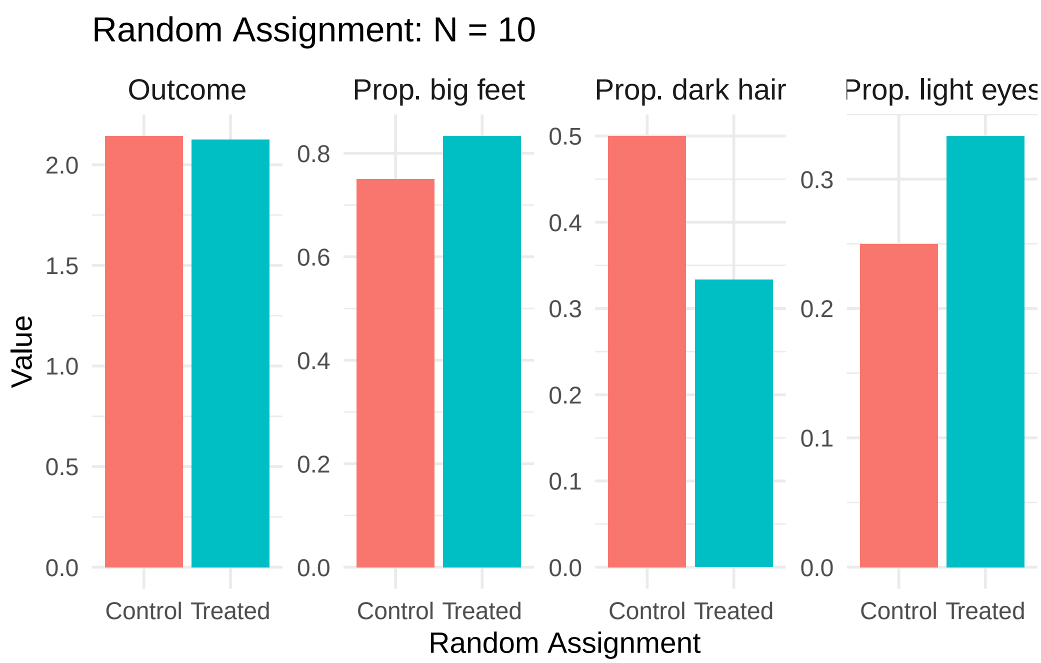
```
# Function to run the random assignment simulation for a given sample size
run_random_assignment <- function(data, sample_size, seed = 181247) {

  # Set seed for reproducibility
  set.seed(seed)

  # Take a random sample from the untreated population
  # (We use untreated only to avoid any prior treatment effects)
  sample_data <- data |>
```

```
    filter(treated == 0) |>
    slice_sample(n = sample_size)

  # RANDOMLY assign treatment - this is the key step!
  # Each person has a 50% chance of being assigned to treatment
  # This assignment is INDEPENDENT of their characteristics
  random_treatment <- sample(c(1, 0), size = sample_size, replace = TRUE)

  # Add the random treatment assignment to our sample
  sample_data <- sample_data |>
    mutate(rand_treat = random_treatment)

  # Calculate means for each randomly assigned group
  sample_data |>
    group_by(rand_treat) |>
    summarise(
      Outcome = mean(outcome),
      `Prop. light eyes` = mean(eye_light),
      `Prop. dark hair` = mean(hair_dark),
      `Prop. big feet` = mean(big_feet),
      .groups = "drop"
    ) |>
    mutate(sample_size = sample_size)
}

# Run the simulation for different sample sizes
results_n10 <- run_random_assignment(dat, sample_size = 10)
results_n25 <- run_random_assignment(dat, sample_size = 25)
results_n50 <- run_random_assignment(dat, sample_size = 50)
results_n100 <- run_random_assignment(dat, sample_size = 100)
results_n500 <- run_random_assignment(dat, sample_size = 500)


# Combine all results
all_results <- bind_rows(results_n10, results_n25, results_n50,
  results_n100, results_n500)

# Reshape for plotting
all_results_long <- all_results |>
  pivot_longer(
    cols = Outcome:`Prop. big feet`,
    names_to = "variable",
```

```
    values_to = "value"
  ) |>
  mutate(
    group = ifelse(rand_treat == 1, "Treated", "Control"),
    sample_size_label = paste("N =", sample_size)
  )
```

Now let's see what happens when we randomly assign treatment. Starting with a small sample (N=10), we see in Figure 3.5 that random assignment reduces but doesn't eliminate the differences between groups. With such a small sample, random chance can still produce imbalanced groups. With only 10 people, we still see noticeable differences between the randomly assigned treatment and control groups.

Figure 3.5: Random assignment with N=10. Some imbalance remains due to small sample size.



As we increase the sample size to N=25 and N=50, as shown in Figure 3.6 and Figure 3.7, the differences between treatment and control groups shrink. The LLN is at work: larger samples produce more balanced groups in terms of their background characteristics.

By N=500, as show in Figure 3.8, the treatment and control groups are nearly identical across all characteristics. The difference in outcomes between groups is now close to zero—the true causal effect! Any remaining difference is just random variation, not systematic selection bias.

Figure 3.6: Random assignment with N=25.

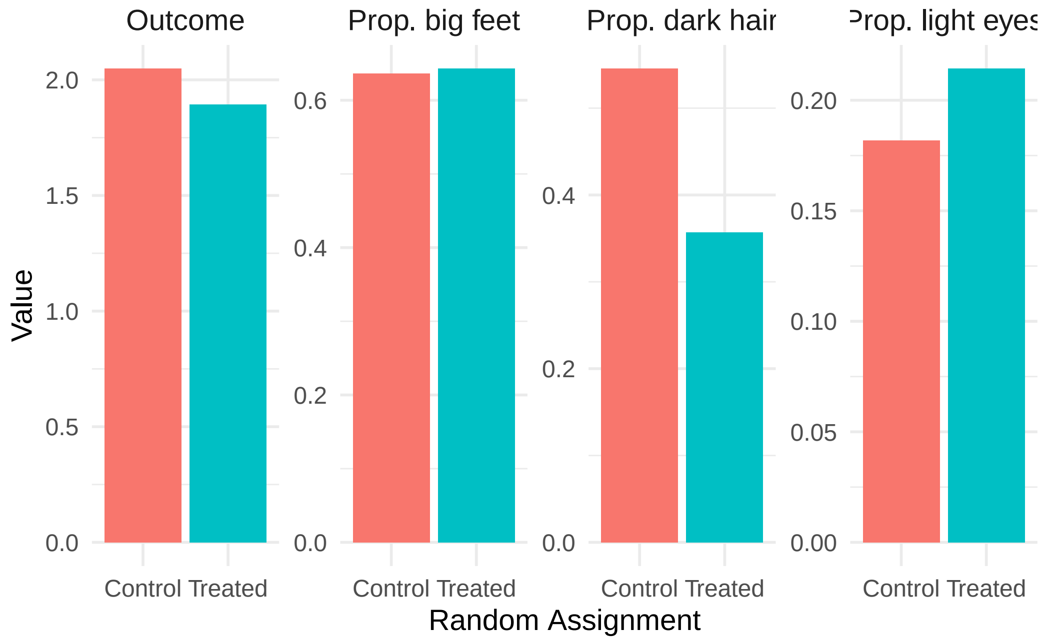


Figure 3.7: Random assignment with N=50.

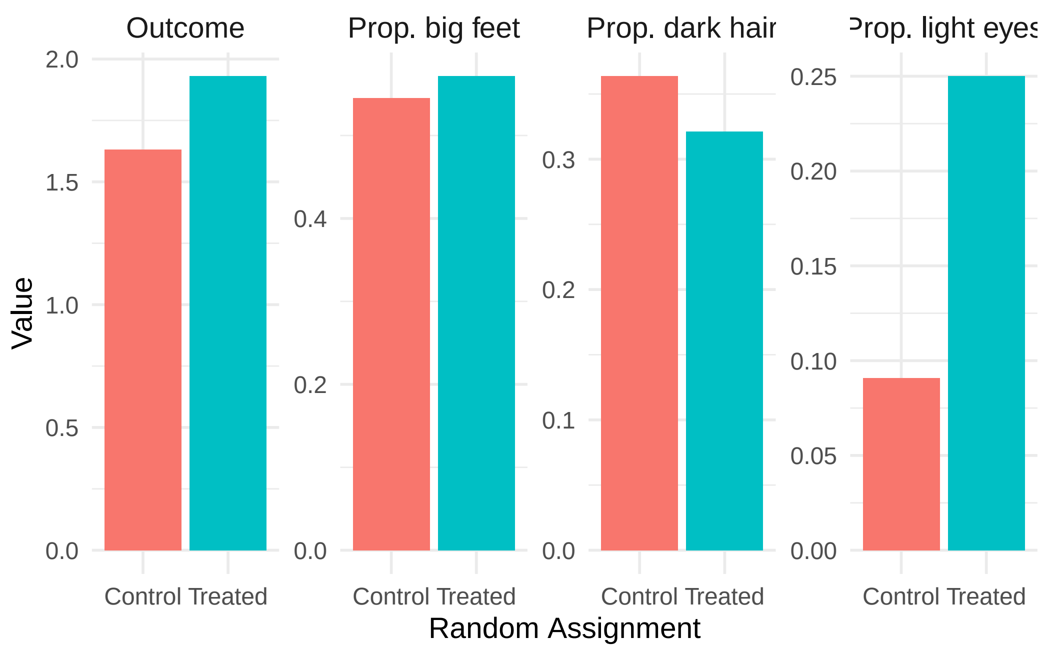
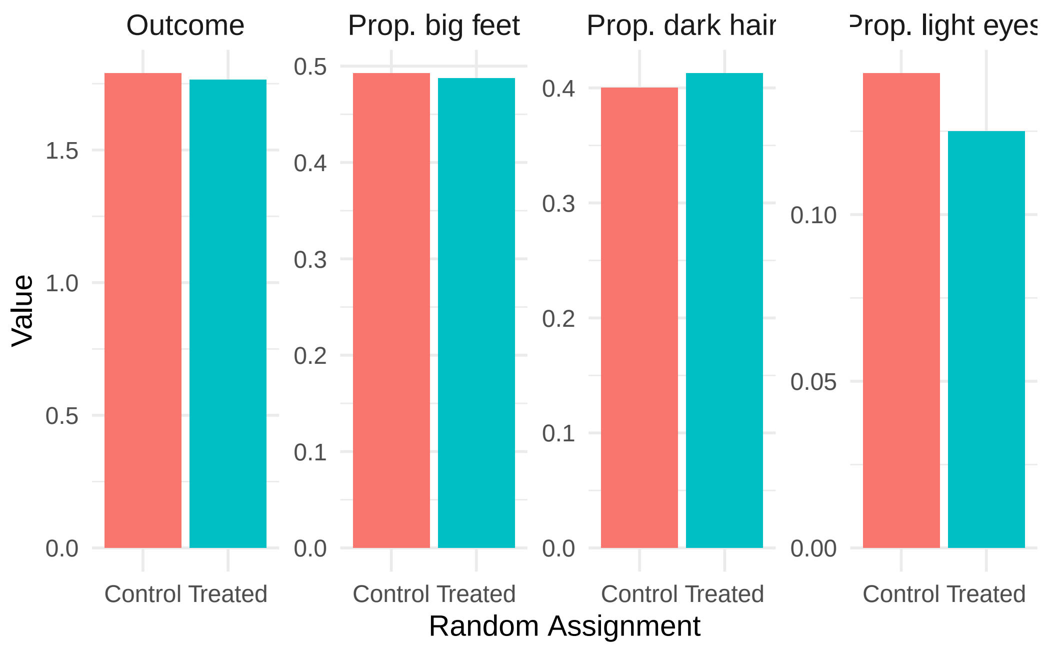


Figure 3.8: Random assignment with N=500. The treatment and control groups are now nearly identical in their characteristics, and the difference in outcomes is close to the true causal effect of zero.



3.6 The Potential Outcomes Framework

Now that we intuitively understand why simple comparisons can be misleading and how randomization addresses this problem, let's formalize these concepts mathematically using the **Potential Outcomes Framework**.

3.6.1 Defining Potential Outcomes

Consider our example of estimating the causal effect of college on wages. For any individual i, we can define two **potential outcomes**:

- Y_{1i}: The wage individual i would earn **if they attend college**
- Y_{0i}: The wage individual i would earn **if they do not attend college**

The causal effect of college for individual i is simply the difference between these potential outcomes:

$$\kappa_i = Y_{1i} - Y_{0i}$$

This framework captures the "multiverse" intuition: Y_{1i} is College Morty's wage, Y_{0i} is Morty C-137's wage, and κ_i is the causal effect of college for Morty.

3.6.2 The Fundamental Problem of Causal Inference

The fundamental problem is that we only observe **one** potential outcome for each individual. If person i attends college, we observe Y_{1i} but never see Y_{0i}. If they don't attend, we observe Y_{0i} but never see Y_{1i}. We never observe both potential outcomes for the same individual.

Let D_i be a treatment indicator:

$$D_i = \begin{cases} 1 & \text{if individual } i \text{ is treated (attends college)} \\ 0 & \text{otherwise} \end{cases}$$

What we actually observe is:

$$Y_i = D_i \cdot Y_{1i} + (1 - D_i) \cdot Y_{0i}$$

That is, we observe the treated outcome for treated individuals and the untreated outcome for untreated individuals—but never both.

3.6.3 Selection Bias Formalized

Now we can formally see why simple comparisons are problematic. Suppose we compare average outcomes between those who are treated ($D_i = 1$) and those who are untreated ($D_i = 0$). The difference in group means is:

$$E[Y_i|D_i = 1] - E[Y_i|D_i = 0]$$

Since we only observe Y_{1i} for the treated and Y_{0i} for the untreated, this becomes:

$$E[Y_{1i}|D_i = 1] - E[Y_{0i}|D_i = 0]$$

Here's the key insight. Let's add and subtract a term that represents the counterfactual—what the treated group's outcome would have been without treatment:

$$E[Y_{1i}|D_i = 1] - E[Y_{0i}|D_i = 0] + \underbrace{\{E[Y_{0i}|D_i = 1] - E[Y_{0i}|D_i = 1]\}}_{=0}$$

Rearranging terms:

$$\underbrace{E[Y_{1i}|D_i = 1] - E[Y_{0i}|D_i = 1]}_{\text{Causal effect for the treated}} + \underbrace{\{E[Y_{0i}|D_i = 1] - E[Y_{0i}|D_i = 0]\}}_{\text{Selection bias}}$$

! The Selection Bias Decomposition

A simple difference in group means equals the causal effect **plus** selection bias:

$$\underbrace{E[Y_i|D_i = 1] - E[Y_i|D_i = 0]}_{\text{Observed difference}} = \underbrace{\kappa}_{\text{Causal effect}} + \underbrace{\{E[Y_{0i}|D_i = 1] - E[Y_{0i}|D_i = 0]\}}_{\text{Selection bias}}$$

The selection bias term compares the **untreated potential outcomes** of the treated group to the untreated potential outcomes of the control group. In the health insurance example, this captures differences in baseline health (absent insurance) between those who choose to get insurance and those who don't. If people who get insurance would be healthier even without insurance (perhaps because they're wealthier or more health-conscious), then selection bias is positive, and the simple comparison overstates the causal effect.

3.6.4 How Randomization Eliminates Selection Bias

When treatment is randomly assigned, the treatment and control groups are drawn from the same population. By the Law of Large Numbers, their average characteristics—including their average potential outcomes—converge to each other. Formally:

$$E[Y_{0i}|D_i = 1] = E[Y_{0i}|D_i = 0]$$

This says that the average untreated potential outcome is the same for both groups. Plugging this into our decomposition:

$$E[Y_i|D_i = 1] - E[Y_i|D_i = 0] = \kappa + \{E[Y_{0i}|D_i = 1] - E[Y_{0i}|D_i = 0]\} = \kappa + 0 = \kappa$$

With random assignment, the simple difference in means equals the causal effect. Selection bias disappears.

3.7 The Oregon Health Insurance Experiment

Most econometric research cannot rely on randomization to identify causal effects. We typically work with observational data where treatment is not randomly assigned, and much of this course focuses on methods for addressing selection bias in such settings. However, there have been a few landmark studies that used randomization to study important policy questions. One of the most influential is the **Oregon Health Insurance Experiment**.

3.7.1 Context: Medicaid Expansion

Throughout the 2000s, many U.S. states expanded their Medicaid programs to cover more low-income residents. Most states simply extended coverage to everyone who became newly eligible. But in 2008, Oregon did something unique: facing limited resources, they offered Medicaid coverage through a public lottery.

3.7.2 The Lottery

Oregon's approach created a natural randomized experiment. People who wanted coverage could sign up for the lottery. Winners received the opportunity to apply for the Oregon Health Plan (the state's Medicaid program), while losers remained uninsured. To be eligible, applicants had to be legal Oregon residents aged 19-64, uninsured for the past six months, with income below the federal poverty line, and not otherwise eligible for health insurance.

Because lottery winners were chosen randomly, the comparison between winners and losers provides a valid causal estimate of the effect of health insurance—free from the selection bias that plagues observational studies.

3.7.3 Results

Researchers followed lottery participants and found that lottery winners (relative to losers) experienced:

1. **Increased take-up of Medicaid:** The lottery actually changed insurance status, confirming the treatment "worked"
2. **Increased use of medical resources:** Winners used more hospital and outpatient services
3. **Small improvements in overall health:** Physical health improvements were modest, though mental health and financial security improved substantially

These findings are important because they come from a rigorous RCT rather than observational comparisons. The study provides some of the cleanest evidence we have on what health insurance actually *causes*, as opposed to what it merely correlates with.

> Discussion Question
>
> The Oregon experiment found modest effects of insurance on physical health. Does this mean expanding health insurance is not worthwhile? What other outcomes might matter for policy decisions?

3.8 Summary and Conclusion

This chapter introduced the fundamental concepts underlying causal inference in econometrics. We began with the intuition that simple comparisons between treated and untreated groups can be misleading due to selection bias—the groups may differ in ways beyond just the treatment. The "multiverse" thought experiment illustrated what we really want: *ceteris paribus* comparisons where everything except the treatment is held constant.

Randomized control trials approximate this ideal by randomly assigning treatment, which ensures that treatment and control groups are comparable on average. The Law of Large Numbers guarantees that with large enough samples, random assignment eliminates selection bias.

We then formalized these ideas using the potential outcomes framework, which defines causal effects as the difference between what would happen with and without treatment. The fundamental problem is that we only observe one potential outcome per individual. Simple comparisons of means capture both the causal effect and selection bias, but random assignment makes selection bias vanish.

3.8.1 Key Takeaways

- **Selection bias** arises when treated and untreated groups differ systematically beyond just the treatment—we're comparing apples to oranges
- **Ceteris paribus** ("all else equal") comparisons isolate the causal effect by holding everything constant except the treatment
- **Randomized control trials** eliminate selection bias by ensuring treatment and control groups are drawn from the same population
- The **Law of Large Numbers** guarantees that with large samples, randomly assigned groups have identical average characteristics

- The **potential outcomes framework** formalizes causal effects as $\kappa = Y_{1i} - Y_{0i}$, but we only observe one potential outcome per individual
- Simple differences in means equal: Causal effect + Selection bias. Randomization sets selection bias to zero.

3.9 Check Your Understanding

Note: This section contains interactive content only available in the HTML version.

4 Bivariate Regression and Ordinary Least Squares

Key Questions

- How do we mathematically model the relationship between two variables?
- What is the population regression model and how do we interpret its parameters?
- How do we derive the OLS estimator?
- What assumptions are required for OLS to be unbiased?
- What is omitted variable bias and when does it arise?
- When can we interpret OLS estimates as causal effects?
- How do we measure the precision of our estimates?

Suggested Readings

- Wooldridge (2019), Ch. 2

In Chapter 3, we learned that randomized control trials provide the gold standard for causal inference by eliminating selection bias through random assignment. But most economic questions cannot be answered with experiments. We typically work with observational data where "treatment" is not randomly assigned. This chapter introduces **Ordinary Least Squares (OLS)** regression, the workhorse tool of econometrics. We focus here on the **bivariate** case—a single independent variable—to build intuition before extending to multiple variables.

4.1 Modeling Relationships Between Variables

Suppose we have two variables, y and x. In econometrics, we are often interested in one of several related goals: estimating the **causal effect** of x on y, explaining variation in y using x, or simply describing how y varies with x. OLS is a tool that can help with all of these—though as we'll see, whether we can interpret results causally depends on assumptions that may or may not hold.

But what do we even *mean* when we say we want to study "how x is related to y"? We need a precise, mathematical way to operationalize this question. The most straightforward approach is to ask: what is the *average* value of y for each different value that x can take?

Consider some examples. How can we quantify the gender pay gap? We compare the *average* wage for men versus women. How do we measure the returns to education? We compare the *average* wage of people with different levels of education. In both cases, we are asking about how the average of one variable changes across values of another.

4.1.1 Conditional Expectations

This idea of finding "the average value of y for a given value of x" is exactly what we call the **conditional expectation**. Formally, the conditional expectation of y given x is written as:

$$E[Y|X = x]$$

This notation asks: if we knew that X took on some specific value x, what would we expect Y to be, on average? The conditional expectation is often the central quantity we want to model and estimate in econometrics.

4.1.2 Example: Education and Wages

Suppose X represents education level, which can take values $X = \{\text{hsdeg}, \text{colldeg}, \text{profdeg}\}$ (high school degree, college degree, professional degree), and Y represents hourly wage.

For each education level, we can compute the average wage among people in that group. This will gives us a good idea of how wages are related to education.

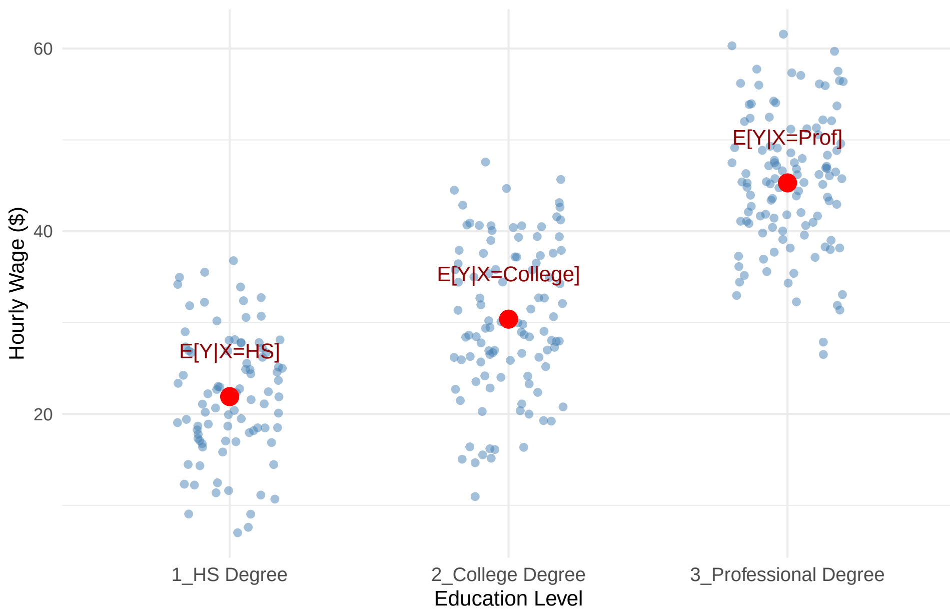
Figure 4.1 is a scatterplot with an indiviudal's wage on the y-axis and their education level on the x-axis, with each point plotted in blue. The red points in Figure 4.1 show the conditional expectations—the average wage for individuals *within each education category,* i.e., $E[Y|X = HS], E[Y|X = College], \text{and} E[Y|X = Prof]$.

These averages tell us how we can expect wages to change when we increase education, which is precisely what we set out to measure.

4.2 The Population Regression Model

Now that we understand what we want to estimate (the conditional expectation), we need a framework for *how* to estimate it. Before we can estimate anything, we must first describe the underlying relationship we believe exists in the population.

Figure 4.1: Wages by education level. Each point represents an individual. Red points show the conditional expectation (average wage) for each education group.



4.2.1 The Bivariate Model

The simplest specification is called the **bivariate** (or simple) regression model:

$$y = \beta_0 + \beta_1 x + \mu \tag{4.1}$$

where:

- y is the **dependent variable** (also called the explained, response, or outcome variable)
- x is the **independent variable** (also called the explanatory, control, or predictor variable)
- μ is the **error term**, which represents all unobserved factors that affect y
- β_0 and β_1 are **population parameters**—the "true" values describing the relationship between x and y in the population

> **i** The Error Term
>
> The error term μ is crucial. It captures everything that affects y other than x. In a wage equation, μ might include ability, motivation, family connections, luck, and countless other factors we cannot observe or measure.

4.2.2 Connecting to Conditional Expectations

How does this regression model relate to our goal of estimating $E[Y|X]$? If we assume that the error term has mean zero conditional on x—that is, $E[\mu|x] = 0$—then:

$$E[y|x] = E[\beta_0 + \beta_1 x + \mu|x] = \beta_0 + \beta_1 x + E[\mu|x] = \beta_0 + \beta_1 x$$

This is exactly the conditional expectation we set out to estimate! The regression line $\beta_0 + \beta_1 x$ traces out the average value of y for each value of x.

4.2.3 Interpreting the Coefficients

The coefficients in Equation 4.1 have specific interpretations:

- β_0 **(the intercept):** The predicted value of y when $x = 0$. Depending on the context, this may or may not have a meaningful interpretation.
- β_1 **(the slope):** The **marginal effect** of x on y. Mathematically, it is the derivative:

$$\frac{dy}{dx} = \beta_1$$

This tells us: when x increases by one unit, y changes by β_1 units, on average.

4.2.4 Example: Education and Wages

A regression model relating years of education to hourly wages might be:

$$wage = \beta_0 + \beta_1 \cdot educ + \mu$$

where *wage* is dollars per hour, *educ* is years of education, and μ captures all other determinants of wages (experience, ability, etc.).

If $\beta_1 = 2.50$, this would mean that each additional year of education is associated with \$2.50 higher hourly wages, on average.

4.3 The Regression Line: Predicted Values and Residuals

The entire goal of linear regression is to use our *sample* of the data from the wider *population* to estimate the values of β_0 and β_1 that are the best possible estimates we can generate. We call these estimates $\hat{\beta}_0$ and $\hat{\beta}_1$, and the method we use to calculate these estimates the **estimator**.

Before we derive estimators for β_0 and β_1, let's introduce some key concepts that will guide our approach.

4.3.1 Fitted Values and the Regression Line

Once we have estimates $\hat{\beta}_0$ and $\hat{\beta}_1$, we can construct the **OLS regression line** (or fitted line):

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

For any value of x, this line gives us $\hat{y}$, the **predicted value** (or **fitted value**) of y. The regression line represents our best guess for the average value of y at each level of x.

4.3.2 Residuals: The Prediction Errors

Of course, our predictions won't be perfect. The **residual** for observation i is the difference between the actual value and the predicted value:

$$\hat{\mu}_i = y_i - \hat{y}_i = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)$$

The residual measures how much our prediction "misses" for each observation. Positive residuals mean we underpredicted; negative residuals mean we overpredicted.

4.3.3 A Numerical Example

Let's see these concepts with actual numbers. The table below shows the first six observations from a regression of MPG on horsepower, along with the predicted values and residuals:

Table 4.1: Example regression data showing actual values, predicted values, and residuals.

	x (Horsepower)	y (MPG)	ŷ (Predicted)	û (Residual)
Mazda RX4	110	21.0	22.59	-1.59

Table 4.1: Example regression data showing actual values, predicted values, and residuals.

	x (Horsepower)	y (MPG)	ŷ (Predicted)	û (Residual)
Mazda RX4 Wag	110	21.0	22.59	-1.59
Datsun 710	93	22.8	23.75	-0.95
Hornet 4 Drive	110	21.4	22.59	-1.19
Hornet Sportabout	175	18.7	18.16	0.54
Valiant	105	18.1	22.93	-4.83

Notice how the residual column equals the actual value minus the predicted value. For the first observation, the car has 110 horsepower and gets 21 MPG. Our regression predicts it should get about 22.59 MPG, so the residual is $21 - 22.59 = -1.59$. The negative residual means we *overpredicted*—the car actually gets slightly worse fuel economy than our model predicted.

4.3.4 The Key Insight: Minimize the Residuals

Here's the crucial idea: we want to choose $\hat{\beta}_0$ and $\hat{\beta}_1$ so that our predictions are as good as possible. In other words, we want to make the residuals as small as possible.

But we can't simply minimize the sum of residuals $\sum \hat{\mu}_i$, because positive and negative residuals would cancel out. Instead, we minimize the **Sum of Squared Residuals (SSR)**:

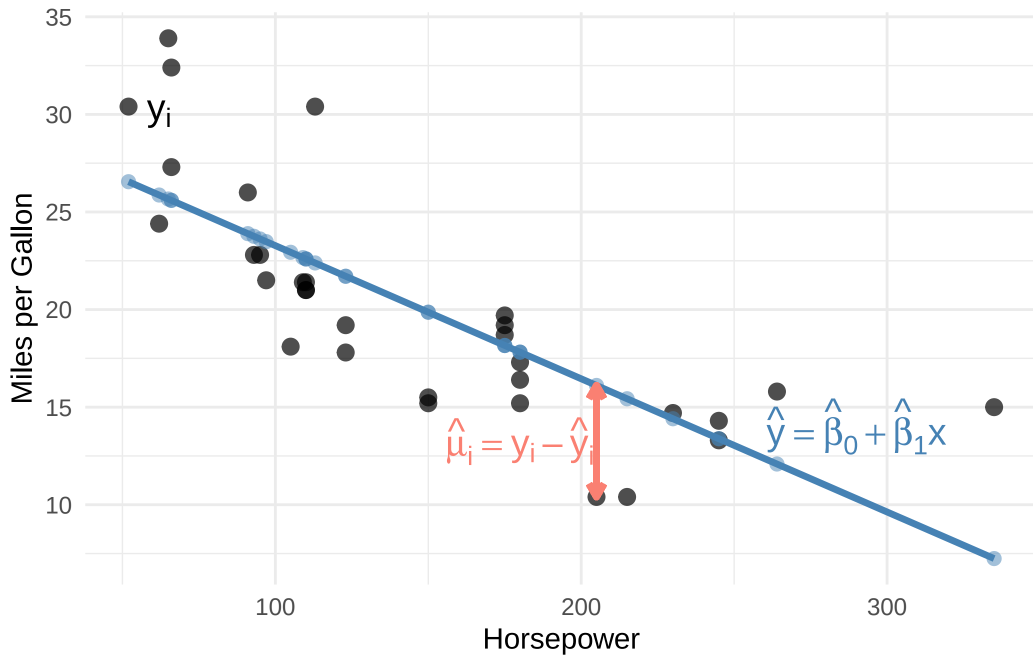
$$SSR = \sum_{i=1}^{n} \hat{\mu}_i^2 = \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

This is exactly where the name "Ordinary Least Squares" comes from: we find the estimates that make the sum of squared residuals as small as possible.

4.4 Deriving the OLS Estimator

Now we turn to the mechanics of estimation. The **population** regression model describes the true relationship, but we never observe the entire population or the true values of β_0 and β_1. Instead, we have a **sample** of data and must *estimate* the population parameters.

Figure 4.2: Components of the regression line. Black points show actual values, the blue line shows predicted values, and the salmon segment illustrates a residual.



4.4.1 The Minimization Problem

Suppose we have a sample of n observations, $\{(x_i, y_i) : i = 1, 2, ..., n\}$. As discussed above, we want to find the values of $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize the sum of squared residuals:

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

This is a calculus optimization problem! The SSR is a function of two unknowns ($\hat{\beta}_0$ and $\hat{\beta}_1$), and we want to find the values that minimize it.

4.4.2 First-Order Conditions

To find the minimum, we take partial derivatives with respect to each parameter, set them equal to zero, and solve. These are called the **first-order conditions**.

Partial derivative with respect to $\hat{\beta}_0$:

$$\frac{\partial SSR}{\partial \hat{\beta}_0} = -2\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

Dividing by -2 and rearranging:

$$\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

Partial derivative with respect to $\hat{\beta}_1$:

$$\frac{\partial SSR}{\partial \hat{\beta}_1} = -2\sum_{i=1}^{n} x_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

Dividing by -2:

$$\sum_{i=1}^{n} x_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

4.4.3 Solving for the Intercept

From the first condition, distributing the summation:

$$\sum_{i=1}^{n} y_i - n\hat{\beta}_0 - \hat{\beta}_1 \sum_{i=1}^{n} x_i = 0$$

Dividing by n and recognizing sample means:

$$\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$$

Solving for the intercept:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \tag{4.2}$$

This tells us that the regression line always passes through the point $(\bar{x}, \bar{y})$—the “center” of the data.

4.4.4 Solving for the Slope

Substituting Equation 4.2 into the second first-order condition and working through the algebra (using summation properties from Appendix B):

$$\sum_{i=1}^{n} x_i[y_i - (\bar{y} - \hat{\beta}_1\bar{x}) - \hat{\beta}_1 x_i] = 0$$

After distributing and rearranging:

$$\sum_{i=1}^{n} x_i(y_i - \bar{y}) = \hat{\beta}_1 \sum_{i=1}^{n} x_i(x_i - \bar{x})$$

Using the summation properties $\sum x_i(y_i - \bar{y}) = \sum(x_i - \bar{x})(y_i - \bar{y})$ and $\sum x_i(x_i - \bar{x}) = \sum(x_i - \bar{x})^2$:

$$\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = \hat{\beta}_1 \sum_{i=1}^{n}(x_i - \bar{x})^2$$

Solving for $\hat{\beta}_1$:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \tag{4.3}$$

4.4.5 Confirming It's a Minimum

How do we know this critical point is a minimum rather than a maximum? We check the second-order conditions. The second partial derivative of SSR with respect to $\hat{\beta}_1$ is:

$$\frac{\partial^2 SSR}{\partial \hat{\beta}_1^2} = 2\sum_{i=1}^{n} x_i^2 > 0$$

Since this is positive (as long as there's variation in x), the SSR function is convex, and our solution is indeed a minimum.

> **! The OLS Slope Formula**
>
> The OLS slope estimate equals the sample covariance of x and y divided by the sample variance of x:

$$\hat{\beta}_1 = \frac{Cov(x,y)}{Var(x)} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

This can also be written in terms of the correlation coefficient:

$$\hat{\beta}_1 = \rho_{xy}\frac{\sigma_y}{\sigma_x}$$

This last expression reveals something important: bivariate OLS is essentially a scaled version of the correlation between x and y. And as we discussed in Chapter 1, correlation does not imply causation! We must be very careful about interpreting bivariate OLS results causally.

4.5 Computing OLS in R

While we derived the formulas by hand, in practice we use software to compute OLS estimates. In R, the `lm()` function (for "linear model") performs OLS regression.

4.5.1 Basic Syntax

```
# General syntax
lm(y ~ x, data = your_data)
```

The formula `y ~ x` specifies that y is the dependent variable and x is the independent variable. An intercept is included by default.

4.5.2 Example: MPG and Horsepower

Let's estimate the relationship between fuel efficiency (MPG) and engine horsepower using the built-in `mtcars` dataset:

```
# Load the data
data(mtcars)

# Estimate the regression
lm_mpg <- lm(mpg ~ hp, data = mtcars)

# View the results
summary(lm_mpg)
```

```
Call:
lm(formula = mpg ~ hp, data = mtcars)

Residuals:
    Min      1Q  Median      3Q     Max
-5.7121 -2.1122 -0.8854  1.5819  8.2360

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 30.09886    1.63392  18.421  < 2e-16 ***
hp          -0.06823    0.01012  -6.742 1.79e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.863 on 30 degrees of freedom
Multiple R-squared:  0.6024,	Adjusted R-squared:  0.5892
F-statistic: 45.46 on 1 and 30 DF,  p-value: 1.788e-07
```

The output shows our estimates: $\hat{\beta}_0 = 30.10$ and $\hat{\beta}_1 = -0.07$. The slope suggests that each additional unit of horsepower is associated with about 0.07 fewer miles per gallon.

4.6 Goodness-of-Fit: R-Squared

How well does our regression line fit the data? The **R-squared** (R^2) provides one measure of goodness-of-fit.

Define:

- **Total Sum of Squares (SST):** $SST = \sum_{i=1}^n (y_i - \bar{y})^2$ — total variation in y
- **Explained Sum of Squares (SSE):** $SSE = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$ — variation explained by the regression
- **Residual Sum of Squares (SSR):** $SSR = \sum_{i=1}^n \hat{\mu}_i^2$ — unexplained variation

The R-squared is:

$$R^2 = \frac{SSE}{SST} = 1 - \frac{SSR}{SST}$$

This measures the fraction of the total variation in y that is explained by the regression. R^2 ranges from 0 (no explanatory power) to 1 (perfect fit).

Interpreting R-Squared

Be careful interpreting R^2, especially in the context of causal inference. A high R^2 does not mean we have a causal estimate—it just means x is correlated with y. Conversely, a low R^2 does not mean our estimate is wrong or unimportant. In cross-sectional data, R^2 values of 0.1 to 0.3 are common and can still represent economically meaningful relationships.

4.7 The Gauss-Markov Assumptions

We've derived the OLS formulas, but how do we know if the estimates are any good? The formulas will always give us *some* numbers, but whether those numbers have useful statistical properties depends on certain conditions being met.

The **Gauss-Markov assumptions** are a set of conditions under which OLS has desirable properties. Understanding these assumptions is crucial: they tell us when we can trust our estimates and when we should be skeptical.

4.7.1 Assumption 1: Linear in Parameters

Gauss-Markov Assumption 1: Linear in Parameters

The population model is: $y = \beta_0 + \beta_1 x + \mu$

This assumption states that the dependent variable y can be written as a linear function of the parameters β_0 and β_1, plus an error term.

What this does NOT mean: This assumption is less restrictive than it might first appear. It does *not* require that y and x have a linear relationship. We can model many non-linear relationships by transforming the variables. For example:

- $y = \beta_0 + \beta_1 x^2 + \mu$ is linear in parameters (we can define $z = x^2$ and estimate $y = \beta_0 + \beta_1 z + \mu$)
- $\log(y) = \beta_0 + \beta_1 x + \mu$ is linear in parameters
- $y = \beta_0 + \beta_1 \log(x) + \mu$ is linear in parameters

What would violate this assumption: Models where parameters enter non-linearly, such as $y = \beta_0 + x^{\beta_1} + \mu$ or $y = \frac{1}{1+e^{-\beta_0-\beta_1 x}} + \mu$ (a logistic function). These require different estimation techniques.

4.7.2 Assumption 2: Random Sampling

> **Gauss-Markov Assumption 2: Random Sampling**
>
> We have a random sample of size n from the population: $\{(x_i, y_i) : i = 1, ..., n\}$

This assumption states that each observation in our dataset is drawn randomly and independently from the population of interest. Each unit in the population has an equal chance of being selected, and the selection of one unit doesn't affect the probability of selecting another.

Why it matters: Random sampling ensures that our sample is representative of the population. If certain types of observations are more likely to be included, our estimates may not generalize to the full population.

When this might be violated:

- **Self-selection:** If people choose whether to participate in a survey, those who respond may differ systematically from those who don't. For example, a wage survey might oversample high earners if low-wage workers are less likely to respond.
- **Convenience sampling:** Surveying only people who are easy to reach (e.g., college students for a study about the general population).
- **Time series data:** Observations collected over time are often correlated with each other. Today's stock price depends on yesterday's stock price, violating independence.
- **Clustered data:** Students within the same classroom, or workers within the same firm, may be more similar to each other than to randomly selected individuals.

4.7.3 Assumption 3: Variation in the Independent Variable

> **Gauss-Markov Assumption 3: Sample Variation in x**
>
> The independent variable has some variation in the sample: $\sum(x_i - \bar{x})^2 > 0$

This assumption simply requires that x is not constant in our sample. If everyone in our sample has the same value of x, we cannot estimate how changes in x relate to changes in y.

Why it matters: Look back at our formula for $\hat{\beta}_1$:

$$\hat{\beta}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

If all x_i equal $\bar{x}$, the denominator is zero, and the formula is undefined. Intuitively, if everyone has 12 years of education, we have no way to estimate what would happen with 13 years of education.

In practice: This assumption is almost always satisfied. It would only fail in unusual circumstances, like if you accidentally collected a sample where everyone happened to have the exact same value of x. However, having *little* variation in x (even if not zero) can lead to imprecise estimates, as we'll see when discussing variance.

4.7.4 Assumption 4: Zero Conditional Mean

> **i** Gauss-Markov Assumption 4: Zero Conditional Mean
>
> The expected value of the error term is zero for any value of x: $E[\mu|x] = 0$

This is the most important and most frequently violated assumption. It states that the average value of all the unobserved factors affecting y is the same regardless of the value of x.

Understanding the assumption: Let's break this down carefully. The error term μ contains everything that affects y other than x. The zero conditional mean assumption says that if we could collect all people with $x = 10$ and compute the average of their μ values, it would be zero. And if we collected all people with $x = 20$ and averaged their μ values, that would also be zero. The average of unobservables is the same at every level of x.

Example: Education and Wages

Consider the regression:

$$wage = \beta_0 + \beta_1 \cdot educ + \mu$$

The error term μ includes ability, motivation, family connections, and other unobserved factors. The zero conditional mean assumption requires:

$$E[\mu|educ = 8] = E[\mu|educ = 12] = E[\mu|educ = 16] = 0$$

Is this plausible? Probably not. People with higher ability tend to get more education (because school is easier for them, or because they enjoy learning). So among people with 16 years of education, average ability is probably higher than among people with 8 years. This means $E[\mu|educ = 16] > E[\mu|educ = 8]$, violating the assumption.

Why it matters: When this assumption fails, OLS is **biased**. The estimate $\hat{\beta}_1$ will systematically overshoot or undershoot the true β_1. In the education example, if high-ability people get more education, our estimate of the "return to education" will partly capture the "return to ability," making education look more valuable than it really is.

This is the essence of **omitted variable bias**, which we'll explore in detail shortly.

4.7.5 Assumption 5: Homoskedasticity

> **Gauss-Markov Assumption 5: Homoskedasticity**
>
> The variance of the error term is constant across all values of x: $Var(\mu|x) = \sigma^2$

This assumption states that the spread of the error term is the same regardless of the value of x. The "noise" in our data is equally noisy everywhere.

Understanding the assumption: While Assumption 4 concerns the *mean* of μ at different values of x, Assumption 5 concerns the *variance*. Homoskedasticity (from Greek: "same scatter") means the scatter of points around the regression line is constant. **Heteroskedasticity** (different scatter) occurs when the variance of μ depends on x.

Example: Income and Spending

Consider a regression of consumption spending on income. For low-income households, spending is relatively constrained—there's not much room for variation because most income goes to necessities. For high-income households, there's much more discretion in spending patterns, leading to greater variation. This would create heteroskedasticity: $Var(\mu|income)$ increases with income.

```
`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
```

Why it matters: Unlike Assumption 4, violating homoskedasticity does **not** make OLS biased. Your estimate of β_1 is still centered on the true value. However, heteroskedasticity affects the *variance* and *standard errors* of the estimator. The usual standard error formulas become incorrect, which can lead to invalid hypothesis tests and confidence intervals.

We'll discuss how to handle heteroskedasticity later in the course using **robust standard errors**.

4.7.6 Summary: Which Assumptions Matter for What?

Assumption	Needed for Unbiasedness?	Needed for Valid Standard Errors?
1. Linear in parameters	Yes	Yes

Assumption	Needed for Unbiasedness?	Needed for Valid Standard Errors?
2. Random sampling	Yes	Yes
3. Variation in x	Yes (otherwise undefined)	Yes
4. Zero conditional mean	**Yes**	Yes
5. Homoskedasticity	No	**Yes**

The key insight: Assumptions 1-4 are required for OLS to be **unbiased** (on average, we hit the target). Assumption 5 is additionally required for the usual standard error formulas to be valid and for OLS to be **BLUE** (Best Linear Unbiased Estimator).

4.8 Unbiasedness of OLS

Before discussing unbiasedness, let's revisit an important distinction we introduced in Appendix B: the difference between the **population** and the **sample**.

4.8.1 Population Parameters vs. Sample Estimates

The **population** represents the entire group we're interested in studying. In our wage-education example, the population might be all workers in the U.S. economy. The population regression model describes the *true* relationship in this population:

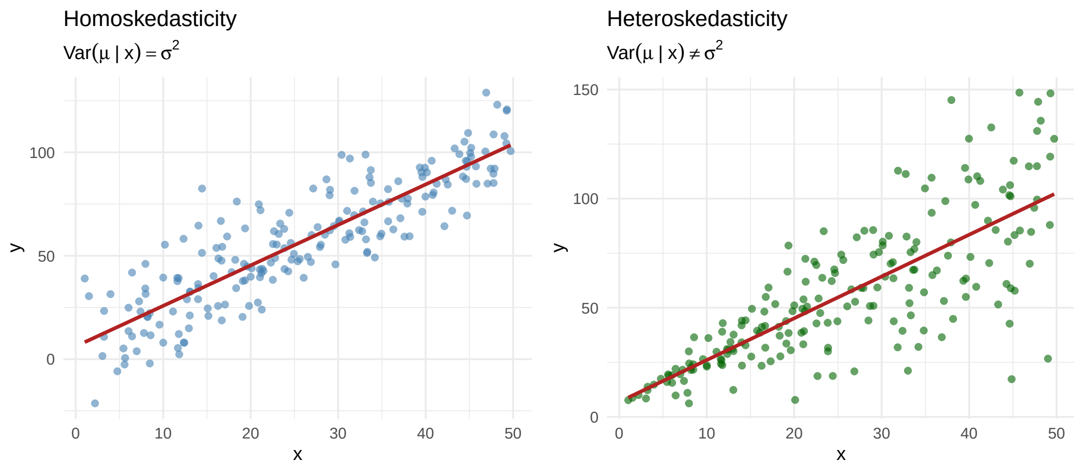
$$y = \beta_0 + \beta_1 x + \mu$$

The parameters β_0 and β_1 are **population parameters**—fixed, unknown constants that describe how x and y are related for everyone in the population. We never observe these true values directly.

Instead, we collect a **sample**—a subset of observations from the population. Using this sample, we compute **estimates** $\hat{\beta}_0$ and $\hat{\beta}_1$. These estimates are our best guesses for the unknown population parameters.

Here's the key insight: *if we drew a different sample, we would get different estimates.* Your classmate analyzing a different random sample of workers would compute different values of $\hat{\beta}_0$ and $\hat{\beta}_1$ than you would. This randomness in the estimates is called **sampling variability**.

Figure 4.3: Left: Homoskedasticity—the spread of points around the line is constant. Right: Heteroskedasticity—the spread increases with x.



4.8.2 What Does Unbiasedness Mean?

Given that estimates vary from sample to sample, how do we know if our estimation procedure is any good? One desirable property is **unbiasedness**.

An estimator $\hat{\theta}$ is **unbiased** if its expected value equals the true population parameter:

$$E[\hat{\theta}] = \theta$$

Unbiasedness does *not* mean that any particular estimate exactly equals the truth. Rather, it means that if we could:

1. Draw a random sample from the population
2. Compute $\hat{\beta}_1$ from that sample
3. Repeat steps 1-2 many, many times
4. Average all the $\hat{\beta}_1$ values we computed

…then that average would equal the true β_1.

In other words, our estimation procedure doesn't systematically overshoot or undershoot the target. Individual estimates might be too high or too low, but on average, we get it right.

4.8.3 Proof Sketch

Starting from our formula for $\hat{\beta}_1$ and substituting the true model $y_i = \beta_0 + \beta_1 x_i + \mu_i$:

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^{n}(x_i - \bar{x})\mu_i}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Taking expectations and using the zero conditional mean assumption ($E[\mu_i|x_i] = 0$):

$$E[\hat{\beta}_1] = \beta_1 + \frac{\sum_{i=1}^{n}(x_i - \bar{x}) \cdot 0}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \beta_1$$

Thus, OLS is unbiased when the Gauss-Markov assumptions hold.

4.8.4 Simulation: Unbiasedness in Action

Show Code: Unbiasedness Simulation

```
# Create a "population" with known parameters
set.seed(1248)
n_pop <- 1000
x_pop <- sample(1:100, size = n_pop, replace = TRUE)
err_pop <- rnorm(n_pop)

# True model: y = 3.5 + 2.21*x + error
pop_data <- tibble(
  x = x_pop,
  y = 3.5 + 2.21 * x + err_pop
)

# Draw many samples and estimate beta_1 each time
set.seed(812476)
n_simulations <- 500
sample_size <- 25

sim_results <- map_dfr(1:n_simulations, function(i) {
  # Draw a random sample
  sample_data <- slice_sample(pop_data, n = sample_size, replace = TRUE)

  # Estimate OLS
  reg <- lm(y ~ x, data = sample_data)

  tibble(
    iteration = i,
    b1_hat = coef(reg)[2]
  )
})

# Calculate the average of all estimates
mean_b1 <- mean(sim_results$b1_hat)
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<ce>
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
```

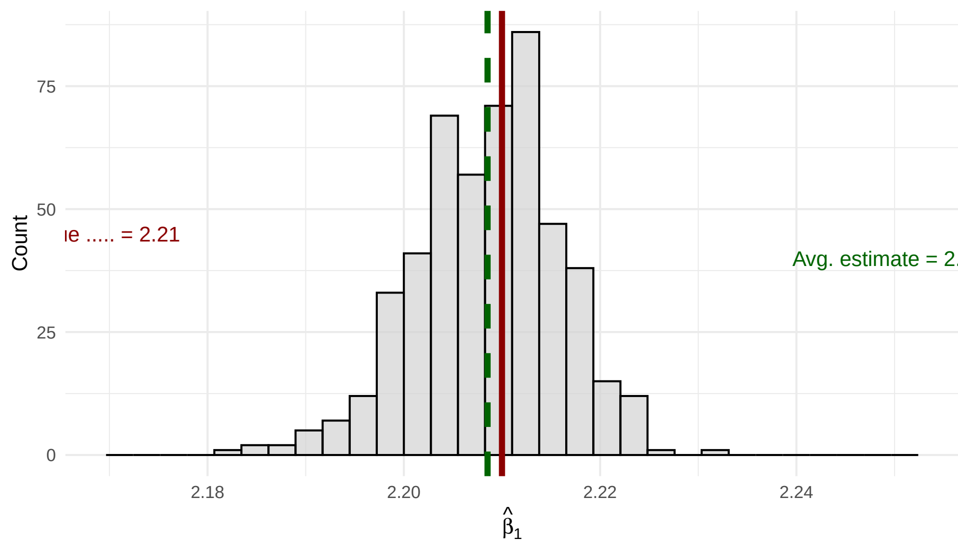
```
<b2>
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<e2>
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<82>
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<81>
```

Figure 4.4: Distribution of OLS slope estimates across 500 samples. The green line shows the average of all estimates, which equals the true value (2.21), demonstrating unbiasedness.



The simulation in Figure 4.4 confirms the theory: while individual estimates vary around the true value, the average across many samples equals the true parameter.

4.9 Omitted Variable Bias

What happens when the zero conditional mean assumption fails? The most common culprit is **omitted variable bias**: we've left out a variable that affects both x and y.

Consider the true model:

$$y = \beta_0 + \beta_1 x + \beta_2 z + \mu$$

If we omit z and estimate only $y = \beta_0 + \beta_1 x + \nu$ (where $\nu = \beta_2 z + \mu$), then whenever z is correlated with x, we have $E[\nu|x] \neq 0$, and our estimate of β_1 is biased.

Example: In the education-wage regression, if ability (z) affects both education choices (x) and wages (y), then omitting ability creates bias. The estimated "return to education" will partly reflect the return to ability.

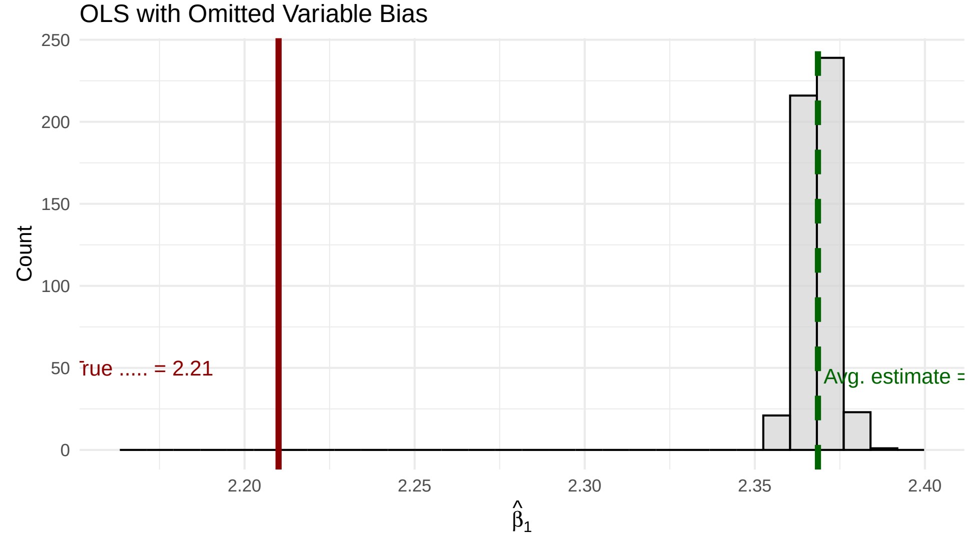
```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<ce>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<b2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<e2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<82>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 2.21' in 'mbcsToSbcs': dot substituted for
<81>
```

4.10 Unbiasedness and Causality: Why This All Matters

Let's step back and connect what we've learned to the bigger picture. In Chapter 3, we discussed how the fundamental challenge of causal inference is selection bias: people who receive "treatment" (more education, a new drug, a policy intervention) may differ systematically from those who don't. This makes it hard to know whether differences in outcomes are caused by the treatment or by those pre-existing differences.

Now we can see the same problem through the lens of regression. **The zero conditional mean assumption is essentially a "no selection bias" assumption.**

Figure 4.5: With omitted variable bias, the average of OLS estimates (green) systematically differs from the true value (red). The bias persists even with many samples.



4.10.1 When Can We Interpret $\hat{\beta}_1$ Causally?

Consider our wage regression:

$$wage = \beta_0 + \beta_1 \cdot educ + \mu$$

When $E[\mu|educ] = 0$, our OLS estimate $\hat{\beta}_1$ is unbiased for β_1. But what does β_1 actually represent?

If the zero conditional mean assumption holds, then β_1 represents the **causal effect** of education on wages: how much wages would change if we could take a person and give them one more year of education, holding everything else constant. In this case, $\hat{\beta}_1$ is an unbiased estimate of this causal effect.

But if $E[\mu|educ] \neq 0$—say, because high-ability people get more education—then β_1 no longer represents a pure causal effect. It conflates the effect of education with the effect of ability. Our estimate $\hat{\beta}_1$ is biased, and we cannot interpret it as the causal effect of education.

4.10.2 Correlation vs. Causation, Revisited

Recall that the OLS slope can be written as:

$$\hat{\beta}_1 = \rho_{xy} \frac{\sigma_y}{\sigma_x}$$

Bivariate OLS is fundamentally measuring the *correlation* between x and y, scaled by their relative standard deviations. Correlation becomes causation **only when the Gauss-Markov assumptions hold**—especially the zero conditional mean assumption.

This is why we emphasize these assumptions so heavily. Without them:

- OLS still gives you numbers
- Those numbers still describe the correlation in your data
- But you **cannot** interpret those numbers as causal effects

> ⚠ The Key Insight
>
> OLS always estimates *something*. Whether that something is a causal effect or merely a correlation depends entirely on whether the assumptions are satisfied. Most of econometrics is about finding clever ways to make the zero conditional mean assumption more plausible.

4.11 Variance of the OLS Estimator

Even when OLS is unbiased, our estimates vary from sample to sample. The **variance** of $\hat{\beta}_1$ tells us how much variation to expect.

Under the Gauss-Markov assumptions:

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{SST_x}$$

where $\sigma^2 = Var(\mu)$ is the error variance and $SST_x = \sum(x_i - \bar{x})^2$ is the total variation in x.

This formula reveals two key insights:

1. **More noise (σ^2) means more variance:** If there's a lot of unexplained variation in y, our estimates are less precise.
2. **More variation in x means less variance:** Spreading out our observations of x gives us more information and more precise estimates. This also implies that larger samples (which have more total variation) yield more precise estimates.

4.12 Standard Errors

The **standard error** of $\hat{\beta}_1$ is the square root of the variance:

$$se(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{SST_x}}$$

where $\hat{\sigma} = \sqrt{\frac{SSR}{n-2}}$ is the estimated standard deviation of the errors.

Standard errors quantify the precision of our estimates and are essential for hypothesis testing and confidence intervals (which we'll cover in the next chapter).

```r
# R reports standard errors automatically
summary(lm_mpg)
```

```
Call:
lm(formula = mpg ~ hp, data = mtcars)

Residuals:
    Min      1Q  Median      3Q     Max 
-5.7121 -2.1122 -0.8854  1.5819  8.2360 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 30.09886    1.63392  18.421  < 2e-16 ***
hp          -0.06823    0.01012  -6.742 1.79e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.863 on 30 degrees of freedom
Multiple R-squared:  0.6024,    Adjusted R-squared:  0.5892 
F-statistic: 45.46 on 1 and 30 DF,  p-value: 1.788e-07
```

The standard error of $\hat{\beta}_1$ is 0.01, shown in the `Std. Error` column.

4.13 OLS is BLUE

Under all five Gauss-Markov assumptions, OLS is **BLUE**:

- **B**est: Lowest variance among all linear unbiased estimators
- **L**inear: The estimator is a linear function of the y_i's
- **U**nbiased: $E[\hat{\beta}] = \beta$
- **E**stimator: It's a rule for using sample data to estimate population parameters

This is a powerful result: among all estimators that are linear and unbiased, OLS has the smallest variance. You cannot do better without either using non-linear methods or accepting some bias.

4.14 Summary and Conclusion

This chapter introduced Ordinary Least Squares regression, the foundational tool of econometrics. We started with the goal of modeling how the conditional expectation $E[Y|X]$ varies with x, and showed how the linear regression model $y = \beta_0 + \beta_1 x + \mu$ provides a framework for this.

We derived the OLS estimator by minimizing the sum of squared residuals, learned to compute regressions in R, and explored the properties of the OLS regression line. We then turned to the statistical properties of OLS, introducing the Gauss-Markov assumptions and proving that OLS is unbiased when these assumptions hold. When the zero conditional mean assumption fails—typically due to omitted variables—our estimates become biased.

Crucially, we connected unbiasedness to causality: the zero conditional mean assumption is what allows us to interpret OLS coefficients as causal effects rather than mere correlations. Without this assumption, OLS still produces numbers, but those numbers cannot be given a causal interpretation.

Finally, we discussed the variance of OLS estimates and the conditions under which OLS is the best linear unbiased estimator.

4.14.1 Key Takeaways

- **The goal** is to estimate the conditional expectation $E[Y|X] = \beta_0 + \beta_1 x$
- **The OLS slope** equals the covariance of x and y divided by the variance of x:

$$\hat{\beta}_1 = \frac{Cov(x, y)}{Var(x)}$$

- **Bivariate OLS is essentially correlation**—be very careful about causal interpretation
- **The Gauss-Markov assumptions** (especially zero conditional mean) are required for unbiasedness

- **Omitted variable bias** arises when we exclude a variable correlated with both x and y
- **Causality requires unbiasedness**—only when $E[\mu|x] = 0$ can we interpret $\hat{\beta}_1$ as a causal effect
- **OLS is BLUE** under the Gauss-Markov assumptions—best among linear unbiased estimators

4.15 Check Your Understanding

4.15.1 Practice: Interpreting Regression Output

The following questions present regression results for you to interpret.

Show Explanation

i. The slope coefficient β_1 is the marginal effect—it tells us how much y changes when x increases by one unit. The intercept β_0 gives the predicted value when $x = 0$.

ii. The zero conditional mean assumption $E[\mu|x] = 0$ states that the expected value of the error term is zero for any given value of x. This is crucial for unbiasedness.

iii. The OLS slope equals the sample covariance of x and y divided by the sample variance of x. This follows directly from the derivation.

iv. Unbiasedness means that if we could repeat the sampling process many times, the *average* of our estimates would equal the true parameter. Individual estimates can still miss the target.

v. Omitted variable bias occurs when we leave out a variable that belongs in the model and is correlated with our included independent variable. This causes $E[\mu|x] \neq 0$.

vi. The zero conditional mean assumption ($E[\mu|x] = 0$) is needed to prove OLS is unbiased, but it is not needed to derive the OLS formulas themselves. The calculus minimization approach works regardless of this assumption—it just tells us which values minimize squared residuals.

vii. As sample size increases, we get more variation in x, so $SST = \sum(x_i - \bar{x})^2$ increases. Since this appears in the denominator, the variance of our estimate decreases. Larger samples give more precise estimates.

Regression Interpretation Questions:

viii. The coefficient on `str` is -2.28, meaning each additional student per teacher is associated with 2.28 fewer points. For 5 additional students: $5 \times (-2.28) = -11.4$ points.

ix. Using the regression equation: $\hat{y} = 698.93 + (-2.28)(20) = 698.93 - 45.6 = 653.33$

x. R-squared measures the fraction of variance in y explained by the regression. Here, 5.1% of test score variation is explained by class size. A low R-squared doesn't make the coefficient meaningless—it just means other factors also matter.

xi. The coefficient is -1.04, so a 10-cent increase predicts: $10 \times (-1.04) = -10.4$ packs per capita.

xii. The negative relationship between price and quantity is exactly what economic theory predicts—the law of demand.

xiii. States with strong anti-smoking policies might have both higher cigarette taxes (raising prices) AND other policies that reduce smoking. This would create omitted variable bias, making price appear more effective than it actually is at reducing consumption.

::::

Note: This section contains interactive content only available in the HTML version.

5 Multivariate Regression

Key Questions

- Why do we need multivariate regression?
- How do we interpret coefficients when there are multiple independent variables?
- What is the zero conditional mean assumption in the multivariate case?
- How does multivariate OLS function as a “matching” estimator?
- What is the omitted variable bias formula and how do we use it?
- What is the bias-variance tradeoff when deciding which controls to include?

Suggested Readings

- Wooldridge (2019), Ch. 3

In Chapter 4, we introduced bivariate OLS and saw that causal interpretation requires the zero conditional mean assumption: $E[\mu|x] = 0$. We also saw that this assumption is frequently violated due to omitted variables—factors in the error term that are correlated with both x and y.

This chapter introduces **multivariate OLS**, which allows us to explicitly control for confounding variables. By including additional variables on the right-hand side of our regression, we can make the zero conditional mean assumption more plausible and get closer to causal estimates.

5.1 The Problem with Bivariate OLS

Let’s revisit why bivariate OLS often fails to give us causal estimates.

Consider a regression of wages on education:

$$wage = \beta_0 + \beta_1 \cdot educ + \mu$$

For $\hat{\beta}_1$ to be an unbiased estimate of the causal effect of education, we need $E[\mu|educ] = 0$. But the error term μ contains everything else that affects wages: ability, family connections, motivation, and countless other factors.

Many of these factors are likely correlated with education. For example, people from wealthier families tend to get more education (better schools, tutors, college is affordable) *and* earn higher wages (family connections, inherited wealth). If family income is in μ and correlated with education, then $E[\mu|educ] \neq 0$, and our estimate is biased.

> **!** The Core Problem
>
> With bivariate OLS, the zero conditional mean assumption requires that *all* factors affecting y (other than x) be uncorrelated with x. This is almost never plausible in observational data.

5.2 The Multivariate Solution

Multivariate OLS offers a solution: **explicitly control** for variables that might be confounders.

If we're worried that family income confounds the education-wage relationship, we can include it in our model:

$$wage = \beta_0 + \beta_1 \cdot educ + \beta_2 \cdot family\_income + \mu$$

Now β_1 represents the effect of education on wages **holding family income constant**. The error term μ no longer contains family income, so that source of bias is eliminated.

5.2.1 The General Multivariate Model

The general multivariate linear regression model with k independent variables is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \mu \tag{5.1}$$

where:

- y is the dependent variable
- $x_1, x_2, \dots, x_k$ are the independent variables
- β_0 is the intercept
- β_j is the coefficient on x_j (for $j = 1, 2, \dots, k$)
- μ is the error term containing all factors not included in the model

5.2.2 Interpreting Coefficients

In multivariate regression, each coefficient has a **ceteris paribus** interpretation:

$$\beta_j = \frac{\partial y}{\partial x_j}$$

This is the change in y associated with a one-unit increase in x_j, **holding all other independent variables constant**.

For example, in the model:

$$wage = \beta_0 + \beta_1 \cdot educ + \beta_2 \cdot family\_income + \mu$$

- β_1: The change in wages for one additional year of education, **holding family income constant**
- β_2: The change in wages for one additional dollar of family income, **holding education constant**

5.2.3 The Zero Conditional Mean Assumption

In multivariate OLS, the zero conditional mean assumption becomes:

$$E[\mu | x_1, x_2, \ldots, x_k] = 0$$

This states that the average of unobserved factors is zero for *all unique combinations* of the independent variables. In other words, everything in μ must be uncorrelated with *all* the variables we've included.

This assumption is much more plausible than its bivariate counterpart. By including relevant control variables, we've removed them from μ, making it more likely that what remains is uncorrelated with our variables of interest.

5.3 Deriving the Multivariate OLS Estimator

The derivation follows the same logic as bivariate OLS: we minimize the sum of squared residuals.

5.3.1 The Minimization Problem

Given a sample of n observations, we want to find values $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ that minimize:

$$SSR = \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{1i} - \hat{\beta}_2 x_{2i} - \dots - \hat{\beta}_k x_{ki})^2$$

5.3.2 First-Order Conditions

Taking partial derivatives with respect to each $\hat{\beta}_j$ and setting them equal to zero yields $k+1$ first-order conditions:

$$\sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{1i} - \dots - \hat{\beta}_k x_{ki}) = 0$$

$$\sum_{i=1}^{n} x_{1i}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{1i} - \dots - \hat{\beta}_k x_{ki}) = 0$$

$$\vdots$$

$$\sum_{i=1}^{n} x_{ki}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{1i} - \dots - \hat{\beta}_k x_{ki}) = 0$$

Solving this system of equations yields the OLS estimates. The algebra is tedious (and typically handled with matrix methods), so we won't derive the explicit formulas here. Fortunately, your computer does this effortlessly.

5.4 Estimating Multivariate OLS in R

In R, we estimate multivariate regression using the same `lm()` function, simply adding variables with the `+` symbol.

5.4.1 Example: MPG, Horsepower, and Weight

Let's estimate the effect of horsepower on fuel efficiency, first without and then with a control for vehicle weight.

```r
# Load data
data(mtcars)

# Bivariate regression
lm_bivariate <- lm(mpg ~ hp, data = mtcars)

# Multivariate regression (adding weight as control)
lm_multivariate <- lm(mpg ~ hp + wt, data = mtcars)
```

Bivariate results:

```r
summary(lm_bivariate)
```

```

Call:
lm(formula = mpg ~ hp, data = mtcars)

Residuals:
    Min      1Q  Median      3Q     Max 
-5.7121 -2.1122 -0.8854  1.5819  8.2360 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 30.09886    1.63392  18.421  < 2e-16 ***
hp          -0.06823    0.01012  -6.742 1.79e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.863 on 30 degrees of freedom
Multiple R-squared:  0.6024,	Adjusted R-squared:  0.5892 
F-statistic: 45.46 on 1 and 30 DF,  p-value: 1.788e-07
```

Multivariate results:

```r
summary(lm_multivariate)
```

```
Call:
lm(formula = mpg ~ hp + wt, data = mtcars)

Residuals:
   Min     1Q Median     3Q    Max 
-3.941 -1.600 -0.182  1.050  5.854 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 37.22727    1.59879  23.285  < 2e-16 ***
hp          -0.03177    0.00903  -3.519  0.00145 ** 
wt          -3.87783    0.63273  -6.129 1.12e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.593 on 29 degrees of freedom
Multiple R-squared:  0.8268,	Adjusted R-squared:  0.8148 
F-statistic: 69.21 on 2 and 29 DF,  p-value: 9.109e-12
```

5.4.2 Interpreting the Results

In the **bivariate** regression, the coefficient on horsepower is -0.068: each additional unit of horsepower is associated with 0.068 fewer miles per gallon.

In the **multivariate** regression (controlling for weight), the coefficient on horsepower drops to -0.032—less than half the size!

What happened? Horsepower and weight are positively correlated (more powerful engines tend to be in heavier vehicles), and weight independently reduces fuel efficiency. The bivariate estimate was conflating the effect of horsepower with the effect of weight. Once we control for weight, we isolate the effect of horsepower alone.

> What Changed?
>
> The coefficient on `hp` dropped from -0.068 to -0.032 when we added `wt`. This suggests that much of what appeared to be the "effect" of horsepower was actually the effect of weight. This is precisely what controlling for confounders accomplishes.

5.5 Multivariate OLS as Matching

There's an intuitive way to understand what multivariate OLS does: it's essentially a **matching** estimator. When we "control for" a variable, we're comparing observations that have similar values of that control variable.

5.5.1 The Returns to Education Example

Suppose we want to estimate the effect of education on wages. A simple comparison of average wages across education levels might be misleading because people with more education tend to work in higher-paying industries. Part of the apparent "return to education" might actually be an "industry premium."

Let's work through this with simulated data to see exactly how controlling for industry works.

```
# Simulate wage data
set.seed(42)
n <- 2000

# Create data where:
# - More educated workers are more likely to be in high-paying industries
# - Industry independently affects wages
wage_data <- tibble(
  # Education: 1 = HS, 2 = Some College, 3 = Bachelor's+
  educ_years = sample(c(12, 14, 16), n, replace = TRUE, prob = c(0.4, 0.3, 0.3)),

  # Higher education -> more likely in professional services (high pay industry)
  ind_professional = rbinom(n, 1, prob = case_when(
    educ_years == 12 ~ 0.20,
    educ_years == 14 ~ 0.40,
    educ_years == 16 ~ 0.65
  )),

  # Wages depend on education AND industry
  wage = 25000 +
    3000 * (educ_years - 12) +     # True education effect: $3000 per year
    15000 * ind_professional +      # Industry premium: $15000
    rnorm(n, 0, 8000)
) |>
  mutate(
    educ_cat = case_when(
```

```
      educ_years == 12 ~ "HS Only",
      educ_years == 14 ~ "Some College",
      educ_years == 16 ~ "Bachelor's+"
    ),
    industry = ifelse(ind_professional == 1, "Professional", "Other")
  )
```

5.5.2 The Naive Comparison

First, let's compute average wages by education level without any controls:

```
naive_means <- wage_data |>
  group_by(educ_cat, educ_years) |>
  summarise(avg_wage = mean(wage), n = n(), .groups = "drop") |>
  arrange(educ_years)

naive_means |>
  knitr::kable(
    col.names = c("Education", "Years", "Avg Wage ($)", "N"),
    digits = 0
  )
```

Education	Years	Avg Wage ($)	N
HS Only	12	28575	829
Some College	14	36961	604
Bachelor's+	16	46341	567

The naive comparison suggests that going from high school to some college is associated with about \$8,386 higher wages, and going from high school to a bachelor's degree is associated with about \$17,766 higher wages.

But wait—the true effect of education in our simulation is only \$3,000 per year (so \$6,000 for 2 extra years and \$12,000 for 4 extra years). The naive estimates are too large! This is because more educated workers are more likely to be in high-paying industries.

5.5.3 Step 1: Compute Within-Industry Differences

To control for industry, we first compute average wages separately for each education-industry combination:

```r
cell_means <- wage_data |>
  group_by(industry, educ_cat, educ_years) |>
  summarise(avg_wage = mean(wage), n = n(), .groups = "drop") |>
  arrange(industry, educ_years)

cell_means |>
  select(industry, educ_cat, avg_wage, n) |>
  knitr::kable(
    col.names = c("Industry", "Education", "Avg Wage ($)", "N"),
    digits = 0
  )
```

Table 5.2: Average wages by education and industry

Industry	Education	Avg Wage ($)	N
Other	HS Only	25275	648
Other	Some College	30351	349
Other	Bachelor's+	36802	199
Professional	HS Only	40388	181
Professional	Some College	46008	255
Professional	Bachelor's+	51499	368

Now let's compute the wage differences **within** each industry:

```r
within_industry <- cell_means |>
  group_by(industry) |>
  mutate(
    diff_from_hs = avg_wage - avg_wage[educ_years == 12]
  ) |>
  filter(educ_years > 12) |>
  select(industry, educ_cat, diff_from_hs, n)

within_industry |>
  knitr::kable(
    col.names = c("Industry", "Education", "Wage Diff vs HS ($)", "N in Cell"),
    digits = 0
  )
```

Industry	Education	Wage Diff vs HS ($)	N in Cell
Other	Some College	5076	349
Other	Bachelor's+	11527	199
Professional	Some College	5620	255
Professional	Bachelor's+	11110	368

Notice that the within-industry differences are much closer to the true values ($6,000 for Some College, $12,000 for Bachelor's+).

5.5.4 Step 2: Compute Weights

To combine these within-industry estimates into a single number, we take a **weighted average**. The weights are based on the number of observations in each industry:

```
# Total workers in each industry
industry_totals <- wage_data |>
  count(industry) |>
  mutate(weight = n / sum(n))

industry_totals |>
  knitr::kable(
    col.names = c("Industry", "N", "Weight"),
    digits = 3
  )
```

Industry	N	Weight
Other	1196	0.598
Professional	804	0.402

5.5.5 Step 3: Compute Weighted Average

Now we combine the within-industry differences using these weights. Let's do this for the Bachelor's+ vs HS comparison:

```
# Get the within-industry differences for Bachelor's+
ba_diffs <- within_industry |>
  filter(educ_cat == "Bachelor's+") |>
  left_join(industry_totals, by = "industry")
```

```r
ba_diffs |>
  select(industry, diff_from_hs, weight) |>
  knitr::kable(
    col.names = c("Industry", "Within-Industry Diff ($)", "Weight"),
    digits = c(0, 0, 3)
  )
```

Industry	Within-Industry Diff ($)	Weight
Other	11527	0.598
Professional	11110	0.402

```r
# Weighted average
weighted_avg <- sum(ba_diffs$diff_from_hs * ba_diffs$weight)
```

The weighted average is:

$$(1.1527 \times 10^4 \times 0.598) + (1.111 \times 10^4 \times 0.402) = 11,360$$

This is much closer to the true effect of $12,000!

5.5.6 The Regression Equivalent

Now let's see what regression gives us:

```r
# Create numeric education variable (years beyond HS)
wage_data <- wage_data |>
  mutate(educ_beyond_hs = educ_years - 12)

# Regression controlling for industry
lm_controlled <- lm(wage ~ educ_beyond_hs + ind_professional, data = wage_data)
summary(lm_controlled)
```

```
Call:
lm(formula = wage ~ educ_beyond_hs + ind_professional, data = wage_data)

Residuals:
     Min       1Q   Median       3Q      Max
```

```
-25629.6  -5261.0     24.9   5383.0  27407.8

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)    
(Intercept)       25179.3      275.7   91.33   <2e-16 ***
educ_beyond_hs     2797.6      117.3   23.84   <2e-16 ***
ind_professional  15180.2      394.9   38.44   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8075 on 1997 degrees of freedom
Multiple R-squared:  0.6093,	Adjusted R-squared:  0.6089 
F-statistic:  1557 on 2 and 1997 DF,  p-value: < 2.2e-16
```

The coefficient on `educ_beyond_hs` is approximately $3,000 per year—almost exactly the true value we built into the simulation!

Compare this to the naive regression without the industry control:

```r
lm_naive <- lm(wage ~ educ_beyond_hs, data = wage_data)
coef(lm_naive)["educ_beyond_hs"]
```

```
educ_beyond_hs 
      4427.053 
```

The naive estimate (4,427) is inflated because it conflates the education effect with the industry premium.

> **Multivariate OLS as Matching**
>
> When you include control variables, OLS is effectively:
>
> 1. **Grouping** observations by similar values of the control variables
> 2. **Computing** the relationship between x and y within each group
> 3. **Taking a weighted average** across groups, where weights reflect group sizes
>
> This is why multivariate regression is sometimes called a "matching" estimator—we're matching people who work in the same industry and comparing their wages based on education.

5.6 Goodness-of-Fit: Adjusted R-Squared

In bivariate OLS, we used R^2 to measure how much variation in y our model explains. The same concept applies to multivariate OLS, but with an important caveat.

5.6.1 The Problem with R-Squared

The R^2 can never decrease when you add more variables—it can only increase or stay the same. This is because adding variables can only improve (or maintain) the fit, even if those variables are irrelevant.

This creates a problem: R^2 mechanically increases as you add more variables, potentially making a bloated model look better than a parsimonious one.

5.6.2 The Adjusted R-Squared

The **adjusted R-squared** penalizes for the number of parameters:

$$\bar{R}^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - k - 1}$$

where n is the sample size and k is the number of independent variables.

The adjusted R^2:

- Is always smaller than the regular R^2
- Can decrease when you add irrelevant variables
- Provides a better comparison between models with different numbers of variables

```r
# Compare R-squared and Adjusted R-squared
c(
  "Bivariate R-sq" = summary(lm_bivariate)$r.squared,
  "Bivariate Adj R-sq" = summary(lm_bivariate)$adj.r.squared,
  "Multivariate R-sq" = summary(lm_multivariate)$r.squared,
  "Multivariate Adj R-sq" = summary(lm_multivariate)$adj.r.squared
)
```

```
       Bivariate R-sq    Bivariate Adj R-sq     Multivariate R-sq
            0.6024373             0.5891853             0.8267855
Multivariate Adj R-sq
            0.8148396
```

5.7 Gauss-Markov Assumptions for Multivariate OLS

The Gauss-Markov assumptions extend naturally to the multivariate case, with one important addition.

5.7.1 Assumption 1: Linear in Parameters

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \mu$$

As in the bivariate case, we can transform variables to model non-linear relationships while maintaining linearity in parameters.

5.7.2 Assumption 2: Random Sampling

We have a random sample $\{(x_{1i}, x_{2i}, \ldots, x_{ki}, y_i) : i = 1, \ldots, n\}$ from the population.

5.7.3 Assumption 3: No Perfect Multicollinearity

This assumption has two parts:

1. **No constant variables:** Each independent variable must have some variation in the sample
2. **No exact linear relationships:** No independent variable can be a perfect linear function of other independent variables

The second part is new and important in multivariate regression.

> ⚠ Perfect Multicollinearity
>
> **Perfect multicollinearity** occurs when one independent variable is an exact linear combination of others. Examples:
>
> - Including both `family_income` and `family_income_thousands` (one is just the other divided by 1000)
> - Including `parent1_income`, `parent2_income`, and `family_income` when `family_income = parent1_income + parent2_income`
> - Including dummy variables for all categories of a categorical variable plus an intercept
>
> When perfect multicollinearity exists, the OLS formulas break down (the system of equations has no unique solution). R will typically drop one of the collinear variables automatically and issue a warning.

5.7.4 Assumption 4: Zero Conditional Mean

$$E[\mu|x_1, x_2, \dots, x_k] = 0$$

The error term has mean zero for all combinations of the independent variables. This can still fail due to:

- Omitted variables correlated with included variables
- Wrong functional form
- Measurement error in independent variables
- Simultaneous causality

5.7.5 Assumption 5: Homoskedasticity

$$Var(\mu|x_1, x_2, \dots, x_k) = \sigma^2$$

The variance of the error term is constant across all combinations of independent variable values.

5.8 The Omitted Variable Bias Formula

Even when we can't include an omitted variable (perhaps because it's unobservable), we can still reason about the **direction** of the bias.

5.8.1 Setup

Suppose the true model is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \mu$$

But we can't observe x_2, so we estimate the misspecified model:

$$\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1$$

where the tilde $\tilde{()}$ indicates estimates from the misspecified model.

5.8.2 The OVB Formula

It can be shown that:

$$\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \tilde{\delta}_1 \tag{5.2}$$

where $\tilde{\delta}_1$ is the coefficient from regressing x_2 on x_1:

$$x_2 = \tilde{\delta}_0 + \tilde{\delta}_1 x_1 + \text{error}$$

Taking expectations:

$$E[\tilde{\beta}_1] = \beta_1 + \beta_2 \tilde{\delta}_1$$

Therefore, the **bias** is:

$$\text{Bias}(\tilde{\beta}_1) = \beta_2 \cdot \tilde{\delta}_1 \tag{5.3}$$

5.8.3 Interpreting the Bias Formula

The bias depends on two factors:

1. **β_2:** The effect of the omitted variable on y
2. **$\tilde{\delta}_1$:** The correlation between the omitted variable (x_2) and the included variable (x_1)

If either factor is zero, there is no bias. But if both are non-zero, bias exists, and its direction depends on the signs:

	$Corr(x_1, x_2) > 0$	$Corr(x_1, x_2) < 0$
$\beta_2 > 0$	Positive bias (overestimate)	Negative bias (underestimate)
$\beta_2 < 0$	Negative bias (underestimate)	Positive bias (overestimate)

5.8.4 Example: Education, Wages, and Ability

Consider estimating the return to education while omitting ability:

- β_2 **(effect of ability on wages):** Positive (more able people earn more)
- $\tilde{\delta}_1$ **(correlation of ability with education):** Positive (more able people get more education)

Therefore:

$$\text{Bias} = (+)(+) = \text{Positive}$$

Our estimate of the return to education is **too large**—it includes part of the return to ability.

5.8.5 Example: Policing, Crime, and Poverty

Consider estimating the effect of policing on crime while omitting local poverty:

- **Effect of poverty on crime (β_2):** Positive (higher poverty → more crime)
- **Correlation of poverty with policing ($\tilde{\delta}_1$):** Positive (high-poverty areas get more policing)

Therefore:

$$\text{Bias} = (+)(+) = \text{Positive}$$

Our estimate suggests policing *increases* crime more than it actually does (or reduces it less than it actually does). This is why naive correlations between police presence and crime can be misleading.

5.9 Variance in Multivariate OLS

Under the Gauss-Markov assumptions, the variance of $\hat{\beta}_j$ is:

$$Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1 - R_j^2)} \tag{5.4}$$

where:

- σ^2 is the error variance
- $SST_j = \sum(x_{ji} - \bar{x}_j)^2$ is the total variation in x_j
- R_j^2 is the R-squared from regressing x_j on all other independent variables

5.9.1 What Increases Variance?

From Equation 5.4, the variance of $\hat{\beta}_j$ increases when:

1. **More noise (σ^2):** Higher error variance means less precise estimates
2. **Less variation in x_j (SST_j):** Less information about how x_j relates to y
3. **Higher R_j^2:** More correlation between x_j and other independent variables

The third point is crucial: when your variable of interest is highly correlated with control variables, your estimate becomes less precise. This is sometimes called **imperfect multicollinearity** or the **multicollinearity problem**.

5.9.2 Standard Errors

The standard error is estimated as:

$$se(\hat{\beta}_j) = \frac{\hat{\sigma}}{\sqrt{SST_j(1 - R_j^2)}}$$

where $\hat{\sigma}^2 = \frac{SSR}{n-k-1}$ is the estimated error variance, and $n - k - 1$ represents the **degrees of freedom**.

5.10 The Bias-Variance Tradeoff

Should you include a variable in your regression? The answer depends on a tradeoff.

5.10.1 The Costs of Omitting a Relevant Variable

If a variable truly belongs in the model ($\beta_2 \neq 0$) and is correlated with x_1, omitting it causes **bias**. Your estimate systematically misses the true value.

5.10.2 The Costs of Including an Irrelevant Variable

What if you include a variable that doesn't belong ($\beta_2 = 0$)?

From the OVB formula:

$$E[\hat{\beta}_1] = \beta_1 + (0) \cdot \tilde{\delta}_1 = \beta_1$$

No bias! Including an irrelevant variable doesn't bias your estimate.

But look at the variance formula. Including an extra variable increases R_1^2 (the correlation between x_1 and other variables), which increases the variance of $\hat{\beta}_1$.

Including irrelevant variables inflates variance without reducing bias.

5.10.3 The Tradeoff

Action	Bias	Variance
Omit relevant variable	Increases	Decreases
Include irrelevant variable	No change	Increases

This is the **bias-variance tradeoff**:

- If you're too conservative (omit too many variables), you risk bias
- If you're too liberal (include everything), you inflate variance

5.10.4 Practical Guidance

There's no perfect solution, but some principles help:

1. **Include variables you have strong theoretical reasons to believe are confounders.** Use economic theory and prior research to guide variable selection.
2. **Be cautious about highly correlated controls.** If two controls are very highly correlated with each other, including both can dramatically inflate variance.
3. **Larger samples help.** As n increases, variance decreases, making the "include it just in case" strategy less costly.
4. **Check how estimates change.** If adding a control substantially changes your coefficient of interest, that control is probably important.

5.11 Simulation: Bias vs. Variance

Let's see the bias-variance tradeoff in action with a simulation.

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Include x ' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Include x ' in 'mbcsToSbcs': dot substituted for <82>
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Include x ' in 'mbcsToSbcs': dot substituted for <82>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Omit x ' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Omit x ' in 'mbcsToSbcs': dot substituted for <82>
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Omit x ' in 'mbcsToSbcs': dot substituted for <82>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Include x ' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Include x ' in 'mbcsToSbcs': dot substituted for <82>
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Include x ' in 'mbcsToSbcs': dot substituted for <82>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Omit x ' in 'mbcsToSbcs': dot substituted for <e2>

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```
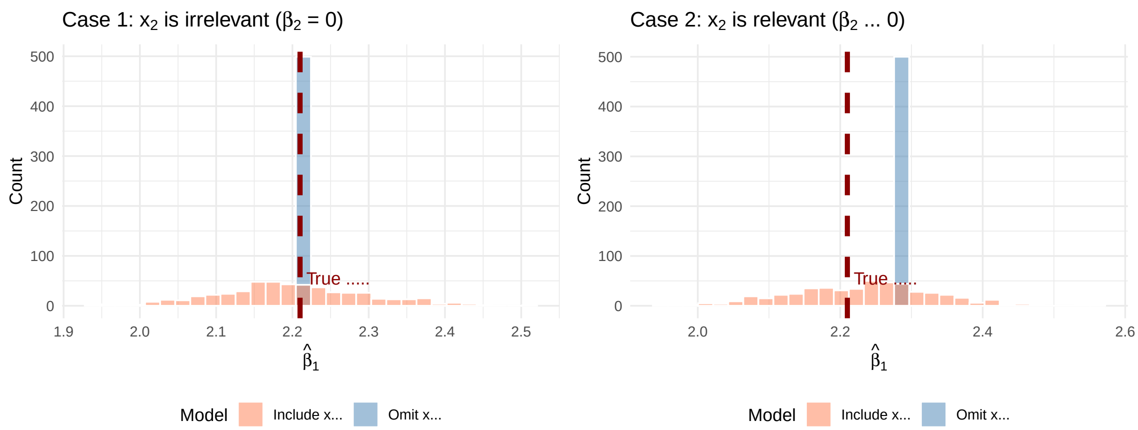

In **Case 1** (left panel), x_2 doesn't actually affect y. Including it (coral) doesn't reduce bias (both distributions are centered on the true value), but it does increase variance (the coral distribution is wider).

In **Case 2** (right panel), x_2 does affect y. Omitting it (blue) causes bias—the distribution is centered away from the true value. Including it (coral) eliminates the bias, even though variance is slightly higher.

Figure 5.1: The bias-variance tradeoff. Left: When x doesn't affect y (=0), including it adds variance without reducing bias. Right: When x does affect y (0), omitting it causes bias.



5.12 Summary

This chapter extended OLS to the multivariate case, allowing us to control for confounding variables.

5.12.1 Key Takeaways

- **Multivariate OLS** includes multiple independent variables:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \mu$$

- **Coefficients have ceteris paribus interpretations:** β_j is the effect of x_j holding all other variables constant
- **The zero conditional mean assumption** now requires $E[\mu|x_1, \ldots, x_k] = 0$—more plausible with good controls
- **Multivariate OLS is like matching:** It compares observations within groups defined by control variables
- **The OVB formula** tells us the direction of bias from omitting a variable:

$$\text{Bias} = \beta_2 \cdot \tilde{\delta}_1$$

- **There's a bias-variance tradeoff:** Omitting relevant variables causes bias; including irrelevant variables inflates variance

- **Perfect multicollinearity** occurs when one variable is an exact linear function of others; it must be avoided

5.13 Check Your Understanding

Note: This section contains interactive content only available in the HTML version.

6 Statistical Inference

Key Questions

- How do we know if our OLS estimate reflects a "real" effect or is just due to random chance?
- What is the sampling distribution of our OLS estimator and why does it matter?
- How do we conduct hypothesis tests using t-statistics and p-values?
- What do confidence intervals tell us about our estimates?
- How do we test multiple hypotheses simultaneously using F-tests?
- How do we read and interpret regression tables in published research?

Suggested Readings

- Wooldridge (2019), Ch. 4

6.1 From Point Estimates to Uncertainty

In the previous chapters, we learned how to estimate the parameters of a regression model using OLS. We discussed how, under certain assumptions, our OLS estimator $\hat{\beta}_j$ is *unbiased*—meaning that on average, across many hypothetical samples, our estimate equals the true population parameter β_j.

But here's the catch: we only ever observe *one* sample. Our single estimate might be too high or too low relative to the truth, simply due to the random variation inherent in sampling. **Statistical inference** is the set of tools that allows us to quantify this uncertainty and make statements about population parameters based on sample data.

The central question of statistical inference is: *Is our single OLS estimate "real," or is it due to random chance?*

6.2 The Sampling Distribution: A Simulation

To understand why uncertainty matters, let's run a simulation. Imagine we have a population where the true relationship between x and y is:

$$y = 2.45 + 0 \cdot x + \mu$$

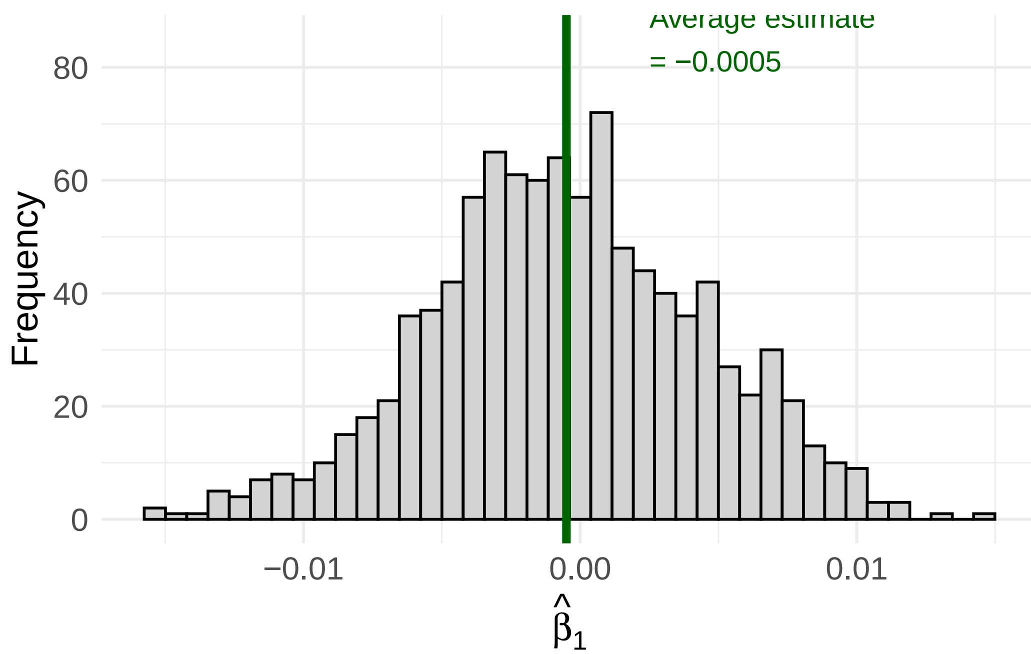
Notice that the true effect of x on y is **exactly zero**. In the population, x has no effect on y whatsoever.

But what happens when we draw samples from this population and estimate OLS regressions? Let's find out.

```
set.seed(124890)

# Create our "population" data
pop_data <- tibble(
  x = sample(1:100, size = 1000, replace = TRUE),
  err = rnorm(1000),
  y = 2.45 + 0 * x + err  # True beta_1 = 0
)

# Number of simulation repetitions
n_sims <- 1000

# Run the simulation
sim_results <- map_dfr(1:n_sims, function(i) {
  # Draw a random sample of 50 observations
  sample_data <- pop_data |>
    slice_sample(n = 50, replace = TRUE)

  # Estimate OLS regression
  reg <- lm(y ~ x, data = sample_data)

  # Store the estimate
  tibble(
    iteration = i,
    b1_hat = coef(reg)["x"]
  )
})

# Calculate the average estimate
```

```r
avg_estimate <- mean(sim_results$b1_hat)

# Create the visualization
ggplot(sim_results, aes(x = b1_hat)) +
  geom_histogram(fill = "lightgrey", color = "black", bins = 40) +
  geom_vline(xintercept = avg_estimate, color = "darkgreen", linewidth = 1.5) +
  annotate("text", x = avg_estimate + 0.003, y = 85,
           label = paste0("Average estimate\n= ", round(avg_estimate, 4)),
           color = "darkgreen", size = 4, hjust = 0) +
  labs(
    x = expression(hat(beta)[1]),
    y = "Frequency"
  ) +
  theme_minimal() +
  theme(
    axis.text = element_text(size = 12),
    axis.title = element_text(size = 14)
  )
```

Figure 6.1: Sampling distribution of ˆ when the true = 0. Each bar represents the frequency of estimates across 1,000 different samples.



Look at what happened! Even though the true $\beta_1 = 0$, we got a *distribution* of different estimates ranging from about -0.02 to +0.02. Some samples produced positive estimates, others produced negative estimates. The average across all samples is very close to zero (as expected from unbiasedness), but any *individual* sample could give us an estimate that deviates from zero.

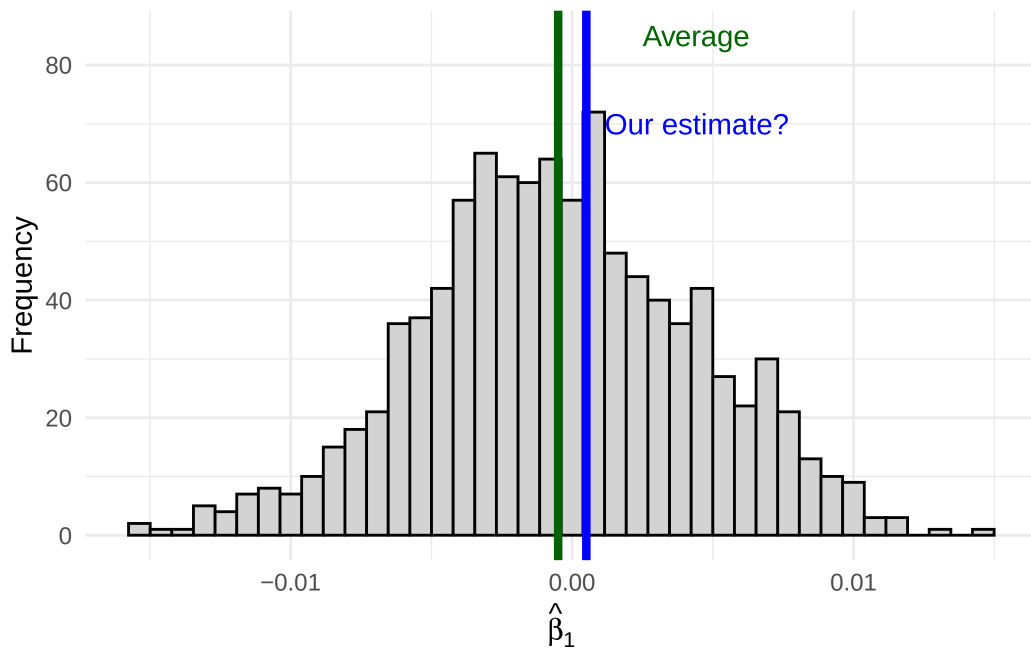
This distribution of estimates across all possible samples is called the **sampling distribution** of $\hat{\beta}_j$.

6.3 Where Does Our Estimate Fall?

Now imagine you only have access to *one* sample—as is always the case in practice. You run your regression and get an estimate of, say, $\hat{\beta}_1 = 0.001$.

How do you know if your estimate comes from here (close to the true value):

Figure 6.2: An estimate close to the average is likely consistent with the null hypothesis.

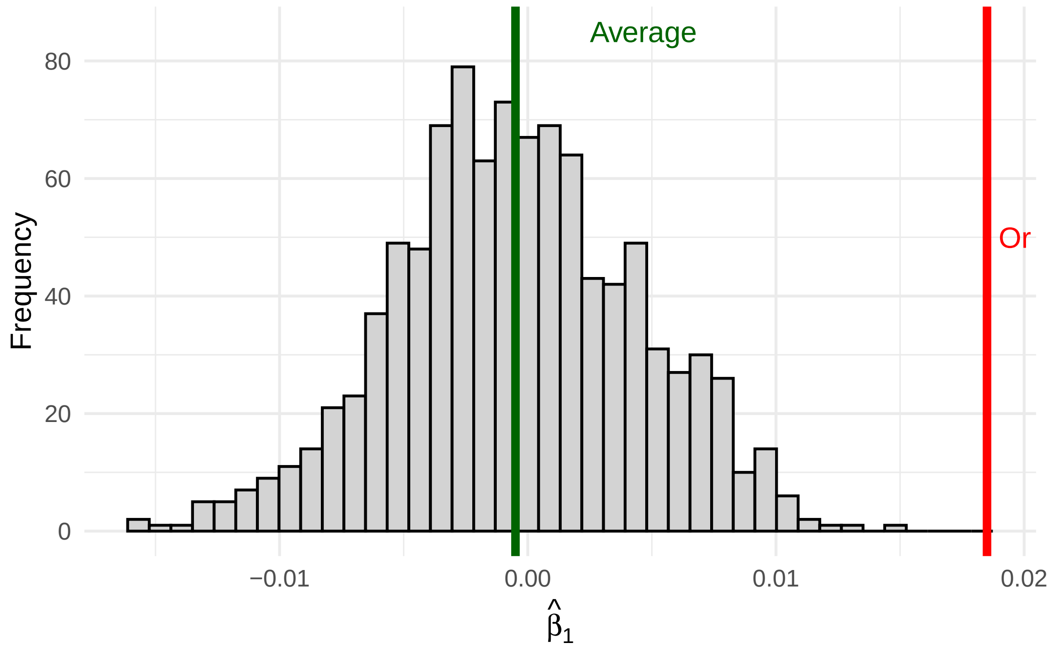


Or from here (far out in the tails):

If your estimate falls in the middle of the distribution where $\beta_1 = 0$, it's consistent with the null hypothesis of no effect. But if your estimate falls way out in the tails, it suggests that maybe the true β_1 isn't zero after all—because getting such an extreme estimate would be very unlikely if β_1 really were zero.

The tools of statistical inference formalize this intuition.

Figure 6.3: An estimate in the tail is unlikely if the true effect is zero—suggesting the true effect might not be zero.



6.4 The Sampling Distribution of $\hat{\beta}_j$

To conduct statistical inference, we need to know the *shape* of the sampling distribution. We've already established two key properties of our OLS estimator (under the Gauss-Markov assumptions):

Expected Value (Unbiasedness):

$$E[\hat{\beta}_j] = \beta_j$$

Variance (under homoskedasticity):

$$Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1 - R_j^2)}$$

where SST_j is the total sum of squares of x_j and R_j^2 is the R-squared from regressing x_j on all other independent variables.

But knowing the mean and variance isn't enough—we need to know the *shape* of the distribution.

6.4.1 The Normality Assumption

To fully characterize the sampling distribution, we make an additional assumption:

! Normality Assumption

The population error μ is independent of the explanatory variables $x_1, x_2, ..., x_k$ and is normally distributed with zero mean and variance σ^2:

$$\mu \sim Normal(0, \sigma^2)$$

Equivalently, we can write:

$$y|\mathbf{x} \sim Normal(\beta_0 + \beta_1 x_1 + ... + \beta_k x_k, \sigma^2)$$

This says that conditional on the independent variables, the outcome y follows a normal distribution centered on the regression line.

Let's visualize what a normal distribution looks like:

```r
set.seed(123)
mu <- 100
sigma <- 15

normal_data <- tibble(
  values = rnorm(n = 1000, mean = mu, sd = sigma)
)

ggplot(normal_data, aes(x = values)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 30,
                 fill = "skyblue",
                 color = "white",
                 alpha = 0.8) +
  stat_function(fun = dnorm,
                args = list(mean = mu, sd = sigma),
                color = "darkred",
                linewidth = 1.2) +
  labs(
    title = "Normal Distribution with Density Curve",
    subtitle = paste0("Generated from N(  = ", mu, ",   = ", sigma, ")"),
    x = "Values (e.g., IQ Scores)",
    y = "Density"
```

```r
  ) +
  theme_minimal()
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
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Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Generated from N(  = 100,   = 15)' in 'mbcsToSbcs': dot
substituted for <83>
```
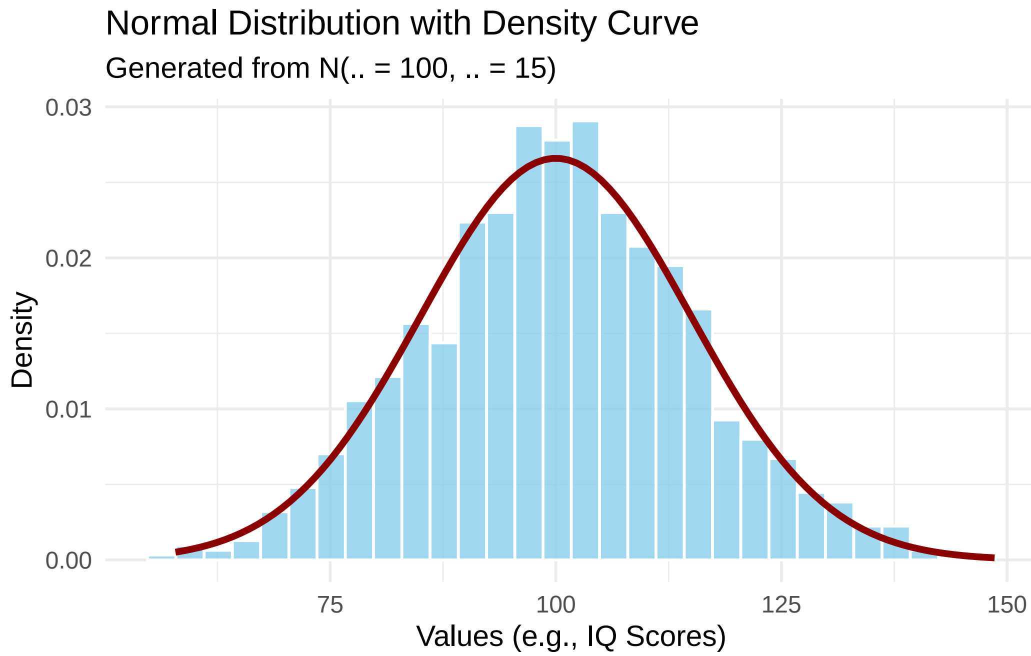

Figure 6.4: The normal (Gaussian) distribution—the classic 'bell curve.' The example shows IQ scores, which are designed to follow N(100, 15).



At first glance, the normality assumption seems quite strong. For many economic variables—like wages, which are bounded below by zero and have a right skew—the distribution is clearly not normal. However, there's good news: as long as our sample size is large enough, the sampling distribution of $\hat{\beta}_j$ will be approximately normal *even if the errors aren't normally distributed*. This result is called **asymptotic normality** and follows from the Central Limit Theorem.

6.4.2 The Normal Sampling Distribution

Under the normality assumption (or with large samples via asymptotic normality), our OLS estimator follows a normal distribution:

$$\hat{\beta}_j \sim Normal\left[\beta_j, Var(\hat{\beta}_j)\right]$$

Let's verify this with a simulation. We'll draw many samples and show that the distribution of estimates indeed looks normal:

```r
# Overlay a normal density on our simulation results
ggplot(sim_results, aes(x = b1_hat)) +
  geom_histogram(aes(y = after_stat(density)),
                 fill = "lightgrey", color = "black", bins = 40) +
  stat_function(
    fun = dnorm,
    args = list(mean = mean(sim_results$b1_hat),
                sd = sd(sim_results$b1_hat)),
    color = "darkred",
    linewidth = 1.2
  ) +
  geom_vline(xintercept = 0, color = "darkgreen", linewidth = 1.2, linetype = "dashed") +
  annotate("text", x = 0.012, y = 55,
           label = "True   = 0", color = "darkgreen", size = 4) +
  labs(
    x = expression(hat(beta)[1]),
    y = "Density"
  ) +
  theme_minimal()
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 0' in 'mbcsToSbcs': dot substituted for <ce>

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conversion failure on 'True   = 0' in 'mbcsToSbcs': dot substituted for <81>
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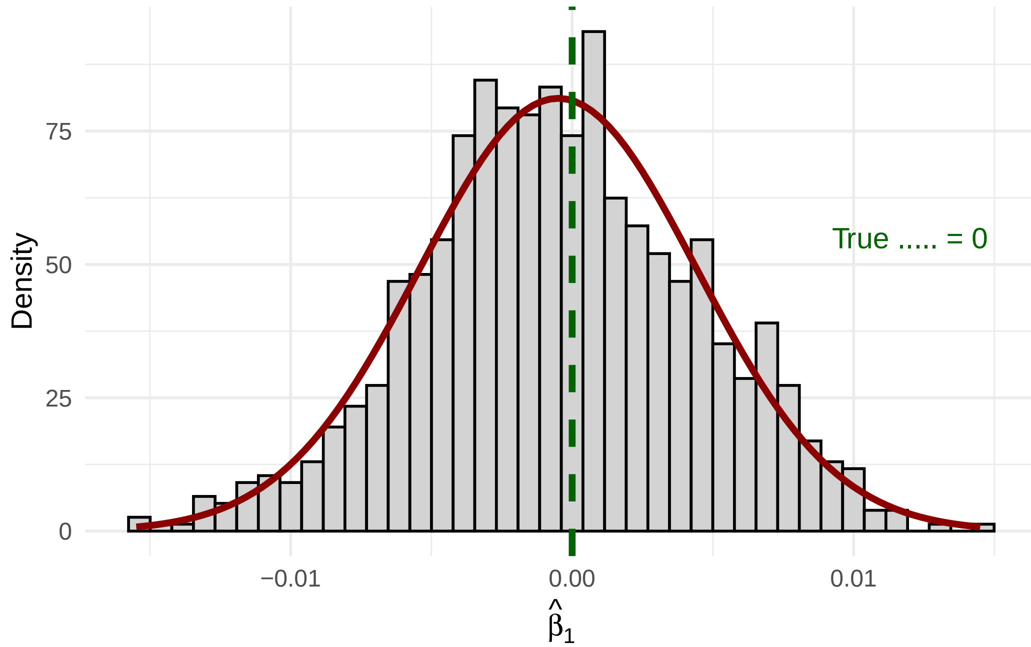
Figure 6.5: The sampling distribution of ˆ follows a normal distribution (red curve) centered on the true value.



The histogram of our 1,000 simulated estimates matches the theoretical normal curve almost perfectly!

Quick Check: This section contains interactive content only available in the HTML version.

6.5 Hypothesis Testing

Now that we understand the sampling distribution, we can use it to make inferences about population parameters. The key idea is **hypothesis testing**: we hypothesize about the value of β_j in the population, then use our sample estimate to evaluate whether the hypothesis is plausible.

6.5.1 The Null and Alternative Hypotheses

In econometrics, we're usually interested in whether a variable has *any* effect on the outcome. We formalize this as:

Null Hypothesis: $H_0 : \beta_j = 0$

The null hypothesis states that the variable x_j has no effect on y in the population (after controlling for other variables).

Alternative Hypothesis: We test the null against one of:

- $H_1 : \beta_j \neq 0$ (two-sided test, most common)
- $H_1 : \beta_j > 0$ (one-sided, if we expect a positive effect)
- $H_1 : \beta_j < 0$ (one-sided, if we expect a negative effect)

For example, consider the wage equation:

$$wage = \beta_0 + \beta_1(educ) + \beta_2(experience) + \beta_3(tenure) + \mu$$

The null hypothesis $H_0 : \beta_2 = 0$ states that, after controlling for education and tenure, experience has no effect on wages.

6.5.2 The t-Statistic

To test our hypothesis, we use the **t-statistic**. Under the null hypothesis, the following quantity follows a t-distribution:

$$\frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} \sim t_{n-k-1}$$

where $n - k - 1$ are the degrees of freedom (sample size minus the number of parameters).

When testing $H_0 : \beta_j = 0$, the t-statistic simplifies to:

$$t_{\hat{\beta}_j} = \frac{\hat{\beta}_j}{se(\hat{\beta}_j)}$$

This is simply the ratio of our estimate to its standard error. The t-stat tells us: *how many standard errors away from zero is our estimate?*

6.5.3 Properties of the t-Statistic

The t-statistic has several useful properties:

1. **Same sign as the estimate:** Since $se(\hat{\beta}_j) > 0$, the t-stat has the same sign as $\hat{\beta}_j$
2. **Magnitude matters:** As $\hat{\beta}_j$ grows in magnitude, so does $t_{\hat{\beta}_j}$
3. **Signal-to-noise ratio:** The t-stat measures how large our estimate is *relative to the noise* (uncertainty) in our data

6.5.4 Visualizing the Hypothesis Test

Let's visualize what the t-test is doing. Under the null hypothesis $H_0 : \beta_j = 0$, we assume that the distribution of possible estimates is centered on zero:

```
set.seed(42)
df_value <- 100  # degrees of freedom

t_sample <- tibble(
  t_value = rt(10000, df = df_value)
)

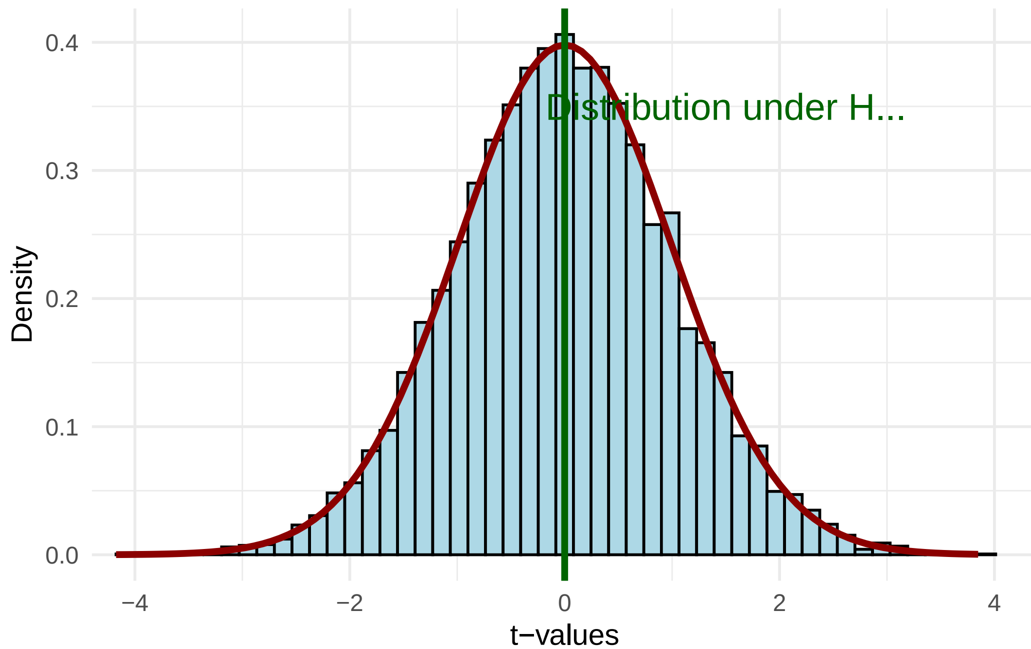
ggplot(t_sample, aes(x = t_value)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 50,
                 fill = "lightblue",
                 color = "black") +
  stat_function(fun = dt,
                args = list(df = df_value),
                color = "darkred",
                linewidth = 1.2) +
  geom_vline(xintercept = 0, color = "darkgreen", linewidth = 1.2) +
  annotate("text", x = 1.5, y = 0.35,
           label = "Distribution under H ",
           color = "darkgreen", size = 5) +
  labs(x = "t-values", y = "Density") +
  coord_cartesian(xlim = c(-4, 4)) +
  theme_minimal()
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Distribution under H ' in 'mbcsToSbcs': dot substituted
for <e2>
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Distribution under H ' in 'mbcsToSbcs': dot substituted
for <82>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Distribution under H ' in 'mbcsToSbcs': dot substituted
for <80>
```

Figure 6.6: The t-distribution under the null hypothesis H :   = 0. Most estimates will fall near zero.



Now suppose we get a t-statistic from our sample. Is it close to zero (consistent with H_0) or far from zero (evidence against H_0)?

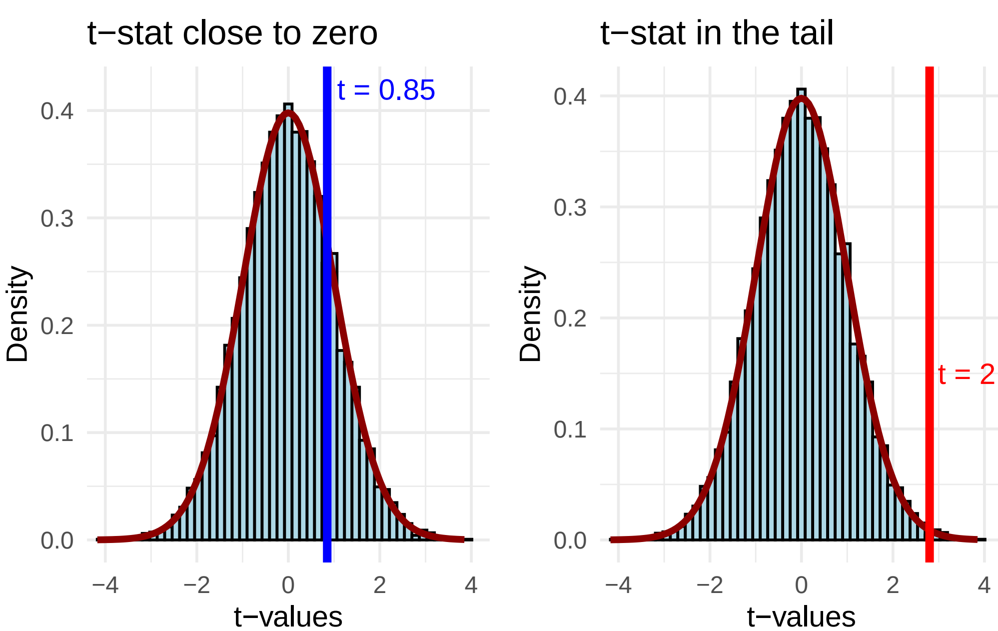
```
p_close <- ggplot(t_sample, aes(x = t_value)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 50, fill = "lightblue", color = "black") +
  stat_function(fun = dt, args = list(df = df_value),
                color = "darkred", linewidth = 1.2) +
  geom_vline(xintercept = 0.85, color = "blue", linewidth = 1.5) +
  annotate("text", x = 0.85, y = 0.42, label = "t = 0.85",
           color = "blue", size = 4, hjust = -0.1) +
  labs(x = "t-values", y = "Density",
```

```
       title = "t-stat close to zero") +
  coord_cartesian(xlim = c(-4, 4)) +
  theme_minimal()

p_far <- ggplot(t_sample, aes(x = t_value)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 50, fill = "lightblue", color = "black") +
  stat_function(fun = dt, args = list(df = df_value),
                color = "darkred", linewidth = 1.2) +
  geom_vline(xintercept = 2.8, color = "red", linewidth = 1.5) +
  annotate("text", x = 2.8, y = 0.15, label = "t = 2.8",
           color = "red", size = 4, hjust = -0.1) +
  labs(x = "t-values", y = "Density",
       title = "t-stat in the tail") +
  coord_cartesian(xlim = c(-4, 4)) +
  theme_minimal()

p_close + p_far
```

Figure 6.7: Two possible t-statistics: one at 0.85 (consistent with H) and one at 2.8 (in the tail, suggesting we reject H).



A t-statistic of 0.85 falls well within the "body" of the distribution—values like this occur frequently when H_0 is true. But a t-statistic of 2.8 is out in the tail—such extreme values are rare when H_0 is true.

6.5.5 Critical Values and Rejection Regions

To formalize our decision rule, we need to define:

1. **Significance level (α):** The probability of rejecting H_0 when it's actually true (Type I error). Common choices: 10%, 5%, or 1%.
2. **Critical value (c):** The threshold value of the t-statistic beyond which we reject H_0.

For a two-sided test at the 5% level, we reject H_0 if $|t_{\hat{\beta}_j}| > c$, where c is the critical value that leaves 2.5% of the distribution in each tail.

```
# Calculate critical value for 5% significance, two-sided
alpha <- 0.05
c_val <- qt(alpha/2, df = df_value, lower.tail = FALSE)

# Create shading data for rejection regions
t_grid <- seq(-4, 4, length.out = 500)
t_dens <- dt(t_grid, df = df_value)

shade_data <- tibble(t_value = t_grid, density = t_dens) |>
  mutate(
    rejection = t_value < -c_val | t_value > c_val
  )

ggplot() +
  # Full distribution
  geom_area(data = shade_data |> filter(!rejection),
            aes(x = t_value, y = density),
            fill = "lightblue", alpha = 0.7) +
  # Rejection regions
  geom_area(data = shade_data |> filter(rejection),
            aes(x = t_value, y = density),
            fill = "red", alpha = 0.5) +
  # Distribution curve
  stat_function(fun = dt, args = list(df = df_value),
                color = "darkblue", linewidth = 1.2) +
  # Critical value lines
  geom_vline(xintercept = c(-c_val, c_val),
             color = "darkgoldenrod", linewidth = 1.2, linetype = "dashed") +
  # Annotations
```

```r
  annotate("text", x = -c_val, y = 0.42,
           label = paste0("-", round(c_val, 2)),
           color = "darkgoldenrod", size = 4, hjust = 1.1) +
  annotate("text", x = c_val, y = 0.42,
           label = paste0("+", round(c_val, 2)),
           color = "darkgoldenrod", size = 4, hjust = -0.1) +
  annotate("text", x = 0, y = 0.2,
           label = "Fail to reject H \n(95%)",
           color = "darkblue", size = 4) +
  annotate("text", x = 3.2, y = 0.05,
           label = "Reject H \n(2.5%)",
           color = "red", size = 3) +
  annotate("text", x = -3.2, y = 0.05,
           label = "Reject H \n(2.5%)",
           color = "red", size = 3) +
  labs(x = "t-values", y = "Density") +
  coord_cartesian(xlim = c(-4, 4)) +
  theme_minimal()
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Fail to reject H ' in 'mbcsToSbcs': dot substituted for
<e2>
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Fail to reject H ' in 'mbcsToSbcs': dot substituted for
<82>
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Fail to reject H ' in 'mbcsToSbcs': dot substituted for
<80>
```

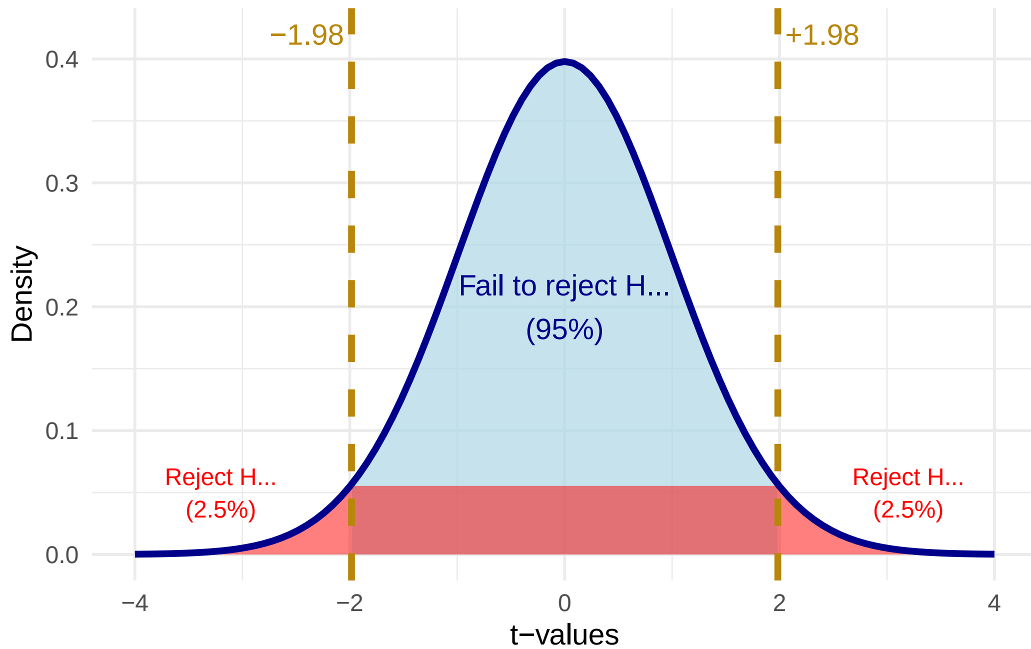
```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Reject H ' in 'mbcsToSbcs': dot substituted for <e2>
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Reject H ' in 'mbcsToSbcs': dot substituted for <82>
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Reject H ' in 'mbcsToSbcs': dot substituted for <80>
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Reject H ' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Reject H ' in 'mbcsToSbcs': dot substituted for <82>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Reject H ' in 'mbcsToSbcs': dot substituted for <80>
```

Figure 6.8: Critical values for a two-sided test at the 5% significance level. We reject H if |t| > 1.96.



6.5.6 Hypothesis Testing Example

Let's work through a complete example. We'll create some simulated housing data and test whether lot area affects sale price:

```r
# Create simulated housing data
set.seed(2025)
n <- 500

housing_data <- tibble(
```

```r
  lot_area = runif(n, 5000, 20000),
  pool_area = rbinom(n, 1, 0.15) * runif(n, 200, 600),
  garage_area = runif(n, 200, 800),
  year_built = sample(1960:2020, n, replace = TRUE),
  year_remod = pmax(year_built, sample(1980:2023, n, replace = TRUE)),
  # True relationship: lot_area has effect of $1.50 per sq ft
  sale_price = -2500000 + 1.50 * lot_area + 80 * pool_area +
               150 * garage_area + 500 * year_built +
               900 * year_remod + rnorm(n, 0, 50000)
)

# Estimate the regression
reg_housing <- lm(sale_price ~ lot_area + pool_area + garage_area +
                    year_built + year_remod,
                  data = housing_data)

summary(reg_housing)
```

```

Call:
lm(formula = sale_price ~ lot_area + pool_area + garage_area +
    year_built + year_remod, data = housing_data)

Residuals:
    Min      1Q  Median      3Q     Max
-134582  -32117     138   32584  167861

Coefficients:
                 Estimate    Std. Error t value             Pr(>|t|)
(Intercept) -2369345.4532   407348.8976  -5.817         0.0000000108 ***
lot_area           1.5528        0.5207   2.982             0.003002 **
pool_area         81.5522       14.6980   5.549         0.0000000470 ***
garage_area      167.0837       13.4931  12.383 < 0.0000000000000002 ***
year_built       554.2853      143.8278   3.854             0.000132 ***
year_remod       775.7472      223.5861   3.470             0.000567 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 51130 on 494 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.3329
F-statistic: 50.81 on 5 and 494 DF,  p-value: < 0.00000000000000022
```

To test $H_0 : \beta_1 = 0$ (lot area has no effect):

```r
# Extract the coefficient and standard error for lot_area
b1_hat <- coef(reg_housing)["lot_area"]
se_b1 <- summary(reg_housing)$coefficients["lot_area", "Std. Error"]

# Calculate the t-statistic
t_stat <- b1_hat / se_b1

# Find the critical value at 5% significance
sig_level <- 0.05
n_obs <- nobs(reg_housing)
k <- 5  # number of independent variables
df <- n_obs - k - 1

t_crit <- qt(sig_level / 2, df = df, lower.tail = FALSE)

# Display results
cat("Estimate:", round(b1_hat, 4), "\n")
```

```
Estimate: 1.5528
```

```r
cat("Standard Error:", round(se_b1, 4), "\n")
```

```
Standard Error: 0.5207
```

```r
cat("t-statistic:", round(t_stat, 2), "\n")
```

```
t-statistic: 2.98
```

```r
cat("Critical value (5%):", round(t_crit, 2), "\n")
```

```
Critical value (5%): 1.96
```

```r
cat("Reject H ?", abs(t_stat) > t_crit, "\n")
```

```
Reject H ? TRUE
```

Since $|t_{\hat{\beta}_1}|$ exceeds the critical value of approximately 1.96, we **reject the null hypothesis** that lot area has no effect on sale price. The estimate is **statistically significant at the 5% level**.

6.5.7 Rule of Thumb

With modern datasets that have hundreds or thousands of observations, the critical value for a two-sided 5% test is essentially always **1.96   2**.

This gives us a handy rule of thumb:

> The "Rule of 2"
>
> An estimate is statistically significant at the 5% level if:
>
> $$|\hat{\beta}_j| > 2 \times se(\hat{\beta}_j)$$
>
> In other words, if your estimate is more than twice as large as its standard error, you can reject the null hypothesis of no effect at the 5% level.

6.6 P-Values

While the t-test with a fixed significance level is useful, it has limitations. Saying an estimate is "significant at 5%" doesn't tell us *how strongly* we reject the null. Was it a narrow rejection or a "knock out"?

The **p-value** answers this question:

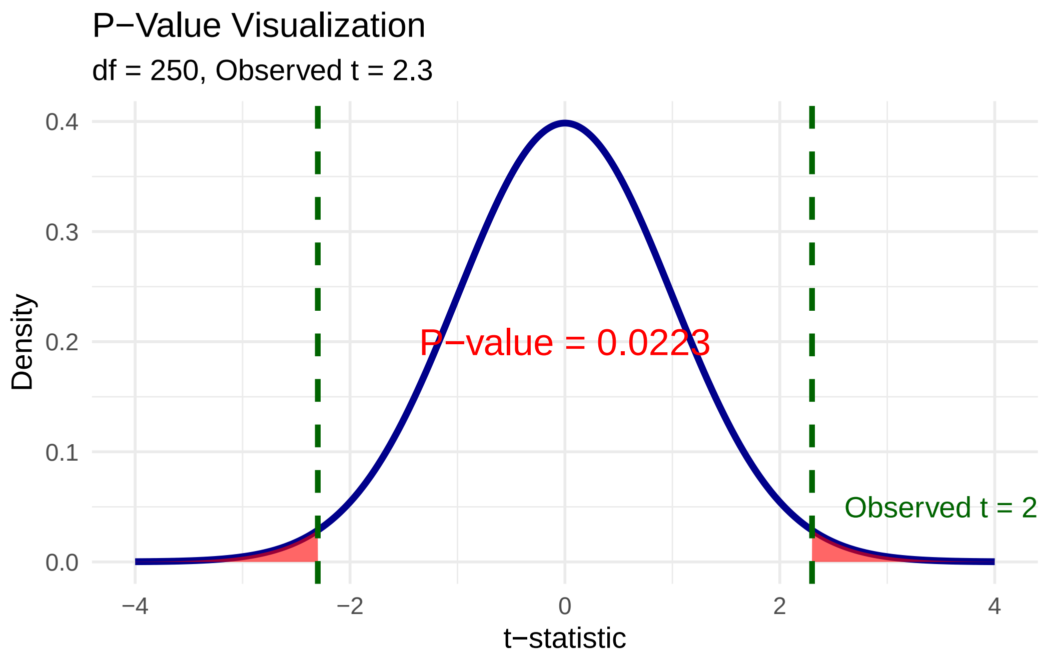
> **!** Definition: P-Value
>
> The p-value is the smallest significance level at which we would reject the null hypothesis. Equivalently: the p-value is the probability of observing a t-statistic as extreme (or more extreme) as the one we calculated, *if the null hypothesis were true.*

6.6.1 Visualizing P-Values

```r
# Parameters
df_viz <- 250
observed_t <- 2.3

# Calculate p-value
p_val <- 2 * pt(abs(observed_t), df = df_viz, lower.tail = FALSE)

# Create the visualization
```

```r
t_grid <- seq(-4, 4, length.out = 500)
t_dens <- dt(t_grid, df = df_viz)

shade_data <- tibble(t_value = t_grid, density = t_dens) |>
  mutate(
    in_tail = abs(t_value) >= abs(observed_t)
  )

ggplot() +
  # Main curve
  geom_line(data = shade_data, aes(x = t_value, y = density),
            color = "darkblue", linewidth = 1.2) +
  # Shaded tails (p-value area)
  geom_area(data = shade_data |> filter(in_tail & t_value > 0),
            aes(x = t_value, y = density),
            fill = "red", alpha = 0.6) +
  geom_area(data = shade_data |> filter(in_tail & t_value < 0),
            aes(x = t_value, y = density),
            fill = "red", alpha = 0.6) +
  # Observed t-statistic lines
  geom_vline(xintercept = c(-observed_t, observed_t),
             color = "darkgreen", linewidth = 1, linetype = "dashed") +
  # Annotations
  annotate("text", x = observed_t + 0.3, y = 0.05,
           label = paste0("Observed t = ", observed_t),
           color = "darkgreen", size = 4, hjust = 0) +
  annotate("text", x = 0, y = 0.2,
           label = paste0("P-value = ", round(p_val, 4)),
           color = "red", size = 5) +
  labs(
    title = "P-Value Visualization",
    subtitle = paste0("df = ", df_viz, ", Observed t = ", observed_t),
    x = "t-statistic",
    y = "Density"
  ) +
  theme_minimal()
```

Figure 6.9: The p-value is the total shaded area in both tails beyond the observed t-statistic.



6.6.2 Interpreting P-Values

The p-value tells us the probability of getting our estimate (or something more extreme) if the null were true:

- **p = 0.035:** There's a 3.5% chance of getting our estimate if $\beta_j = 0$. That's quite unlikely—evidence against H_0.
- **p = 0.35:** There's a 35% chance of getting our estimate if $\beta_j = 0$. That's not unusual at all—no evidence against H_0.

Common decision rules:

- $p < 0.10$: Reject H_0 at 10% level (weak evidence against H_0)
- $p < 0.05$: Reject H_0 at 5% level (moderate evidence against H_0)
- $p < 0.01$: Reject H_0 at 1% level (strong evidence against H_0)

6.6.3 A Note on Language

When we *cannot* reject the null hypothesis, we say:

> "We **fail to reject** the null at the x% level."

We do **not** say we "accept" the null. Why? Because there are many possible values of β_j that we would also fail to reject. Failing to reject $\beta_j = 0$ doesn't mean β_j actually equals zero—it just means our data aren't precise enough to distinguish β_j from zero.

6.6.4 Statistical vs. Practical Significance

⚠ Important Distinction

Statistical significance and **practical (economic) significance** are not the same thing!
A coefficient can be statistically significant but practically unimportant. This is especially common in very large samples, where even tiny effects can be detected.

For example, suppose we find that an additional year of education increases wages by \$0.50 per year, and this is statistically significant with $p < 0.001$. Statistically, we're very confident the effect isn't zero. But practically, a 50-cent annual raise is economically trivial.

Always interpret the *magnitude* of coefficients, not just their statistical significance.

Quick Check: This section contains interactive content only available in the HTML version.

6.7 Confidence Intervals

Another way to quantify uncertainty is through **confidence intervals**. While hypothesis tests ask "is β_j different from zero?", confidence intervals ask "what range of values is β_j likely to fall within?"

6.7.1 Computing a Confidence Interval

A confidence interval at the $(1 - \alpha)$ level is:

$$\hat{\beta}_j \pm c_\alpha \times se(\hat{\beta}_j)$$

where c_α is the critical value at significance level α.

For a 95% confidence interval with large samples:

$$\hat{\beta}_j \pm 1.96 \times se(\hat{\beta}_j)$$

6.7.2 Example

Using our housing regression:

```
# 95% confidence interval for lot_area coefficient
lower <- b1_hat - 1.96 * se_b1
upper <- b1_hat + 1.96 * se_b1

cat("Point estimate:", round(b1_hat, 4), "\n")
```

```
Point estimate: 1.5528
```

```
cat("95% CI: [", round(lower, 4), ",", round(upper, 4), "]\n")
```

```
95% CI: [ 0.5323 , 2.5733 ]
```

```
# Or use confint() function
confint(reg_housing, "lot_area", level = 0.95)
```

```
             2.5 %  97.5 %
lot_area 0.5298262 2.57581
```

6.7.3 Interpreting Confidence Intervals

> "We are 95% confident that the true effect of lot area on sale price is between \$0.53 and \$2.57 per square foot."

More precisely: if we drew 100 different samples and computed 100 different confidence intervals, about 95 of them would contain the true β_j.

Key insight: If the confidence interval includes zero, the estimate is *not* statistically significant at that level.

6.7.4 Simulation: Understanding Confidence Intervals

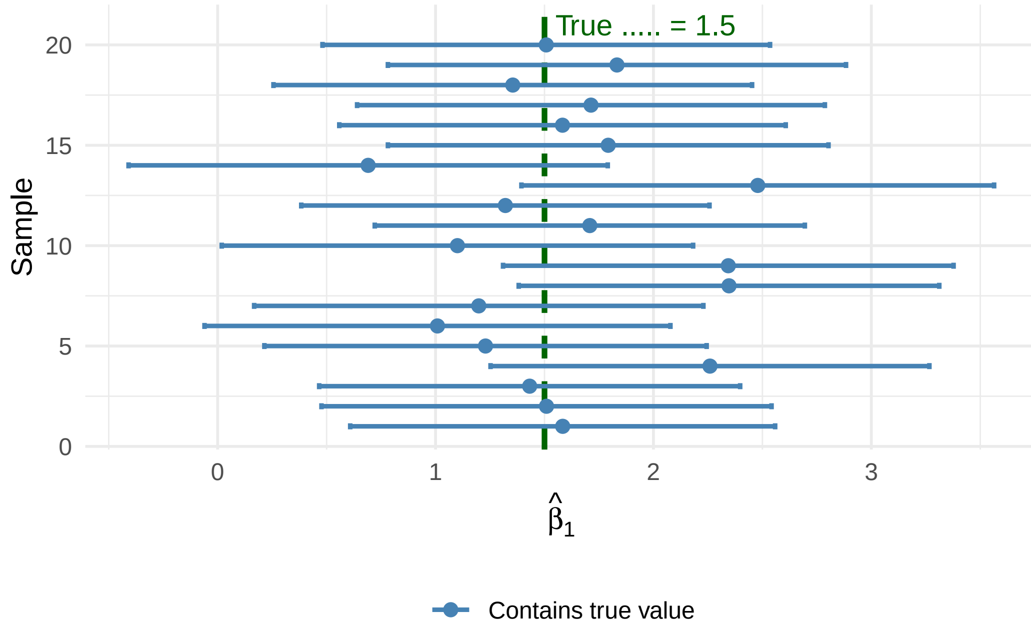
Let's run a simulation to see what "95% confidence" really means:

```r
set.seed(456)

# True parameter value
true_beta1 <- 1.50

# Generate 20 samples and compute CIs
n_samples <- 20

ci_data <- map_dfr(1:n_samples, function(i) {
  # Generate new sample
  sample_data <- tibble(
    lot_area = runif(500, 5000, 20000),
    pool_area = rbinom(500, 1, 0.15) * runif(500, 200, 600),
    garage_area = runif(500, 200, 800),
    year_built = sample(1960:2020, 500, replace = TRUE),
    year_remod = pmax(year_built, sample(1980:2023, 500, replace = TRUE)),
    sale_price = -2500000 + true_beta1 * lot_area + 80 * pool_area +
                 150 * garage_area + 500 * year_built +
                 900 * year_remod + rnorm(500, 0, 50000)
  )

  # Fit model
  reg <- lm(sale_price ~ lot_area + pool_area + garage_area +
              year_built + year_remod, data = sample_data)

  # Get CI
  ci <- confint(reg, "lot_area", level = 0.95)

  tibble(
    sample = i,
    estimate = coef(reg)["lot_area"],
    lower = ci[1],
    upper = ci[2],
    covers_true = lower <= true_beta1 & upper >= true_beta1
  )
})

ggplot(ci_data, aes(y = sample)) +
  geom_vline(xintercept = true_beta1, linetype = "dashed",
             color = "darkgreen", linewidth = 1) +
  geom_errorbarh(aes(xmin = lower, xmax = upper,
                     color = covers_true),
```

```r
                 height = 0.3, linewidth = 0.8) +
  geom_point(aes(x = estimate, color = covers_true), size = 2) +
  scale_color_manual(values = c("TRUE" = "steelblue", "FALSE" = "red"),
                     labels = c("TRUE" = "Contains true value",
                                "FALSE" = "Misses true value")) +
  annotate("text", x = true_beta1 + 0.05, y = 21,
           label = paste0("True   = ", true_beta1),
           color = "darkgreen", size = 4, hjust = 0) +
  labs(
    x = expression(hat(beta)[1]),
    y = "Sample",
    color = NULL
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 1.5' in 'mbcsToSbcs': dot substituted for <ce>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 1.5' in 'mbcsToSbcs': dot substituted for <b2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 1.5' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 1.5' in 'mbcsToSbcs': dot substituted for <82>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'True   = 1.5' in 'mbcsToSbcs': dot substituted for <81>
```

Figure 6.10: Twenty 95% confidence intervals from different samples. The true = 1.50 (dashed line). About 95% of intervals capture the true value.



6.8 T-Tests, P-Values, and CIs: A Comparison

These three approaches are closely related but answer slightly different questions:

Method	Question Answered
T-statistic	Is my estimate large relative to the noise in the data?
P-value	What's the probability of seeing my estimate if H_0 were true?
Confidence Interval	What range of values is plausible for β_j?

All three are mathematically connected:

- If $|t| > 1.96$, then p $<$ 0.05, and the 95% CI excludes zero
- If p $<$ 0.05, then $|t| > 1.96$, and the 95% CI excludes zero
- If the 95% CI excludes zero, then p $<$ 0.05 and $|t| > 1.96$

6.9 Reading Regression Tables

Now that we understand statistical inference, we can properly interpret the regression tables you'll encounter in academic papers.

6.9.1 From R Output to Publication Format

When you run `summary()` on a regression in R, you get detailed but messy output. Academic papers present this information in a standardized, clean format using tables.

```
# Standard R output
summary(reg_housing)
```

```

Call:
lm(formula = sale_price ~ lot_area + pool_area + garage_area +
    year_built + year_remod, data = housing_data)

Residuals:
    Min      1Q  Median      3Q     Max
-134582  -32117     138   32584  167861

Coefficients:
                 Estimate    Std. Error t value             Pr(>|t|)
(Intercept) -2369345.4532   407348.8976  -5.817         0.0000000108 ***
lot_area           1.5528        0.5207   2.982             0.003002 **
pool_area         81.5522       14.6980   5.549         0.0000000470 ***
garage_area      167.0837       13.4931  12.383 < 0.0000000000000002 ***
year_built       554.2853      143.8278   3.854             0.000132 ***
year_remod       775.7472      223.5861   3.470             0.000567 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 51130 on 494 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.3329
F-statistic: 50.81 on 5 and 494 DF,  p-value: < 0.00000000000000022
```

In a publication, this same information would appear as:

6.9.2 Decoding the Table Structure

Rows are variables: The top indicates the dependent variable. Rows below show independent variables and the constant (intercept).

The main numbers are coefficients: Each cell contains $\hat{\beta}_j$.

Numbers in parentheses are standard errors: These are $se(\hat{\beta}_j)$.

Table 6.2: The Effect of Lot Area on Home Sale Price

	Sale Price
(Intercept)	−2 369 345.453***
	(407 348.898)
Lot Area (sq ft)	1.553***
	(0.521)
Pool Area	81.552***
	(14.698)
Garage Area	167.084***
	(13.493)
Year Built	554.285***
	(143.828)
Year Remodeled	775.747***
	(223.586)
Num.Obs.	500
R2	0.340
R2 Adj.	0.333

* p <0.1, ** p <0.05, *** p <0.01
Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Stars indicate significance levels:

- * = significant at 10% level ($p < 0.10$)
- ** = significant at 5% level ($p < 0.05$)
- *** = significant at 1% level ($p < 0.01$)

Bottom statistics: Sample size (N), R-squared, and sometimes other diagnostics.

6.9.3 Multiple Columns Show Robustness

Most regression tables have multiple columns, each representing a different specification. This lets readers see how estimates change as controls are added:

```r
# Three specifications with increasing controls
reg1 <- lm(sale_price ~ lot_area, data = housing_data)
reg2 <- lm(sale_price ~ lot_area + pool_area + garage_area, data = housing_data)
reg3 <- lm(sale_price ~ lot_area + pool_area + garage_area +
             year_built + year_remod, data = housing_data)
```

Notice how the coefficient on Lot Area changes across specifications. In column (1), without controls, the estimate is different from column (3). This pattern often reveals omitted variable bias in simpler specifications.

6.9.4 What to Look for in Regression Tables

1. **What is the coefficient on the variable of interest?** How do we interpret it?
2. **Is it statistically significant?** At what level?
3. **How does the coefficient change across specifications?** Does it remain stable or shift dramatically?
4. **Are the sample size and R-squared reasonable?** Very small N or very low R^2 might raise concerns.

Practice: Reading a Regression Table

Consider this table examining the effect of stock performance on CEO salary:

Is the effect of Return on Stock statistically significant at the 5% level in Column (1)? Select answer… Yes No

What happens to the Adjusted R^2 when we add log(Sales) as a control? Select answer… It increases substantially It decreases It stays about the same

Table 6.3: The Effect of Lot Area on Home Sale Price: Multiple Specifications

	(1)	(2)	(3)
(Intercept)	385 464.906***	288 761.744***	−2 369 345.453***
	(8462.611)	(10 454.349)	(407 348.898)
Lot Area (sq ft)	1.038	1.472***	1.553***
	(0.635)	(0.545)	(0.521)
Pool Area		86.081***	81.552***
		(15.352)	(14.698)
Garage Area		170.687***	167.084***
		(14.112)	(13.493)
Year Built			554.285***
			(143.828)
Year Remodeled			775.747***
			(223.586)
Num.Obs.	500	500	500
R2	0.005	0.274	0.340
R2 Adj.	0.003	0.269	0.333

* p <0.1, ** p <0.05, *** p <0.01
Standard errors in parentheses. * p < 0.1, ** p < 0.05, *** p < 0.01

Table 6.4: Effect of Stock Performance on CEO Salary

	(1)	(2)
(Intercept)	6.806***	4.788***
	(0.041)	(0.234)
Return on Stock (%)	0.001	0.001
	(0.001)	(0.001)
Log(Sales)		0.287***
		(0.033)
Num.Obs.	209	209
R2 Adj.	−0.004	0.264

* p <0.1, ** p <0.05, *** p <0.01

Based on this table, which variable appears to be a more important determinant of CEO salary? Select answer... Return on Stock Company Sales Both are equally important

6.10 F-Tests: Testing Multiple Hypotheses

Sometimes we want to test hypotheses about *multiple* coefficients simultaneously. For example: "Do experience AND tenure jointly affect wages?"

The **F-test** allows us to test such joint hypotheses.

6.10.1 Why Not Just Use Multiple t-Tests?

You might think: "Why not just run separate t-tests on each coefficient?"

There are two problems:

1. **No restrictions on other parameters:** A t-test on β_2 puts no restrictions on β_3. If the variables are correlated, this can be misleading.
2. **Multiple comparisons problem:** If you run many tests at the 5% level, you'll reject *some* true null hypotheses just by chance. With 20 tests, you'd expect about 1 false rejection even if all nulls are true!

The F-test controls for these issues.

6.10.2 Setting Up the F-Test

Consider the wage model:

$$\log(wage) = \beta_0 + \beta_1(education) + \beta_2(experience) + \beta_3(tenure) + \beta_4(age) + \mu$$

Suppose we want to test whether on-the-job training variables (experience and tenure) matter at all:

$$H_0 : \beta_2 = \beta_3 = 0$$
$$H_1 : H_0 \text{ is not true}$$

The alternative is satisfied if *either* $\beta_2 \neq 0$ or $\beta_3 \neq 0$ (or both).

6.10.3 Computing the F-Statistic

The F-test compares two models:

Unrestricted Model: The full model with all variables

$$UR : \log(\widehat{wage}) = \hat{\beta}_0 + \hat{\beta}_1(educ) + \hat{\beta}_2(exper) + \hat{\beta}_3(tenure) + \hat{\beta}_4(age)$$

Restricted Model: The model assuming the null is true (dropping the restricted variables)

$$R : \log(\widehat{wage}) = \hat{\beta}_0 + \hat{\beta}_1(educ) + \hat{\beta}_4(age)$$

The F-statistic is:

$$F = \frac{(SSR_R - SSR_{UR})/q}{SSR_{UR}/df_{UR}}$$

where:

- SSR_R = Sum of squared residuals from restricted model
- SSR_{UR} = Sum of squared residuals from unrestricted model
- q = Number of restrictions (variables dropped)
- df_{UR} = Degrees of freedom in unrestricted model $(n - k - 1)$

6.10.4 F-Test Example

Let's test whether year built and year remodeled jointly affect home prices:

```r
# Unrestricted model (full model)
reg_unrestricted <- lm(sale_price ~ lot_area + pool_area + garage_area +
                         year_built + year_remod,
                       data = housing_data)

# Restricted model (dropping year variables)
reg_restricted <- lm(sale_price ~ lot_area + pool_area + garage_area,
                     data = housing_data)

# Compute F-statistic manually
ssr_r <- sum(resid(reg_restricted)^2)
ssr_ur <- sum(resid(reg_unrestricted)^2)
df_ur <- df.residual(reg_unrestricted)
q <- 2  # number of restrictions
```

```r
f_numerator <- (ssr_r - ssr_ur) / q
f_denominator <- ssr_ur / df_ur
f_stat <- f_numerator / f_denominator

# Critical value at 5% significance
f_crit <- qf(0.05, df1 = q, df2 = df_ur, lower.tail = FALSE)

cat("F-statistic:", round(f_stat, 2), "\n")
```

```
F-statistic: 24.69
```

```r
cat("Critical value (5%):", round(f_crit, 2), "\n")
```

```
Critical value (5%): 3.01
```

```r
cat("Reject H ?", f_stat > f_crit, "\n")
```

```
Reject H ? TRUE
```

Since the F-statistic exceeds the critical value, we **reject the null hypothesis**. Year built and year remodeled are **jointly statistically significant**—they collectively improve the model's fit.

6.10.5 Using R's anova() Function

In practice, you'll use R's `anova()` function to perform F-tests:

```r
anova(reg_restricted, reg_unrestricted)
```

```
Analysis of Variance Table

Model 1: sale_price ~ lot_area + pool_area + garage_area
Model 2: sale_price ~ lot_area + pool_area + garage_area + year_built +
    year_remod
  Res.Df           RSS Df    Sum of Sq      F           Pr(>F)
1    496 1420348267818
2    494 1291292854097  2 129055413721 24.686 0.00000000006048 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The output shows the F-statistic, degrees of freedom, and p-value directly.

6.10.6 The F-Statistic in Terms of R^2

The F-statistic can also be written in terms of R-squared values:

$$F = \frac{(R_{UR}^2 - R_R^2)/q}{(1 - R_{UR}^2)/df_{UR}}$$

This shows that the F-test is essentially asking: *Does the unrestricted model fit the data significantly better than the restricted model?*

If R_{UR}^2 is much larger than R_R^2, the F-statistic will be large, and we'll reject the null.

> **Properties of the F-Test**
>
> - The F-statistic is always non-negative, so F-tests are always **one-sided**
> - If the null is rejected, we say the variables are **jointly statistically significant**
> - We cannot determine *which* individual variable is significant—only that at least one is
> - If variables are jointly insignificant, it provides justification for dropping them from the model

6.11 Chapter Summary

> **Key Takeaways**
>
> 1. **Statistical inference** allows us to move from sample estimates to statements about population parameters, accounting for sampling variability.
> 2. The **sampling distribution** of $\hat{\beta}_j$ describes how our estimates would vary across repeated samples. Under the normality assumption (or with large samples), it follows a normal distribution.
> 3. **Hypothesis testing** uses the t-statistic to determine whether our estimate is "far enough" from zero to reject the null hypothesis: $t = \hat{\beta}_j / se(\hat{\beta}_j)$
> 4. **P-values** tell us the probability of seeing our data if the null were true. Smaller p-values provide stronger evidence against H_0.
> 5. **Confidence intervals** provide a range of plausible values for the population parameter. A 95% CI excludes zero if and only if the estimate is significant at the 5% level.

6. **Statistical significance practical significance.** Always consider the magnitude of effects, not just their p-values.
7. **F-tests** allow us to test multiple hypotheses simultaneously, avoiding the multiple comparisons problem.
8. **Regression tables** in academic papers present coefficients, standard errors (in parentheses), and significance stars. Multiple columns show robustness across specifications.

6.12 Practice Exercises

Note: This section contains interactive content only available in the HTML version.

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A Intro to R and RStudio

Key Questions

- How do I install R and RStudio on my computer?
- What are the main components of the RStudio interface?
- How do I write and run R code?
- What are objects and how do I use them?
- How do I install and load R packages?

Suggested Readings

- Wickham, Çetinkaya-Rundel, and Grolemund (2023), Ch. 1-3

This appendix will walk you through installing R and RStudio, understanding the RStudio interface, and writing your first lines of R code. By the end of this chapter, you will be ready to start working with data and running econometric analyses.

A.1 Installing R

To get started with R, you first need to download and install the base R software. R is available for free from CRAN (the Comprehensive R Archive Network). Navigate to https://cloud.r-project.org/ and click the download link for your operating system (Windows, macOS, or Linux). Follow the installation instructions provided for your system.

Once the installation is complete, you will have R installed on your computer. However, we will not typically interact with R directly. Instead, we will use RStudio, an integrated development environment (IDE) that makes working with R much more convenient.

A.2 Installing RStudio

RStudio is a powerful IDE designed specifically for working with R. It provides a user-friendly interface with helpful features like syntax highlighting, code completion, and integrated help documentation. You can download RStudio Desktop for free from https://posit.co/download/

rstudio-desktop/. The website should automatically detect your operating system and show you the appropriate download link. Click the link and follow the installation instructions.

> **!** Only Use RStudio
>
> After installing both R and RStudio, you will only ever need to open RStudio. RStudio runs R in the background, so you don't need to open the base R application separately. Feel free to delete the shortcut for base R from your desktop if you wish. When you open an R script file (`.R`) for the first time, be sure to set RStudio as the default program to open such files.

A.3 The RStudio Interface

When you first open RStudio, you will see an interface divided into several panels (or "panes"), as seen in Figure A.1. Each panel serves a different purpose in your workflow. Understanding these panels is essential for working efficiently in R.

A.3.1 The Script Editor

The Script Editor is where you write your R code in script files. A script file is simply a text file containing R code that can be saved, edited, and run again later. Working in script files is essential for reproducibility—you can always go back and see exactly what you did, make changes, and re-run your analysis. To open a new script file, go to File → New File → R Script, or use the keyboard shortcut Cmd+Shift+N (Mac) or Ctrl+Shift+N (Windows/Linux).

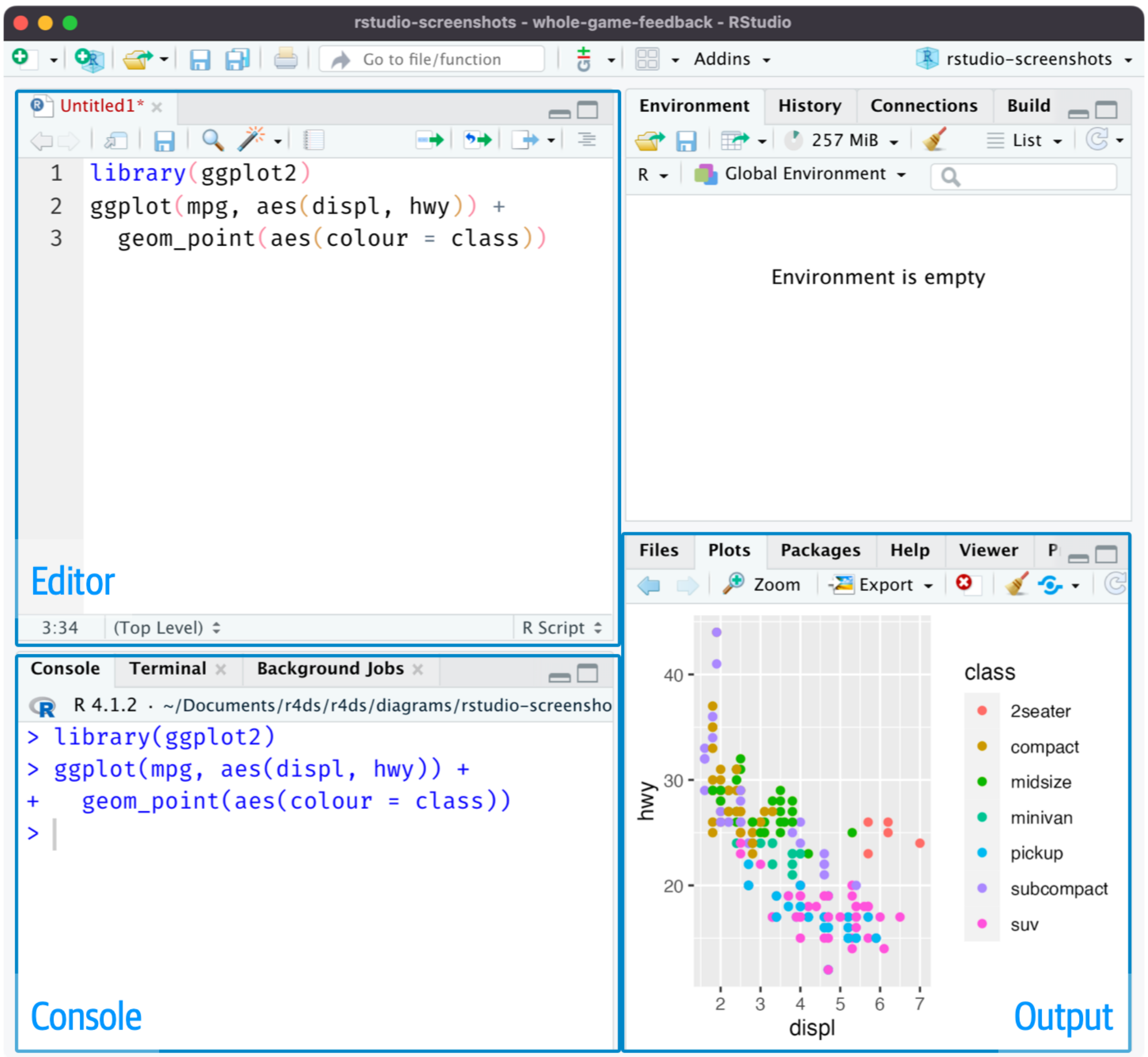
A.3.2 The Console

The Console is where R actually executes your code. When you run code from a script, it gets sent to the console line by line. You can also type code directly into the console and press Enter to run it immediately. However, code typed directly into the console is not saved, so this approach is generally only recommended for quick tests or exploratory work. Output from many functions, including regression results, will be printed to the console.

A.3.3 The Environment Panel

The Environment panel displays all the "objects" you have created in your current R session. These objects might include data frames, vectors, or stored values. You can see each object's name along with a brief summary of its contents. This panel is useful for keeping track of what data and variables you currently have available.

Figure A.1: The RStudio interface with its four main panels labeled.



A.3.4 The Plots and Output Panel

This panel serves multiple purposes. When you create visualizations (like scatter plots or histograms), they will appear here. This panel also contains tabs for browsing files on your computer, viewing installed packages, and accessing R's built-in help documentation.

> Customizing Your Layout
>
> You can hide or show different panels under the "View" menu. This is useful when you want more screen space for a particular task, such as focusing on your script while writing code.

A.4 Basic Coding in R

Now that RStudio is set up, let's write some code! We'll start with simple operations in the console to get familiar with R's syntax.

A.4.1 Arithmetic Operations

R can function as a sophisticated calculator. Try typing the following expressions in the console and pressing Enter:

```r
2 + 2
```

```
[1] 4
```

```r
20 * 5 / 87
```

```
[1] 1.149425
```

```r
sin(pi / 2)
```

```
[1] 1
```

```r
sqrt(6)
```

```
[1] 2.44949
```

As you can see, R can handle basic arithmetic as well as mathematical functions like `sin()` and `sqrt()`. The `pi` you see in the third example is a built-in constant in R representing the mathematical constant π.

A.4.2 Objects and Assignment

R is an "object-oriented language," meaning that data, variables, and nearly everything else is stored as an "object." You create objects using the assignment operator `<-`. The assignment operator takes whatever is on the right side and stores it in the name on the left side.

```r
x <- 2 + 5
```

After running this code, the value `7` is now stored in an object called `x`. Notice that R didn't print anything—it just stored the value. To see what's stored in an object, type its name and press Enter:

```r
x
```

```
[1] 7
```

You can use objects in subsequent calculations, just like variables in algebra:

```r
x + 10
```

```
[1] 17
```

You can also update an object by reassigning it:

```r
x <- x + 50
x
```

```
[1] 57
```

> **Naming Objects**
>
> When creating object names, keep them short but descriptive. Object names cannot contain spaces. Some good examples include: `flight_data`, `reg_controls_1`, `acs_male`, or `X_mat`.

A.4.3 Vectors

A vector is the simplest type of object in R. Vectors contain a sequence of values of the same type (all numbers, all text, etc.). You create vectors using the `c()` function, which stands for "combine":

```
primes <- c(2, 3, 5, 7, 11, 13)
primes
```

```
[1]  2  3  5  7 11 13
```

One powerful feature of R is that arithmetic operations on vectors are applied element-by-element:

```
primes * 2
```

```
[1]  4  6 10 14 22 26
```

```
primes - 1
```

```
[1]  1  2  4  6 10 12
```

You can also perform arithmetic between two vectors of the same length:

```
odds <- c(1, 3, 5, 7, 9, 11)
primes + odds
```

```
[1]  3  6 10 14 20 24
```

A.4.4 Functions

R has a large collection of built-in functions that perform specific tasks. Functions are called using the syntax: `function_name(argument1 = value1, argument2 = value2, ...)`.

For example, the `seq()` function generates a sequence of numbers:

```
bb <- seq(from = 1, to = 30, by = 3)
bb
```

```
 [1]  1  4  7 10 13 16 19 22 25 28
```

Here, `from`, `to`, and `by` are the *arguments* of the function. We set `from = 1` to start at 1, `to = 30` to end at 30, and `by = 3` to increment by 3.

> **Getting Help**
>
> To learn more about any function, type `?` followed by the function name in the console. For example, `?seq` will open the documentation for the `seq()` function.

A.4.5 Data Frames

Data frames are the primary way that structured, tabular data is stored in R. A data frame is essentially a table where each column is a vector and each row is an observation. R comes with several built-in data frames for practice, including `mtcars`, which contains data on various car models.

Use `head()` to view the first few rows of a data frame:

```r
head(mtcars)
```

```
                   mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
Valiant           18.1   6  225 105 2.76 3.460 20.22  1  0    3    1
```

You can explore the structure of a data frame using several helpful functions:

```r
nrow(mtcars)
```

```
[1] 32
```

```r
ncol(mtcars)
```

```
[1] 11
```

```r
colnames(mtcars)
```

```
 [1] "mpg"  "cyl"  "disp" "hp"   "drat" "wt"   "qsec" "vs"   "am"   "gear"
[11] "carb"
```

To access a specific column (variable) within a data frame, use the `$` operator:

```r
head(mtcars$mpg)
```

```
[1] 21.0 21.0 22.8 21.4 18.7 18.1
```

Here, `mtcars$mpg` extracts the `mpg` column from the `mtcars` data frame as a vector. I wrapped it in `head()` to show only the first few values.

A.4.6 Comments

In R, anything preceded by the `#` symbol is treated as a comment and will not be executed. Comments are essential for documenting your code—explaining what each section does, leaving notes for yourself, or marking areas that need revision.

```r
# Find the mean miles per gallon
mean(mtcars$mpg)
```

```
[1] 20.09062
```

Good commenting habits make your code readable to others (and to your future self).

A.5 Working with R Scripts

While the console is useful for quick tests, you should do the vast majority of your work in script files. Scripts allow you to save your code, edit it easily, and reproduce your entire analysis by running the script from top to bottom.

A.5.1 Creating and Running Scripts

To create a new script, go to File → New File → R Script (or use Cmd/Ctrl + Shift + N). Write your code in the script editor, then use the following keyboard shortcuts to run it:

Cmd/Ctrl + Enter executes the current line (or highlighted selection) in the console. The cursor then automatically moves to the next line, making it easy to step through your code line by line.

Figure A.2: Example R script with cursor position shown. Pressing Cmd/Ctrl + Enter will run the complete command that creates the `not_cancelled` object, then move the cursor to the next statement.

```
library(dplyr)
library(nycflights13)

not_cancelled <- flights |>
  filter(!is.na(dep_delay), !is.na(arr_delay))

not_cancelled |>
  group_by(year, month, day) |>
  summarize(mean = mean(dep_delay))
```

Cmd/Ctrl + Shift + S runs the entire script from top to bottom. This is useful for checking that your complete analysis works correctly and produces reproducible results.

A.5.2 Script Best Practices

Following good practices when writing scripts will save you time and frustration. Always start your script with your name and a brief description of the script's purpose in comments. Load all required packages at the beginning of the script using `library()` so anyone reading your code knows what dependencies are needed. Never include `install.packages()` in your script—you only need to install packages once on your computer, and you don't want your script to install packages on someone else's machine without their permission. Save your scripts with short, descriptive names that contain no spaces, such as `assignment_3.R` or `ec_325_data_plotting.R`.

A.6 Working Directories

R uses a concept called the "working directory"—the folder on your computer where R looks for files you want to load and where it saves files you create. You can see your current working directory at the top of the console, or by running:

```
getwd()
```

A good workflow is to save your R script in a dedicated folder for this course (or a subfolder for each week's work), put all related data files in the same folder, and then set your working directory to that location. In RStudio, go to Session → Set Working Directory → To Source File Location to automatically set the working directory to wherever your script is saved.

A.7 R Packages

R comes with a set of "base" packages that provide fundamental functionality. However, the true power of R lies in its vast ecosystem of user-contributed packages. These packages extend R's capabilities for specialized tasks—from data visualization to complex statistical modeling.

A.7.1 Installing Packages

You only need to install a package once on your computer. Use the `install.packages()` function:

```
install.packages("tidyverse")
```

A.7.2 The Tidyverse

The tidyverse is a collection of R packages designed for data science. These packages share a common philosophy and work seamlessly together, making data manipulation and visualization more intuitive. Some of the most commonly used tidyverse packages include `dplyr` (data manipulation), `ggplot2` (visualization), and `readr` (reading data files).

Installing `tidyverse` installs all of these packages at once.

A.7.3 Loading Packages

Once a package is installed, you need to load it each time you start a new R session using `library()`:

```
library(tidyverse)
```

When you load the tidyverse, you'll see a message listing which packages were attached. You may also see messages about "conflicts"—this just means some tidyverse functions have the same name as functions in base R, and the tidyverse versions will take precedence. This is normal and expected behavior.

> **! Remember the Difference**
>
> `install.packages()` downloads and installs a package to your computer (do this once). `library()` loads an already-installed package into your current R session (do this every time you start R).

A.8 Summary and Conclusion

This appendix walked you through the essential first steps for working with R. You installed R and RStudio, learned how the RStudio interface is organized, and wrote your first R code. You now understand how to create and manipulate objects, work with vectors and data frames, write and run scripts, and install packages.

A.8.1 Key Takeaways

- **RStudio is your interface to R:** Always work in RStudio, not base R. The script editor, console, environment, and plots panels each serve distinct purposes in your workflow.
- **Scripts are essential for reproducibility:** Write your code in script files that you can save, edit, and re-run. Use comments to document what your code does.
- **Objects store everything in R:** Use the assignment operator `<-` to create objects. Vectors hold sequences of values; data frames hold structured tabular data.
- **Packages extend R's capabilities:** Install packages once with `install.packages()`, then load them each session with `library()`. The tidyverse is a particularly useful collection of packages for data science.

- **Working directories matter:** Set your working directory to the folder containing your script and data files to keep your analysis organized.

With these foundations in place, you are ready to start loading real data, creating visualizations, and running econometric analyses.

A.9 Check Your Understanding

Note: This section contains interactive content only available in the HTML version.

B Statistics and Probability Review

Key Questions

- What are the properties of the summation operator?
- How do we interpret derivatives and partial derivatives?
- How do we find the maximum or minimum of a function?
- What is the difference between a population and a sample?
- What is a random variable and how do we describe its distribution?
- How do we measure central tendency, variability, and relationships between variables?
- What is the normal distribution and why is it important?

Suggested Readings

- Wooldridge (2019), Appendix A, B, and C

This appendix reviews the essential mathematical and statistical concepts that form the foundation for econometrics. If you've taken introductory statistics or calculus, much of this material will be familiar. However, econometrics uses these tools in specific ways, so it's worth reviewing them with an eye toward how they'll appear throughout the course.

B.1 Basic Mathematical Tools

B.1.1 The Summation Operator

The summation operator provides a compact way to write the sum of a sequence of numbers. If $\{x_i : i = 1, 2, ..., n\}$ is a sequence of n numbers, then the sum of these numbers is written as:

$$x_1 + x_2 + ... + x_n = \sum_{i=1}^{n} x_i$$

The symbol Σ (capital Greek letter sigma) tells us to add up the terms. The expression $i = 1$ below the sigma indicates that we start counting at $i = 1$, and the n above tells us to stop at $i = n$.

B.1.1.1 Properties of Summation

The summation operator has several useful properties that we will use frequently.

Property 1: Summing a constant. If we add a constant c to itself n times, we get n times the constant:

$$\sum_{i=1}^{n} c = nc$$

Property 2: Factoring out constants. We can pull constants outside the summation:

$$\sum_{i=1}^{n} cx_i = c\sum_{i=1}^{n} x_i$$

Property 3: Separating additive terms. Sums can be split across addition:

$$\sum_{i=1}^{n}(ax_i + by_i) = a\sum_{i=1}^{n} x_i + b\sum_{i=1}^{n} y_i$$

> ⚠ Common Mistakes with Summation
>
> These properties do **not** extend to division or multiplication in the ways you might expect. We **cannot** break up divided expressions:
>
> $$\sum_{i=1}^{n} \frac{x_i}{y_i} \neq \frac{\sum_{i=1}^{n} x_i}{\sum_{i=1}^{n} y_i}$$
>
> We **cannot** break up multiplicative expressions:
>
> $$\sum_{i=1}^{n} x_i^2 \neq \left(\sum_{i=1}^{n} x_i\right)^2$$

B.1.1.2 The Sample Mean

Given n numbers $\{x_i : i = 1, 2, ..., n\}$, we can compute their **average** or **mean** value by adding them up and dividing by n. The average value, denoted $\bar{x}$ (read "x-bar"), is written as:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

The average is an example of a **descriptive statistic**: a value that describes or summarizes a characteristic of a set of data points. In this case, the mean describes the **central tendency** of x_i—where the "middle" of the data lies.

B.1.1.3 Useful Summation Results

Two results involving summations will appear repeatedly in this course.

Result 1: Deviations from the mean sum to zero. The sum of the deviations $(x_i - \bar{x})$ from the mean always equals zero:

$$\sum_{i=1}^{n}(x_i - \bar{x}) = 0$$

This can be proven using the properties above:

$$\sum_{i=1}^{n}(x_i - \bar{x}) = \sum_{i=1}^{n} x_i - \sum_{i=1}^{n} \bar{x} = \sum_{i=1}^{n} x_i - n\bar{x} = n\bar{x} - n\bar{x} = 0$$

The last step follows because $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ implies $n\bar{x} = \sum_{i=1}^{n} x_i$.

Result 2: Sum of squared deviations. The sum of squared deviations can be rewritten as:

$$\sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

More generally, for two variables x and y:

$$\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}$$

These identities are useful for computing variances and covariances, which we will discuss shortly.

B.1.2 Derivatives

The derivative $\frac{dy}{dx}$ of a function $y = f(x)$ measures its **instantaneous rate of change**. It tells you how sensitive the function's output is to a small change in its input. In econometrics, derivatives help us interpret how changes in independent variables affect dependent variables.

Here are derivatives of some common functional forms in econometrics:

Function	Derivative
$y = \beta_0 + \beta_1 x$	$\frac{dy}{dx} = \beta_1$
$y = \beta_0 + \beta_1 x + \beta_2 x^2$	$\frac{dy}{dx} = \beta_1 + 2\beta_2 x$
$y = \beta_0 + \frac{\beta_1}{x}$	$\frac{dy}{dx} = -\frac{\beta_1}{x^2}$
$y = \beta_0 + \beta_1 \ln(x)$	$\frac{dy}{dx} = \frac{\beta_1}{x}$

Notice that in the linear case, the derivative is simply the coefficient β_1—the effect of x on y is constant. In the other cases, the derivative depends on the value of x, meaning the effect of x on y varies depending on where you start.

B.1.3 Partial Derivatives

When y is a function of multiple variables, say $y = f(x_1, x_2)$, we use **partial derivatives** to measure how y changes with respect to one variable while holding the others constant.

The partial derivative of y with respect to x_1, holding x_2 constant, is denoted $\frac{\partial y}{\partial x_1}$. Similarly, the partial derivative with respect to x_2, holding x_1 constant, is $\frac{\partial y}{\partial x_2}$.

For example, if $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$, then:

$$\frac{\partial y}{\partial x_1} = \beta_1, \qquad \frac{\partial y}{\partial x_2} = \beta_2$$

The concept of "holding other variables constant" is central to econometrics. When we estimate a regression coefficient, we are trying to measure the partial effect of one variable on the outcome, controlling for other factors.

B.1.4 Optimization: Maxima and Minima

In econometrics, we often need to find the value of a variable that maximizes or minimizes a function. For example, Ordinary Least Squares (OLS) regression finds coefficient estimates by minimizing the sum of squared residuals. Understanding optimization is essential for understanding how these estimates are derived.

B.1.4.1 Finding Critical Points

To find the maximum or minimum of a function $f(x)$, we take the derivative, set it equal to zero, and solve for x. The solutions are called **critical points**.

For example, consider the quadratic function:

$$f(x) = ax^2 + bx + c$$

Taking the derivative and setting it equal to zero:

$$\frac{df}{dx} = 2ax + b = 0$$

Solving for x:

$$x^* = -\frac{b}{2a}$$

This critical point x^* is where the function reaches its maximum or minimum.

B.1.4.2 Determining Maximum vs. Minimum

How do we know if a critical point is a maximum or a minimum? We use the **second derivative test**. The second derivative tells us about the curvature of the function:

$$\frac{d^2f}{dx^2} = 2a$$

The rule is:

- If $\frac{d^2f}{dx^2} > 0$ at the critical point, the function is **concave up** (bowl-shaped), and we have a **minimum**
- If $\frac{d^2f}{dx^2} < 0$ at the critical point, the function is **concave down** (hill-shaped), and we have a **maximum**

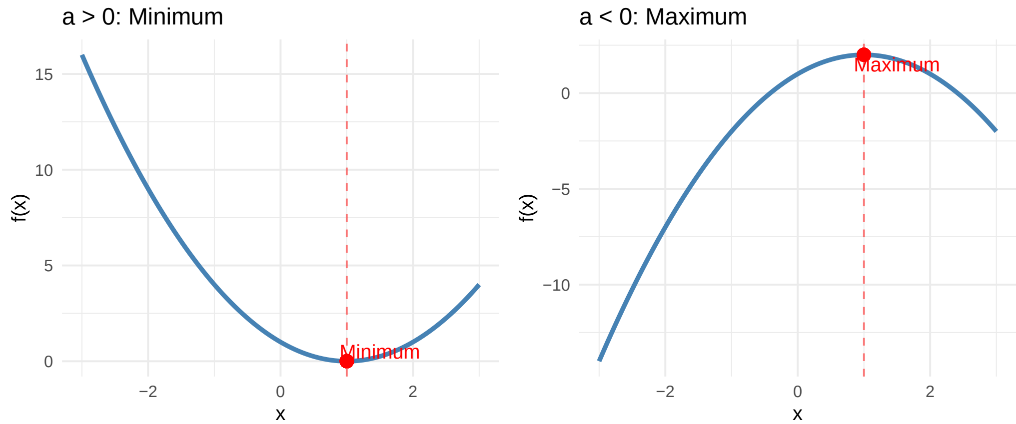
For our quadratic function $f(x) = ax^2 + bx + c$:

- If $a > 0$: The parabola opens upward, and $x^* = -\frac{b}{2a}$ is a **minimum**
- If $a < 0$: The parabola opens downward, and $x^* = -\frac{b}{2a}$ is a **maximum**

```
Warning in geom_point(aes(x = 1, y = 0), color = "red", size = 3): All aesthetics have lengt
i Please consider using `annotate()` or provide this layer with data containing
  a single row.

Warning in geom_point(aes(x = 1, y = 2), color = "red", size = 3): All aesthetics have lengt
i Please consider using `annotate()` or provide this layer with data containing
  a single row.
```

Figure B.1: Left: When a > 0, the parabola opens upward and the critical point is a minimum. Right: When a < 0, the parabola opens downward and the critical point is a maximum.



> **Optimization in OLS**
>
> In OLS regression, we minimize the sum of squared residuals, which is a quadratic function of the coefficients. The second derivative is positive (the function is convex), confirming that the solution is indeed a minimum. This is why setting the first derivative equal to zero gives us the best-fitting line.

B.1.4.3 Optimization with Multiple Variables

When a function depends on multiple variables, we find critical points by setting all partial derivatives equal to zero simultaneously. For a function $f(x_1, x_2)$:

$$\frac{\partial f}{\partial x_1} = 0 \quad \text{and} \quad \frac{\partial f}{\partial x_2} = 0$$

Determining whether a critical point is a maximum, minimum, or neither (a saddle point) requires examining the second partial derivatives, which we will encounter when deriving the OLS estimator.

B.2 Population vs. Sample

Before diving into probability, it's important to understand the distinction between a **population** and a **sample**, as this distinction underlies all of statistical inference.

The **population** is the entire group of individuals, objects, or observations that we are interested in studying. For example, if we want to understand the wages of all U.S. workers, the population is every person employed in the United States. If we want to know the effect of a drug on blood pressure, the population might be all adults with hypertension.

In practice, we almost never observe the entire population. Instead, we collect data on a **sample**—a subset of the population. We might survey 5,000 workers rather than all 160 million, or run a clinical trial with 500 patients rather than everyone with hypertension.

> **!** The Goal of Statistical Inference
>
> The fundamental goal of statistics is to use information from a **sample** to draw conclusions about the **population**. We observe sample statistics (like the sample mean $\bar{x}$) and use them to make inferences about population parameters (like the population mean μ).

This distinction affects our notation:

Concept	Population Parameter	Sample Statistic
Mean	μ	$\bar{x}$
Variance	σ^2	s^2
Standard deviation	σ	s
Correlation	ρ	r

Population parameters are typically denoted with Greek letters and are usually unknown. Sample statistics are computed from our data and are used to estimate the population parameters.

For example, if we want to know the average wage of all U.S. workers (μ), we collect a sample and compute the sample mean ($\bar{x}$). Under certain conditions, $\bar{x}$ is a good estimate of μ. Much of econometrics is about understanding when and how well sample statistics estimate population parameters, and quantifying the uncertainty in those estimates.

B.3 Probability Basics

B.3.1 Random Variables

A **random variable** is a variable whose value is determined by a random phenomenon. It represents a number, but the exact value is determined by chance. Examples include the outcome of a dice roll, whether a coin lands heads or tails, or the percentage of flights that arrive on time.

Most variables we work with in econometrics can be thought of as random variables. A person's wage, years of education, or health status all have elements of randomness—we cannot perfectly predict them in advance.

B.3.2 Probability

Probability quantifies uncertainty. If X is a random variable that takes on k possible values $\{x_1, x_2, ..., x_k\}$, then the probability that X takes a particular value x_j is denoted:

$$p_j = P(X = x_j)$$

Each probability p_j must be between 0 and 1, and the probabilities must sum to 1: $p_1 + p_2 + ... + p_k = 1$.

B.3.3 Probability Distributions

The **probability density function (pdf)**, denoted $f(x)$, gives the probability that a random variable takes on a particular value x. For discrete random variables, $f(x) = P(X = x)$.

The **cumulative distribution function (cdf)**, denoted $F(x)$, gives the probability that the random variable is less than or equal to x:

$$F(x) = P(X \leq x)$$

The cdf is useful for answering questions like "what is the probability that X is below some threshold?"

B.3.4 Expected Values

The **expected value** (or **mathematical expectation**) of a random variable X, denoted $E[X]$, represents the "average" outcome we would expect if we could repeat the random process many times. It is our population "best guess" for the value of X. We often denote it with μ or μ_X.

For a discrete random variable with pdf $f(x)$, the expected value is the weighted average of all possible values, where the weights are the probabilities:

$$E[X] = \sum_{j=1}^{k} x_j f(x_j) = \mu$$

Example: Let X be the outcome of rolling a fair six-sided die. Each value (1 through 6) has probability 1/6. The expected value is:

$$E[X] = \frac{1}{6}(1) + \frac{1}{6}(2) + \frac{1}{6}(3) + \frac{1}{6}(4) + \frac{1}{6}(5) + \frac{1}{6}(6) = 3.5$$

Notice that 3.5 is not a possible outcome of any single roll, but it is the average we would expect over many rolls.

B.3.4.1 Properties of Expected Values

The properties of expected values mirror those of the summation operator:

Property 1: For any constant c, $E[c] = c$.

Property 2: For any constants a and b, $E[aX + b] = aE[X] + b$.

Property 3: If $\{a_1, a_2, ..., a_n\}$ are constants and $\{X_1, X_2, ..., X_n\}$ are random variables, then:

$$E\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i E[X_i]$$

B.4 Measures of Variability

Knowing the central tendency of a variable is useful, but it doesn't tell us everything. Two distributions can have the same mean but look very different if one is tightly clustered around the mean while the other is spread out. We need measures of **variability** or **dispersion**.

B.4.1 Variance

Let X be a random variable with mean $\mu = E[X]$. The **variance** of X, denoted σ^2 or $\text{Var}(X)$, measures how spread out the distribution is:

$$\text{Var}(X) = E[(X - \mu)^2] = \sigma^2$$

The variance is the expected squared deviation from the mean. Squaring serves two purposes: it ensures all deviations are positive (so positive and negative deviations don't cancel out), and it gives more weight to observations far from the mean.

B.4.2 Standard Deviation

The **standard deviation** of X, denoted $\text{sd}(X)$ or σ, is the positive square root of the variance:

$$\text{sd}(X) = \sqrt{\text{Var}(X)} = \sigma$$

The standard deviation is often more interpretable than the variance because it is measured in the same units as X itself. If X is measured in dollars, the standard deviation is also in dollars, whereas the variance would be in "dollars squared."

B.4.3 Covariance

A central concern in econometrics is understanding how variables move together. When X is large, is Y also large? Or is it small? Or is there no relationship at all?

The **covariance** quantifies the linear relationship between two random variables. Let X and Y be random variables with means $\mu_X = E[X]$ and $\mu_Y = E[Y]$. The covariance is defined as:

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

The covariance measures how X and Y move together:

- **Positive covariance:** When X is above its mean, Y tends to be above its mean (and vice versa). The variables move in the same direction.
- **Negative covariance:** When X is above its mean, Y tends to be below its mean. The variables move in opposite directions.
- **Zero covariance:** There is no linear relationship between X and Y.

B.4.4 Correlation Coefficient

The covariance tells us the direction of the relationship, but its magnitude depends on the scales of X and Y, making it hard to interpret. The **correlation coefficient** standardizes the covariance to produce a unitless measure:

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\text{sd}(X) \cdot \text{sd}(Y)} = \rho_{XY}$$

The correlation coefficient is always between -1 and 1:

- $\rho_{XY} = 1$: Perfect positive linear relationship
- $\rho_{XY} = -1$: Perfect negative linear relationship
- $\rho_{XY} = 0$: No linear relationship

Perfect correlations rarely occur in practice. Values of ρ_{XY} closer to 1 or -1 indicate stronger linear relationships.

> **i** Correlation vs. Causation
>
> Remember from Chapter 1: correlation does not imply causation! A strong correlation between X and Y tells us the variables move together, but it does not tell us whether X causes Y, Y causes X, or some third factor causes both.

B.4.5 Conditional Expectations

While correlation measures the overall linear relationship between two variables, we often want to describe how the expected value of one variable changes with another. This is where **conditional expectations** come in.

Let X and Y be random variables. The conditional expectation of Y given that X takes a specific value x is denoted:

$$E[Y|X = x]$$

This asks: what is our "best guess" for Y when we know that X equals x?

Example: What is the expected wage (Y) for people with a college degree ($X = 1$) versus those without ($X = 0$)? $E[Y|X = 1]$ and $E[Y|X = 0]$ answer these questions.

Conditional expectations are fundamental to regression analysis. When we estimate a regression of Y on X, we are essentially modeling how $E[Y|X]$ changes with X.

B.5 Distributions

B.5.1 The Normal Distribution

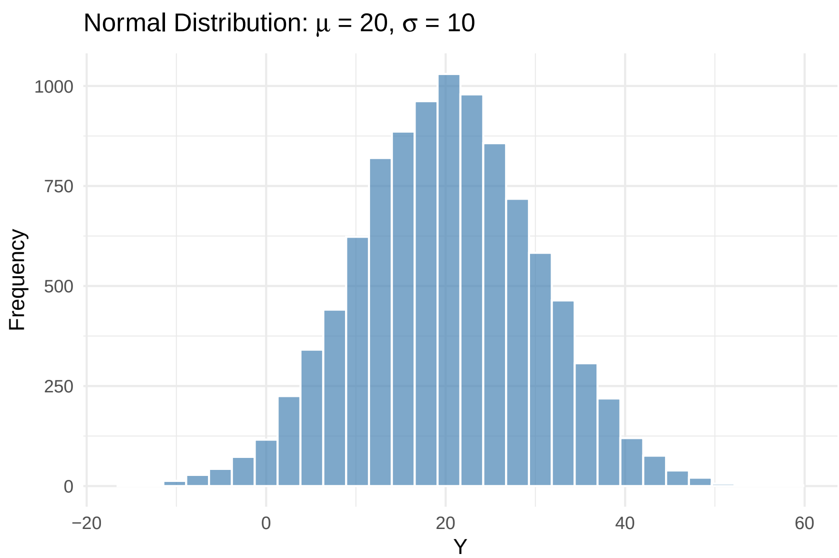
The **normal distribution** (also called the Gaussian distribution) is the most important distribution in statistics and econometrics. Many naturally occurring phenomena are approximately normally distributed, and many statistical procedures assume normality.

A random variable X with mean μ and variance σ^2 that follows a normal distribution is denoted:

$$X \sim N(\mu, \sigma^2)$$

The normal distribution has a characteristic "bell curve" shape: symmetric around the mean, with most observations clustered near the center and fewer observations in the tails.

Figure B.2: A histogram of 10,000 draws from a normal distribution with mean 20 and standard deviation 10.



Two key facts about the normal distribution:

- Approximately **68%** of observations lie within one standard deviation of the mean (between $\mu - \sigma$ and $\mu + \sigma$)
- Approximately **95%** of observations lie within two standard deviations of the mean (between $\mu - 2\sigma$ and $\mu + 2\sigma$)

B.5.2 The Standard Normal Distribution

The **standard normal distribution** is a special case of the normal distribution with mean 0 and variance 1:

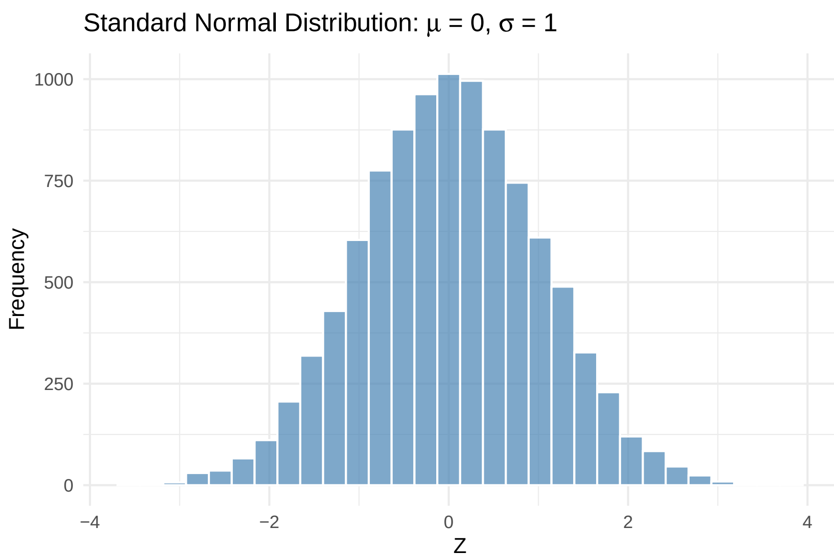
$$Z \sim N(0, 1)$$

The standard normal is important because any normal random variable can be converted to a standard normal through **standardization**. If $X \sim N(\mu, \sigma^2)$, then:

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

Standardization subtracts the mean (centering the distribution at zero) and divides by the standard deviation (scaling it to have unit variance).

Figure B.3: A histogram of 10,000 draws from a standard normal distribution (mean 0, standard deviation 1).



The standard normal is what z-tables (or statistical software) use to compute probabilities. Once you standardize a variable, you can look up the probability of observing values in any range.

B.6 Summary

This appendix reviewed the mathematical and statistical foundations needed for econometrics. We covered the summation operator and its properties, which are essential for understanding formulas throughout the course. We introduced derivatives and partial derivatives, which help us interpret how changes in one variable affect another, and learned how to find and classify maxima and minima of functions. We distinguished between populations and samples—a fundamental concept for statistical inference. We then turned to probability, defining random variables, expected values, variance, covariance, and correlation. Finally, we introduced the normal distribution, which plays a central role in statistical inference.

B.6.1 Key Formulas

Concept	Formula
Sample mean	$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$
Critical point (quadratic)	$x^* = -\frac{b}{2a}$ for $f(x) = ax^2 + bx + c$
Second derivative test	Max if $f''(x^*) < 0$; Min if $f''(x^*) > 0$
Expected value	$E[X] = \sum_{j=1}^{k} x_j f(x_j) = \mu$
Variance	$\text{Var}(X) = E[(X-\mu)^2] = \sigma^2$
Standard deviation	$\text{sd}(X) = \sqrt{\text{Var}(X)} = \sigma$
Covariance	$\text{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$
Correlation	$\rho_{XY} = \frac{\text{Cov}(X,Y)}{\text{sd}(X)\cdot\text{sd}(Y)}$
Standardization	$Z = \frac{X-\mu}{\sigma}$

B.7 Check Your Understanding

Note: This section contains interactive content only available in the HTML version.